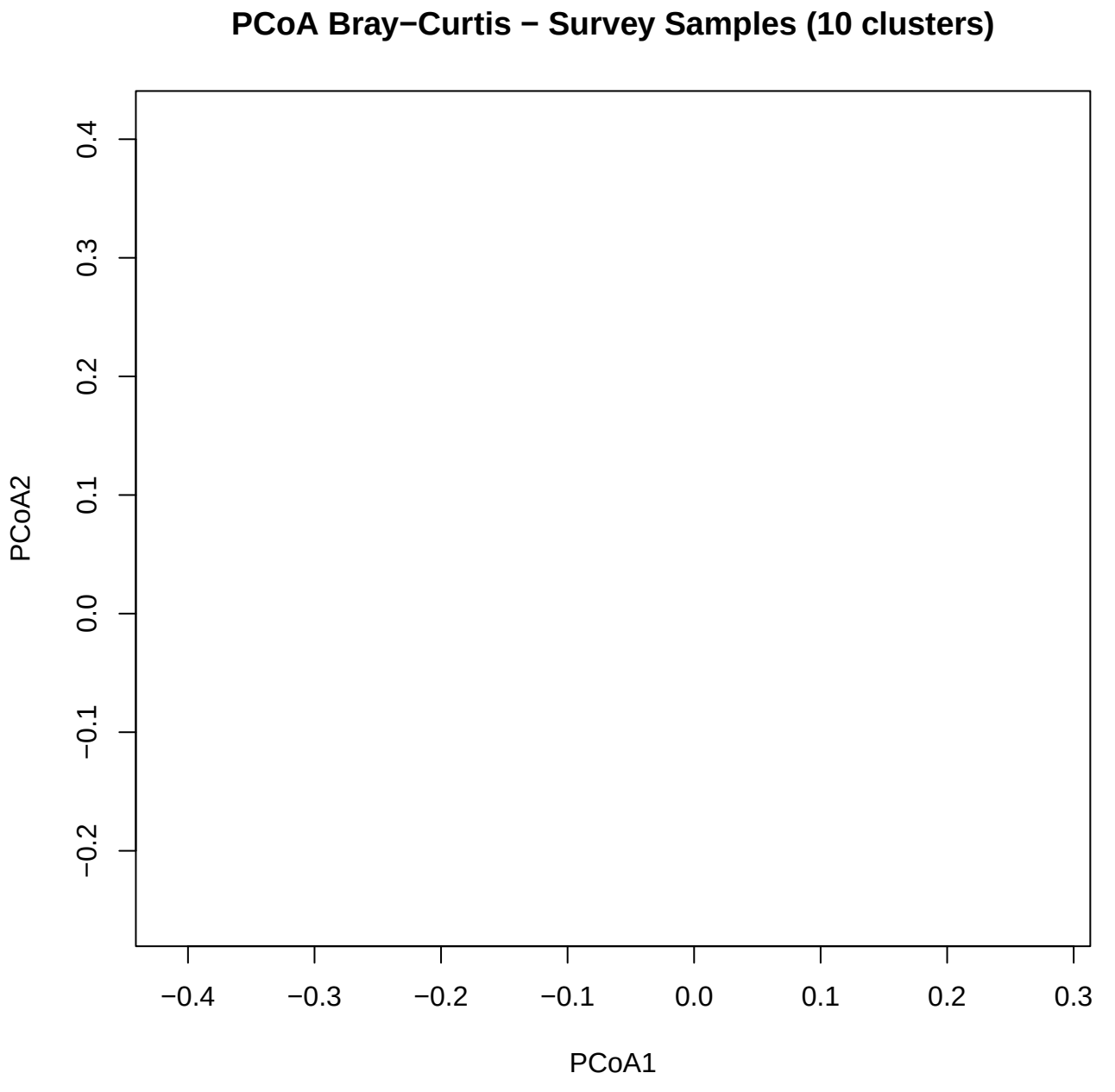
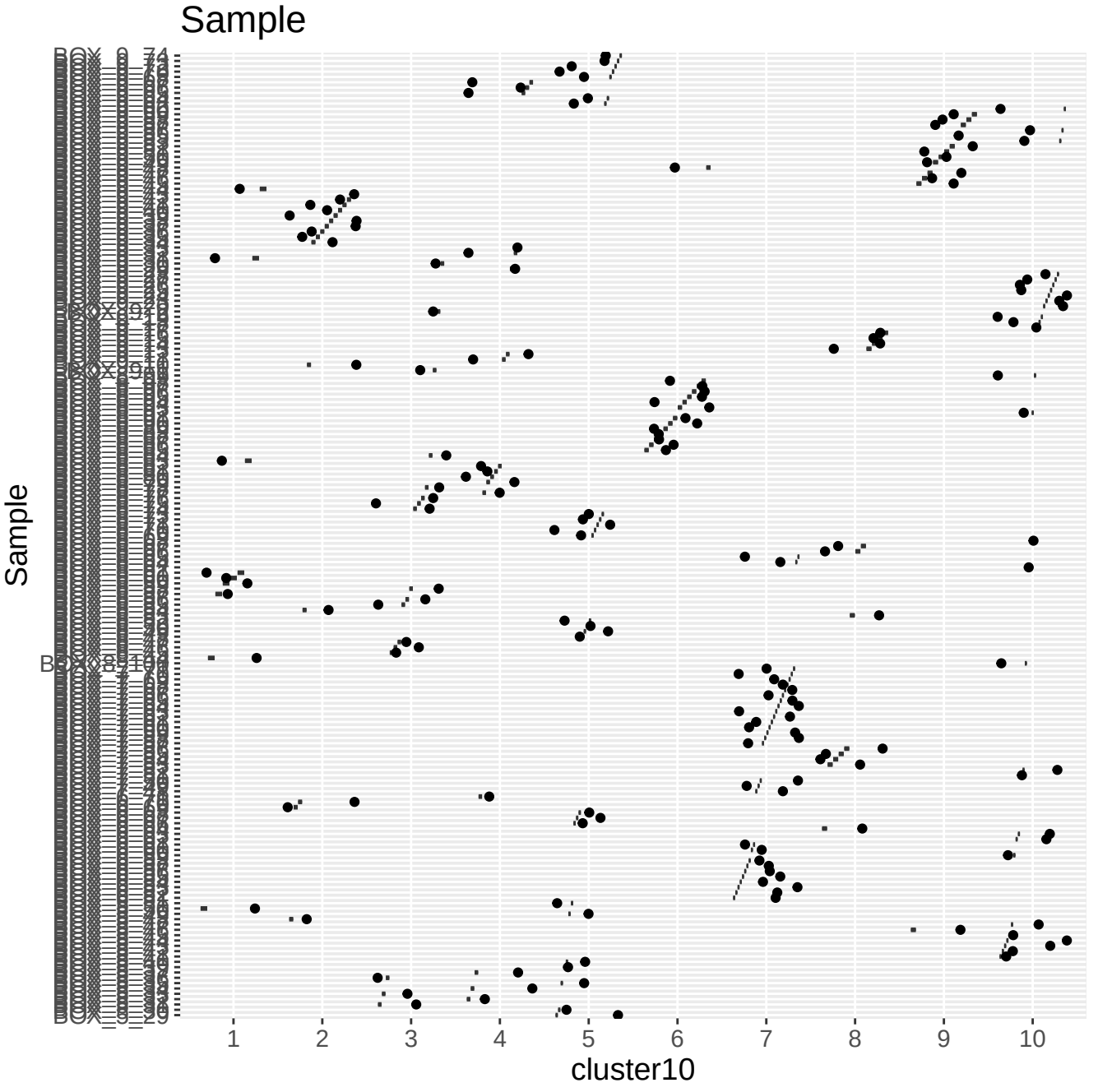
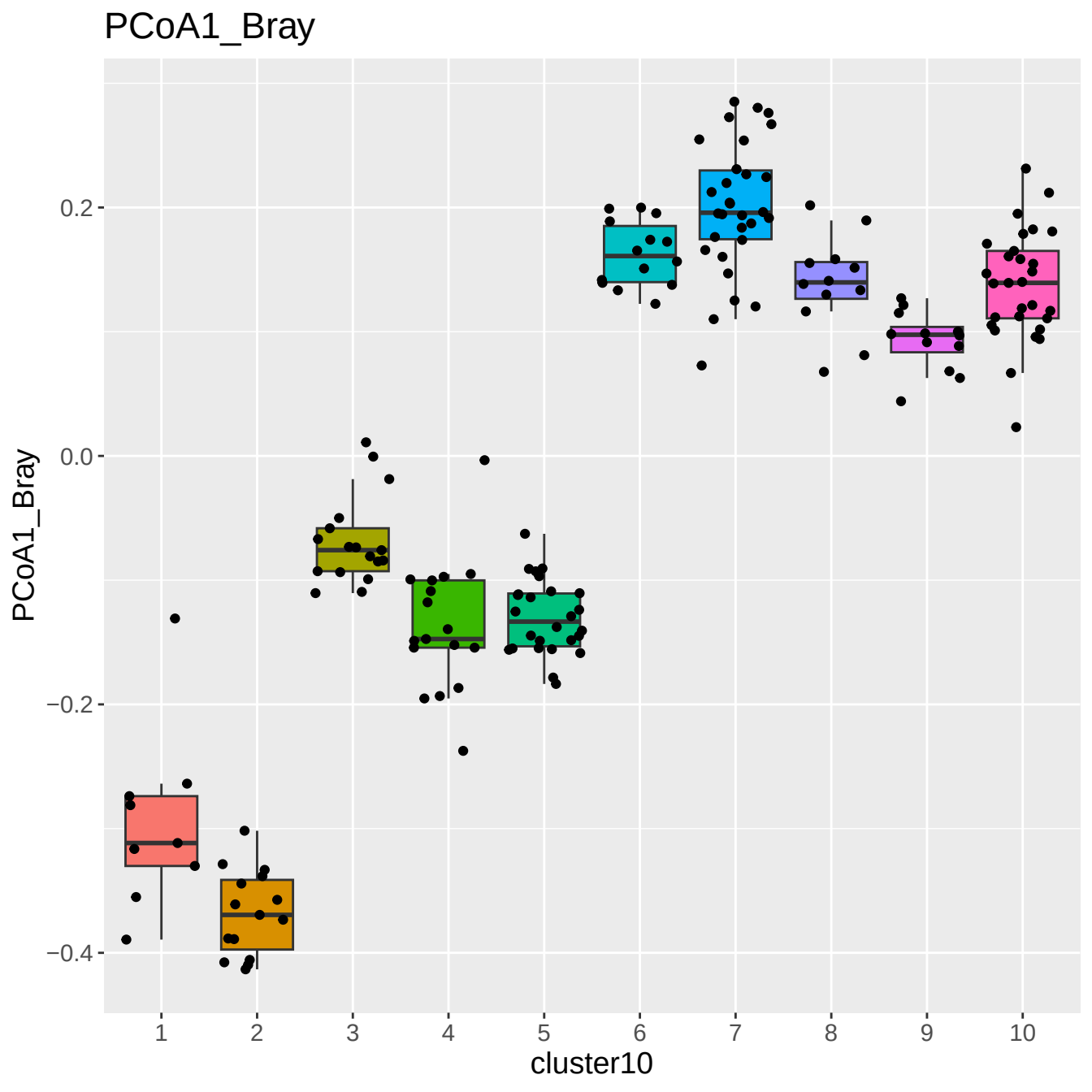
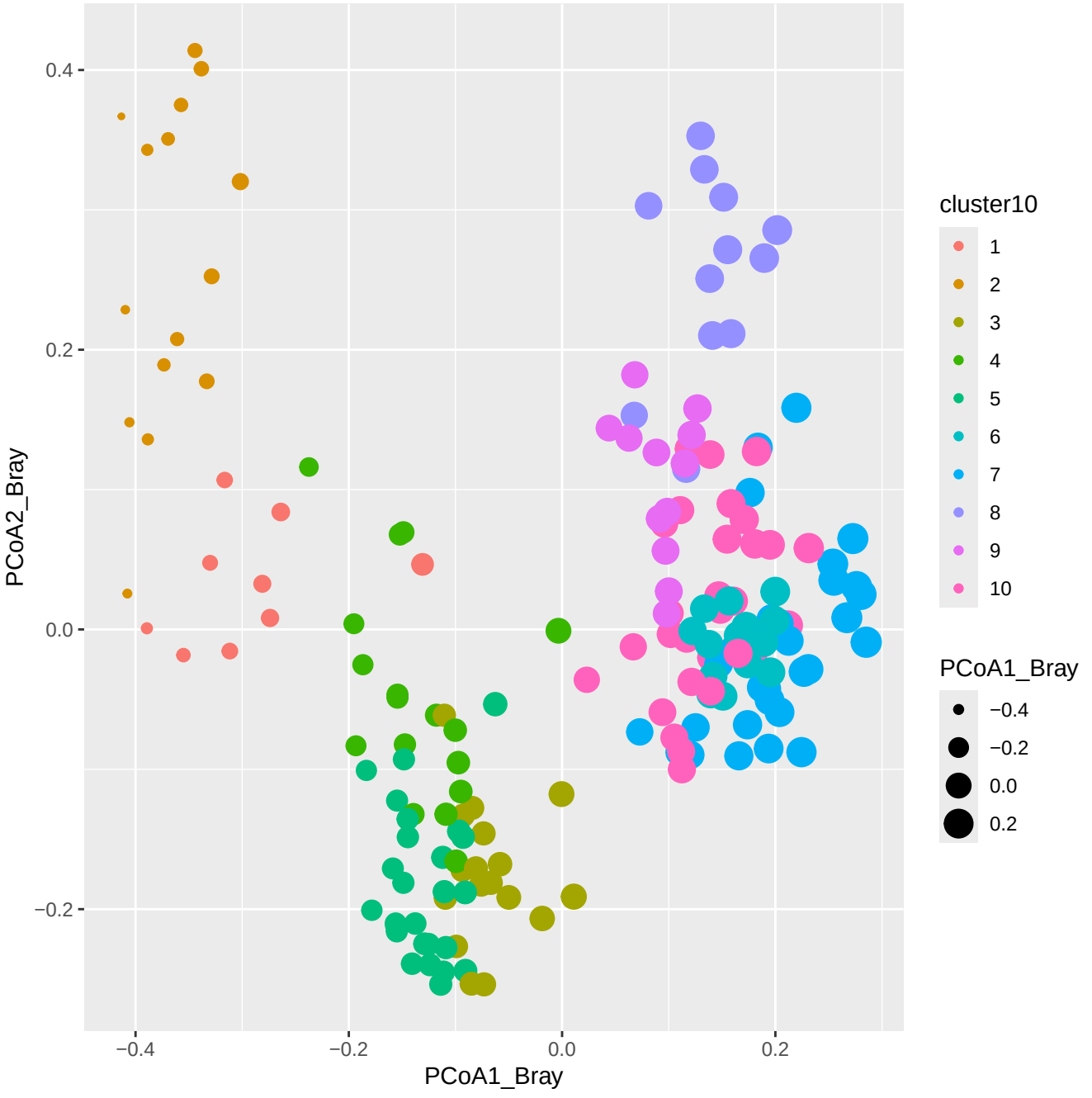
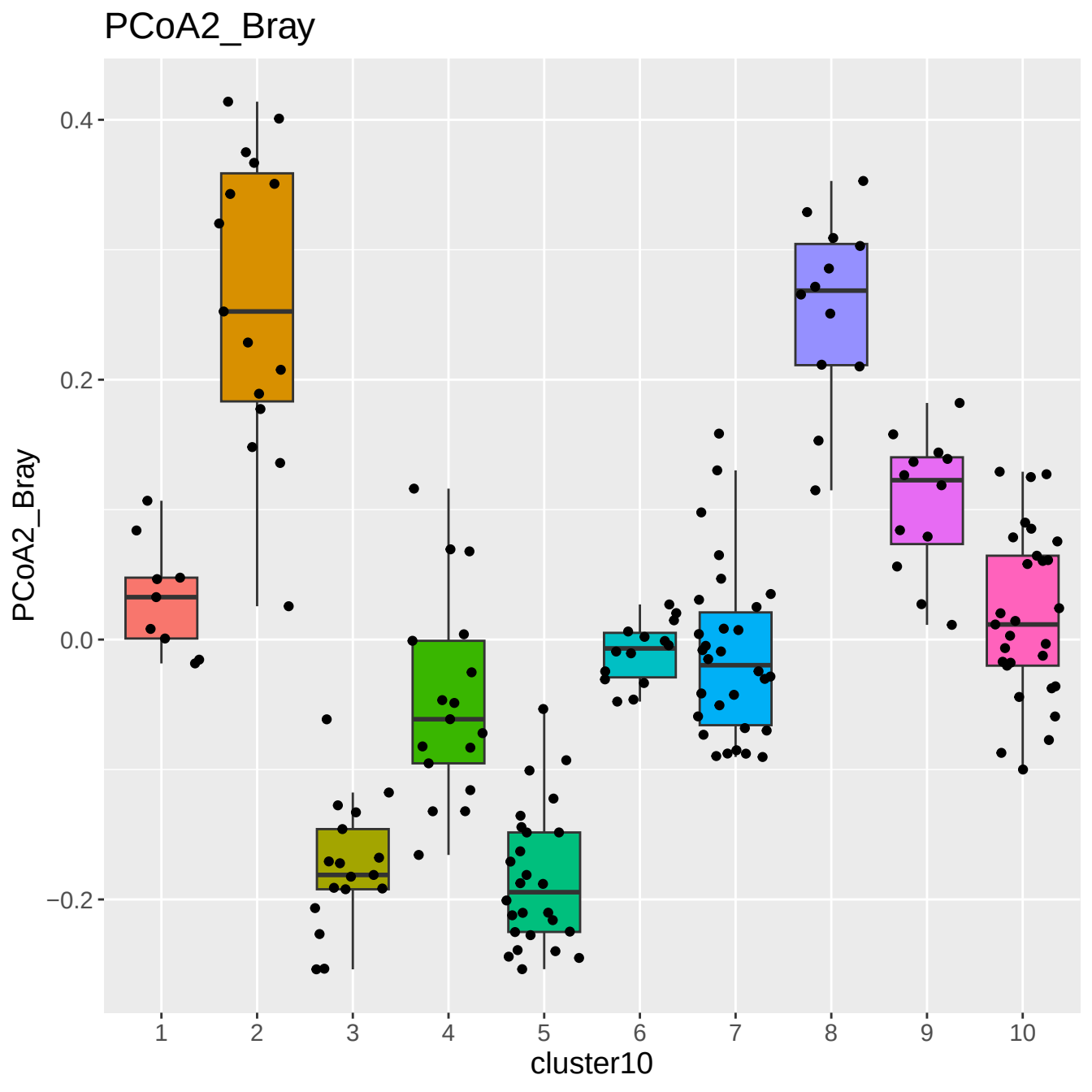

PCoA Bray−Curtis − Survey Samples (10 clusters)
PCoA2
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
PCoA1

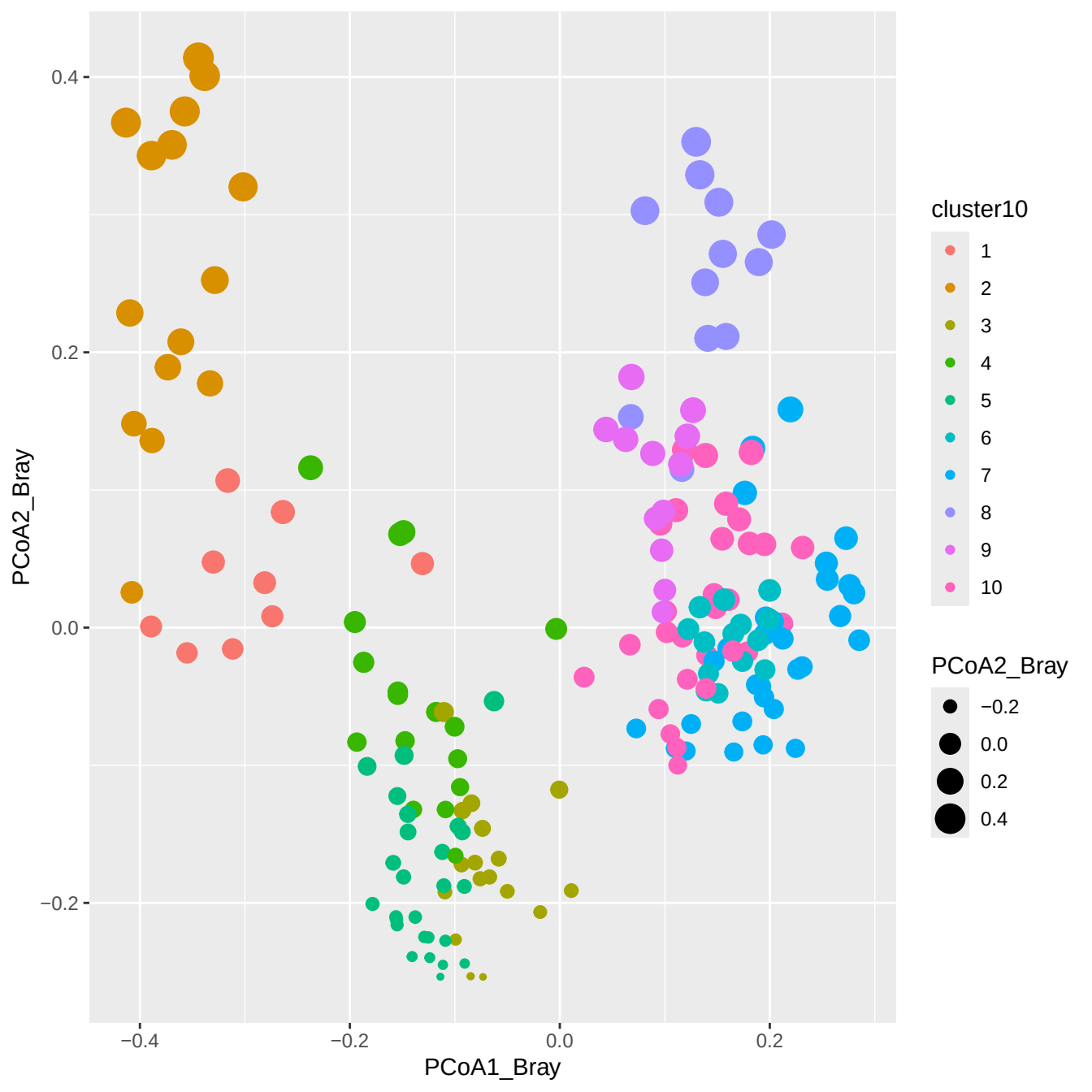

Sample
Sample
cluster10
1
2
3
4
5
6
7
8
9
10

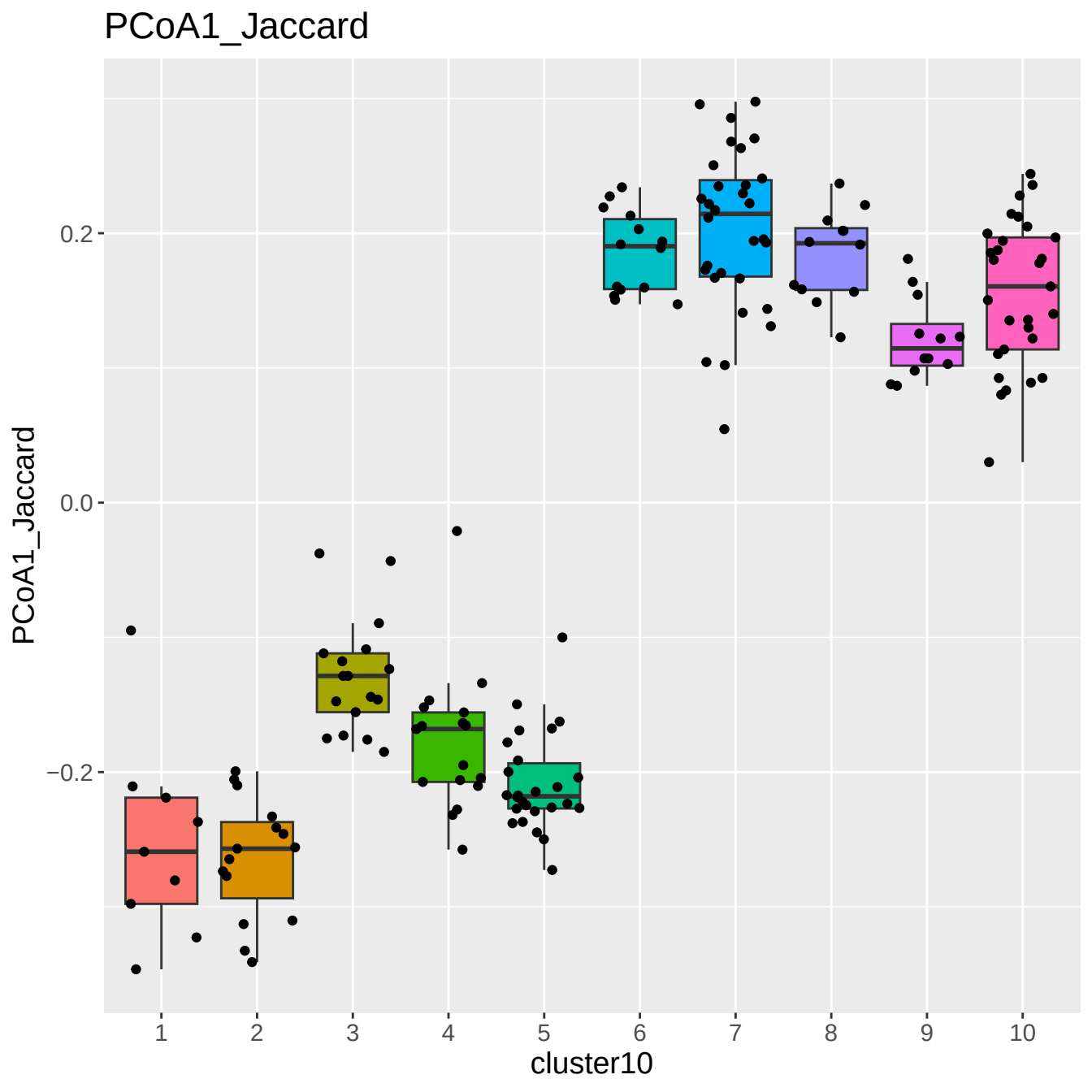
| | | |

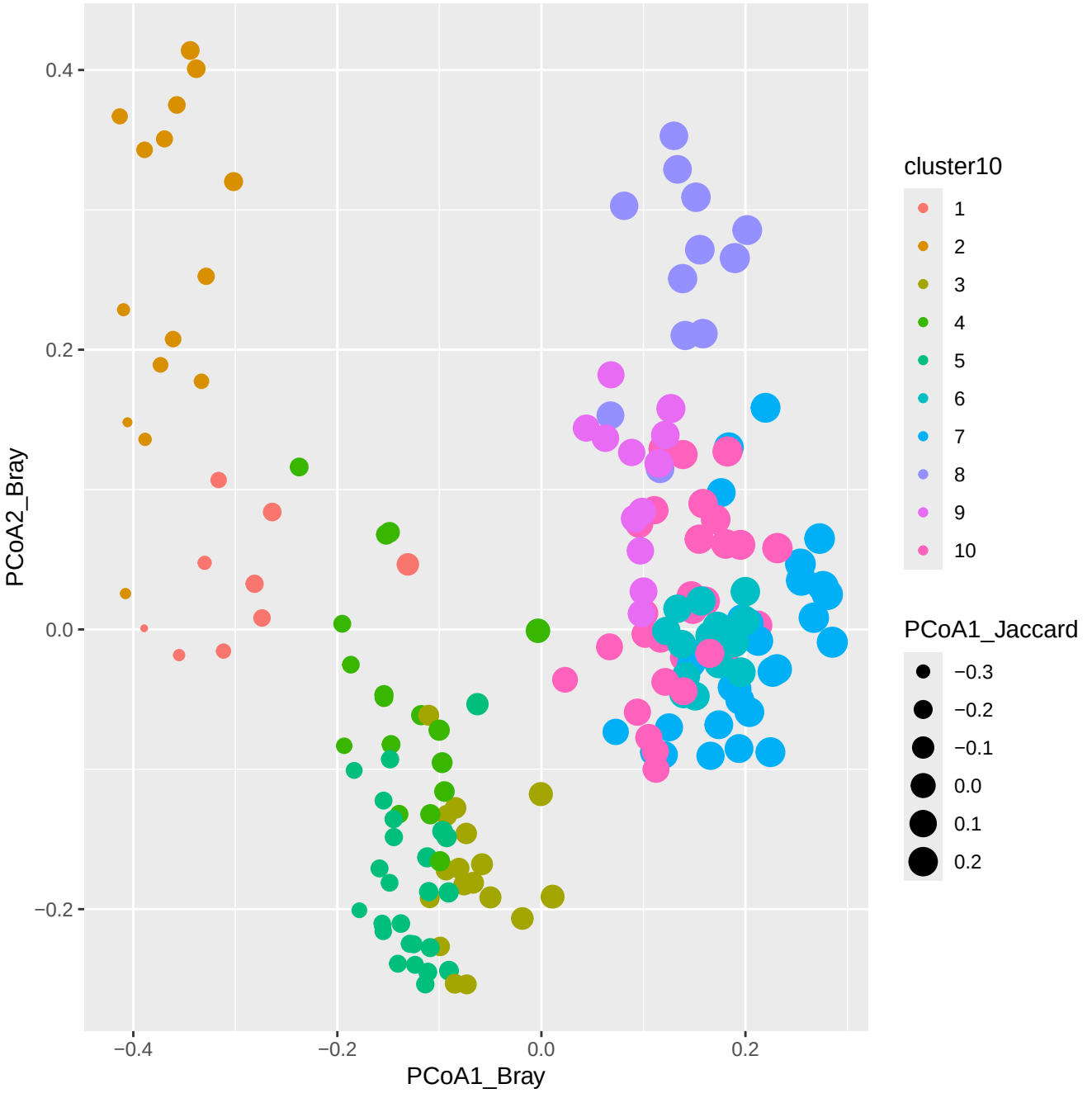

PCoA1_Bray
0.2
0.0
−0.2
−0.4
PCoA1_Bray
1
2
3
4
5
6
7
8
9
10
cluster10

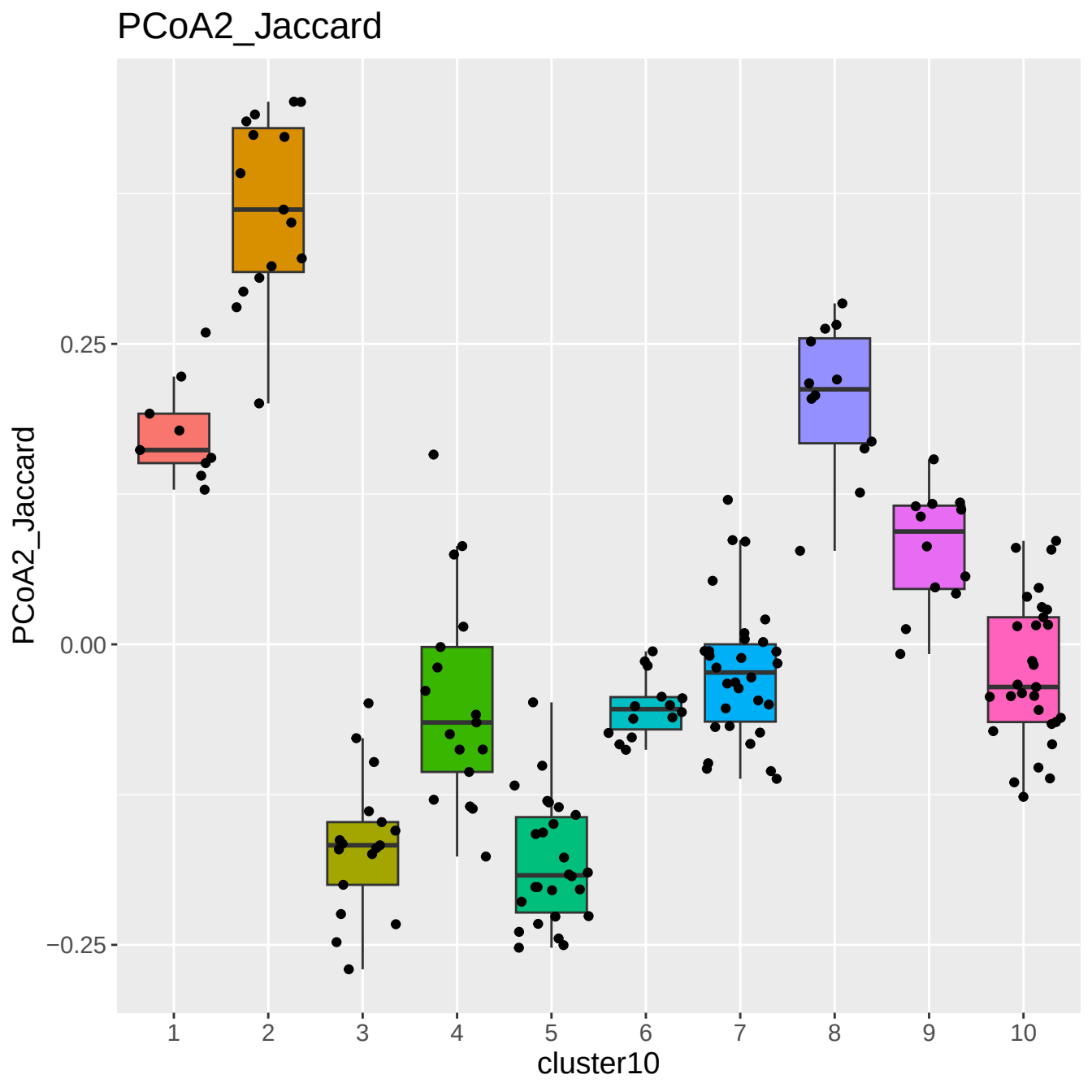

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
PCoA1_Bray
−0.4
−0.2
0.0
0.2

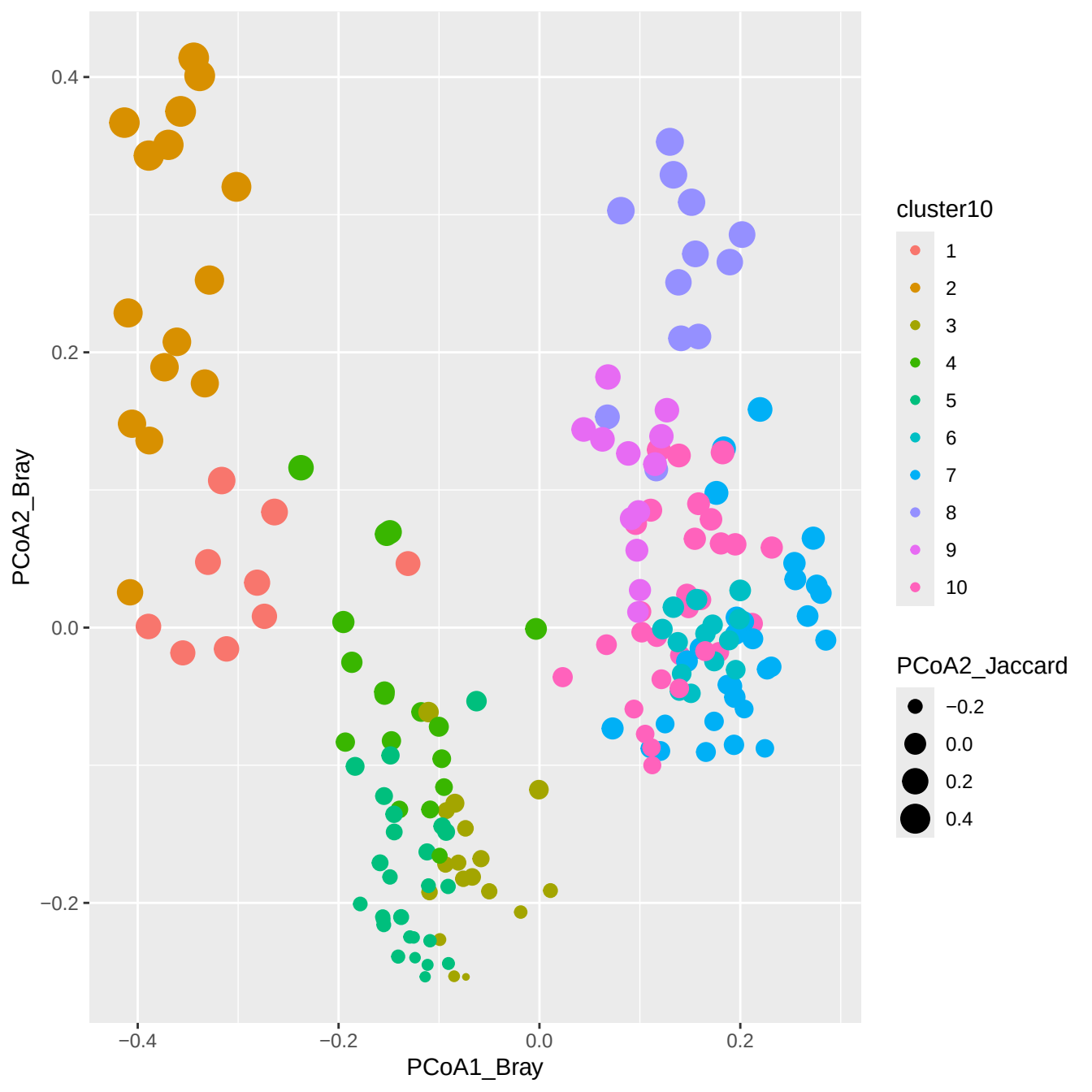

PCoA2_Bray
0.4
0.2
0.0
−0.2
PCoA2_Bray
1
2
3
4
5
6
7
8
9
10
cluster10

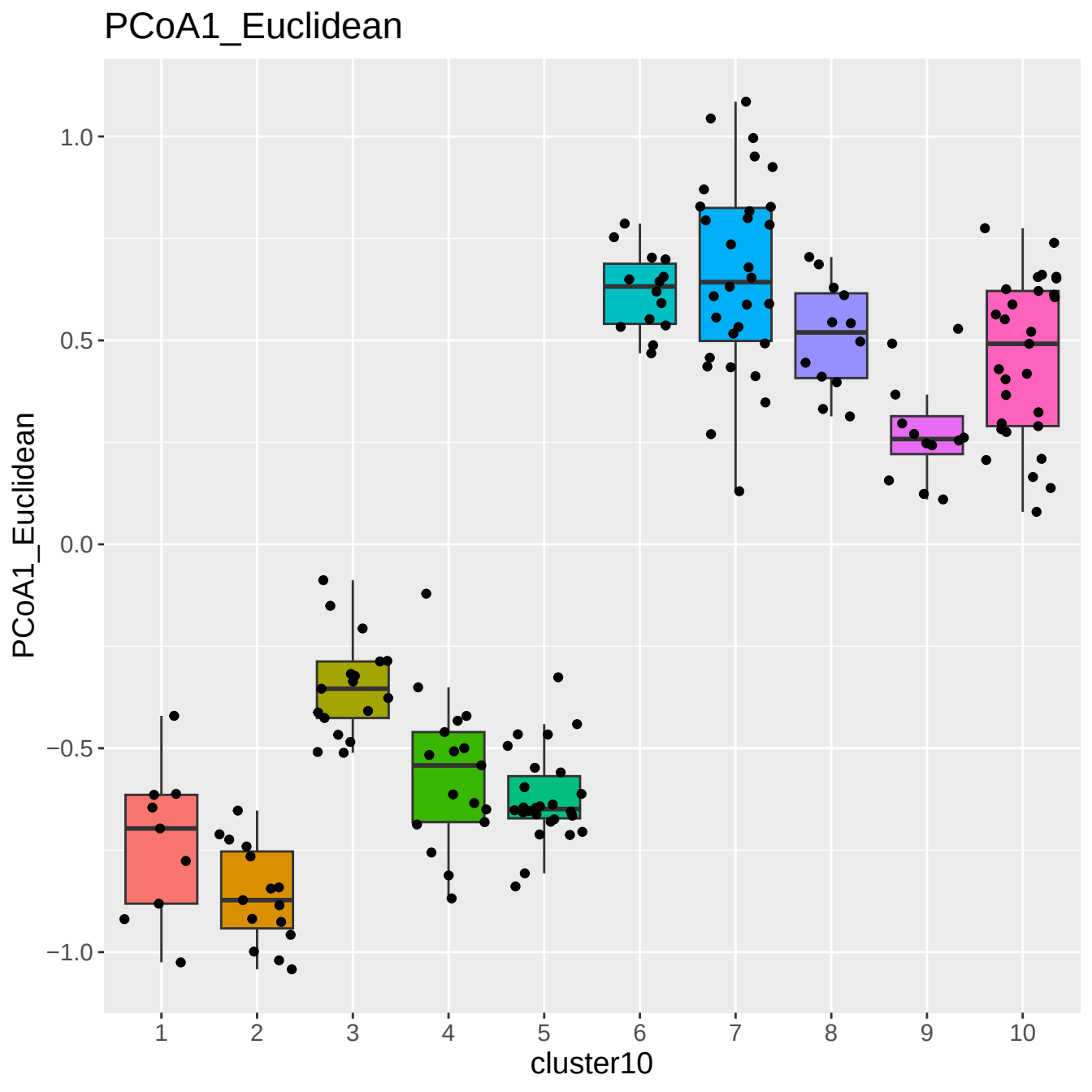

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
PCoA2_Bray
−0.2
0.0
0.2
0.4

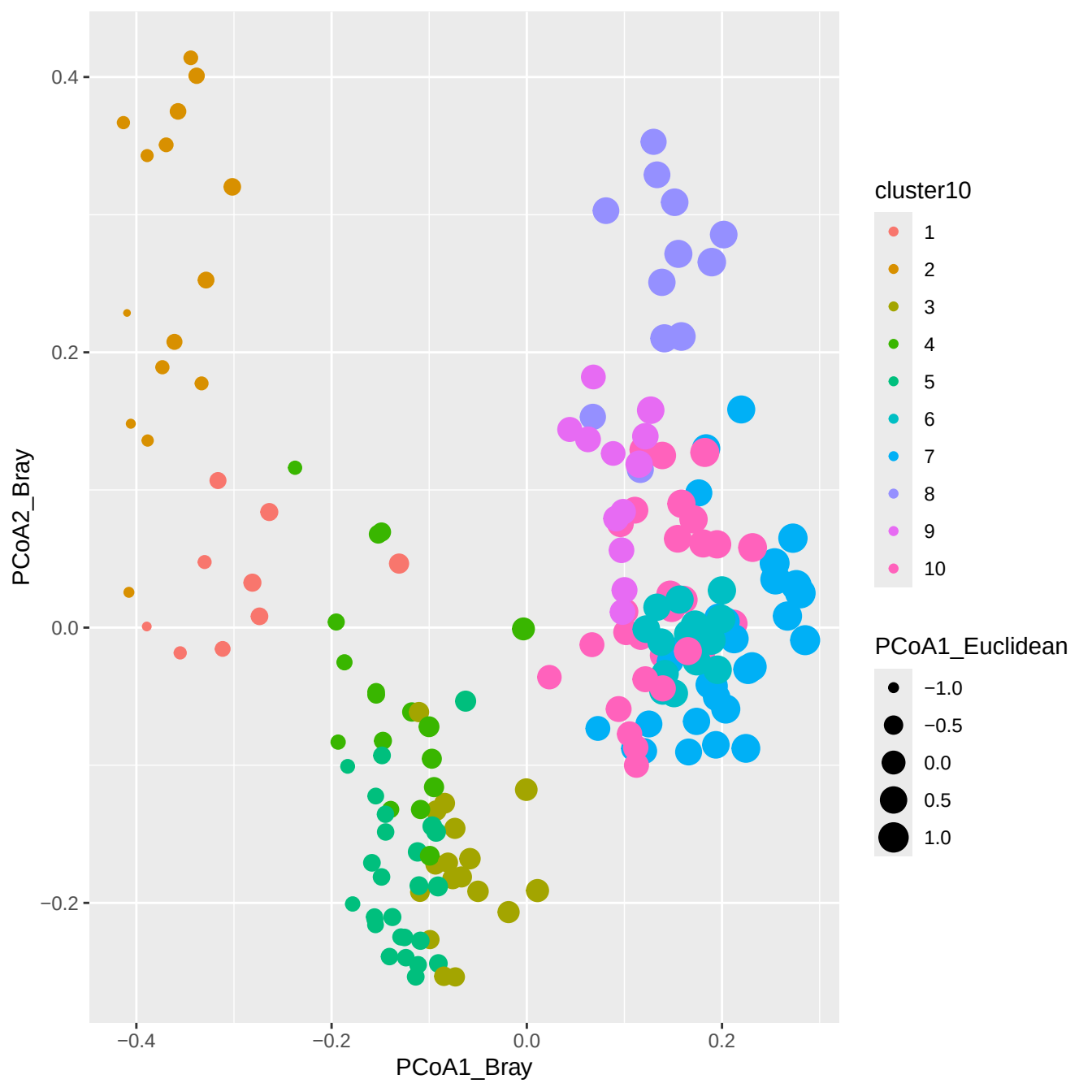

PCoA1_Jaccard
0.2
0.0
−0.2
PCoA1_Jaccard
1
2
3
4
5
6
7
8
9
10
cluster10

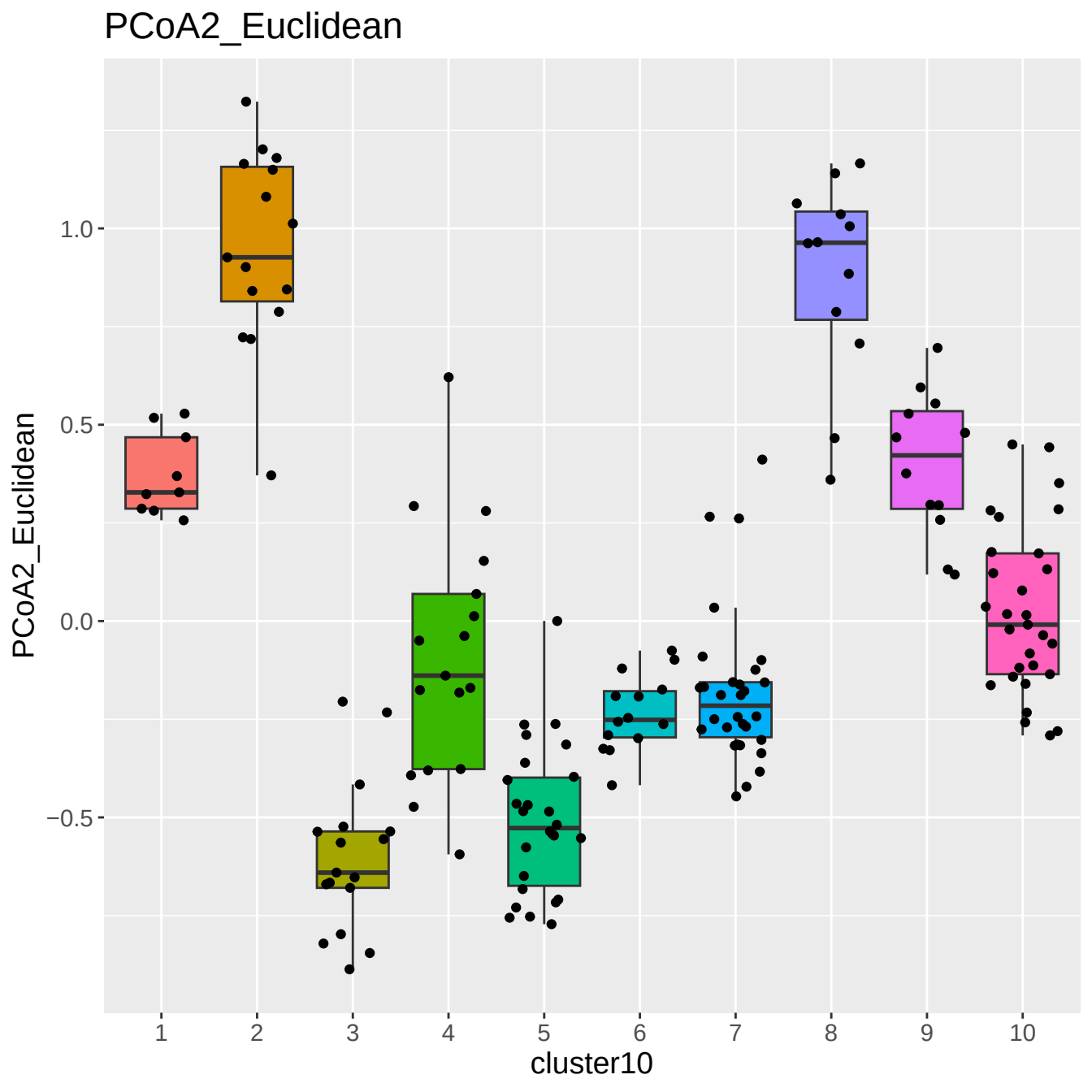

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
PCoA1_Jaccard
−0.3
−0.2
−0.1
0.0
0.1
0.2

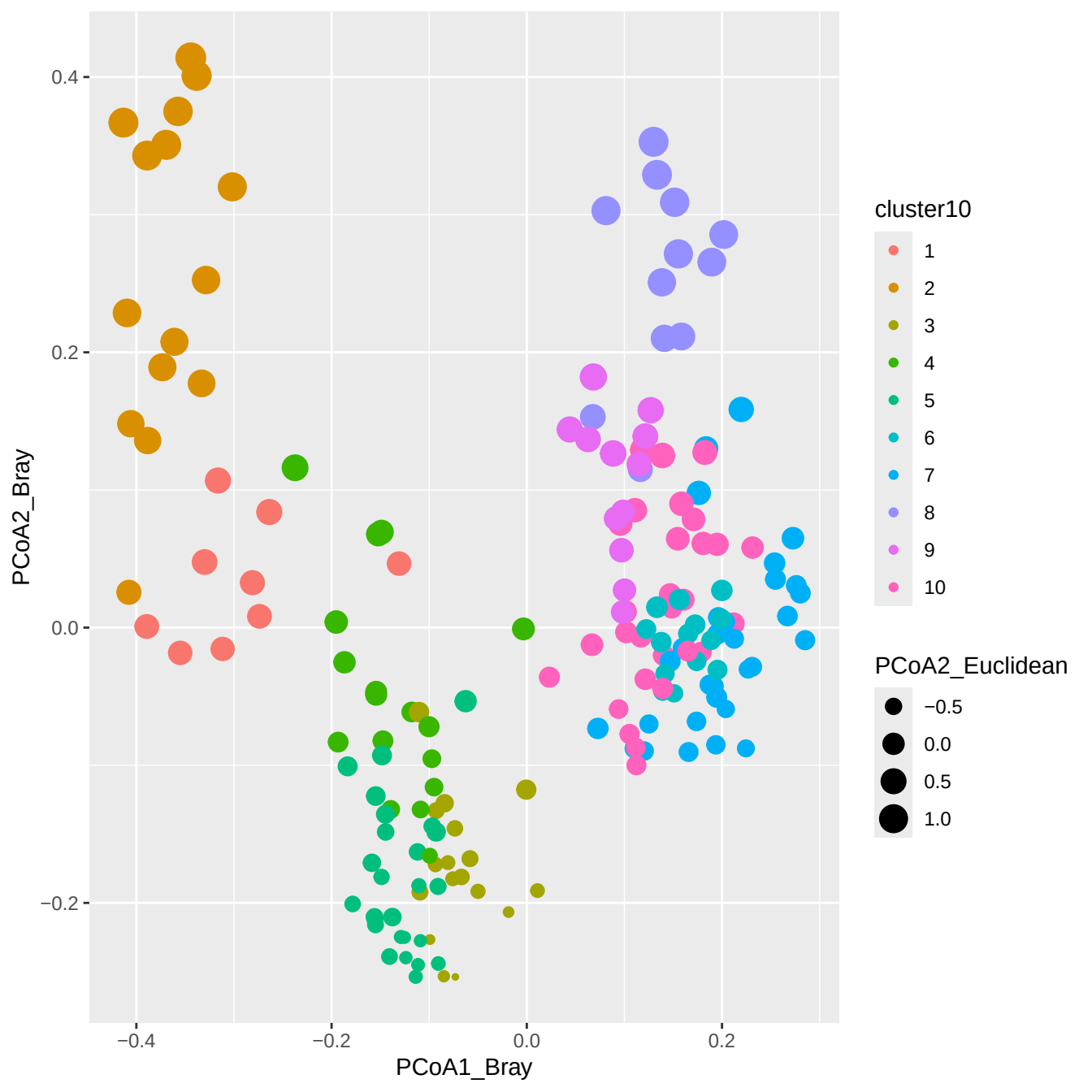

PCoA2_Jaccard
0.25
0.00
−0.25
PCoA2_Jaccard
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
PCoA2_Jaccard
−0.2
0.0
0.2
0.4


PCoA1_Euclidean
1.0
0.5
0.0
−0.5
−1.0
PCoA1_Euclidean
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
PCoA1_Euclidean
−1.0
−0.5
0.0
0.5
1.0


PCoA2_Euclidean
1.0
0.5
0.0
−0.5
PCoA2_Euclidean
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
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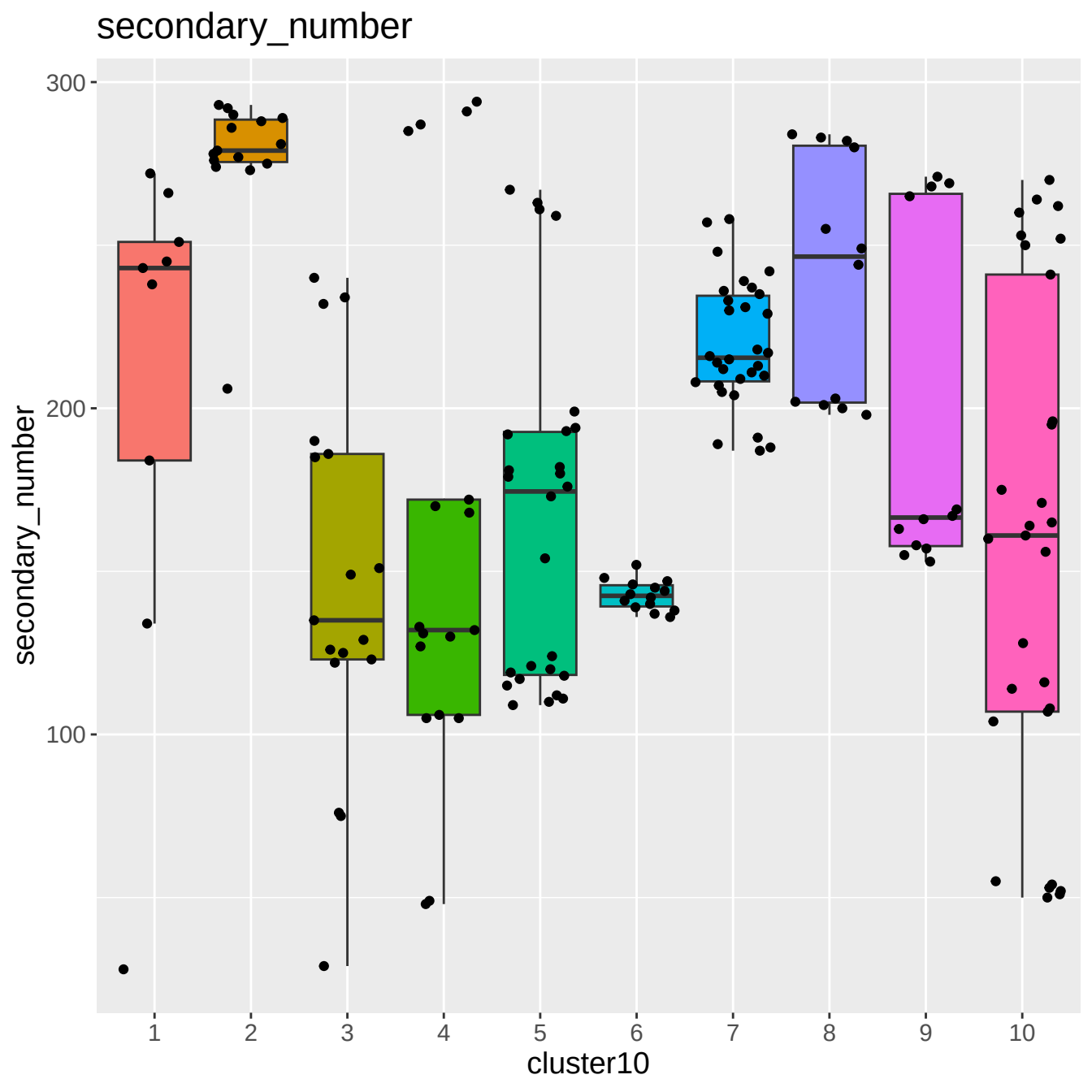
5
6
7
8
9
10
PCoA2_Euclidean
−0.5
0.0
0.5
1.0


secondary_number
300
200
100
secondary_number
1
2
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4
5
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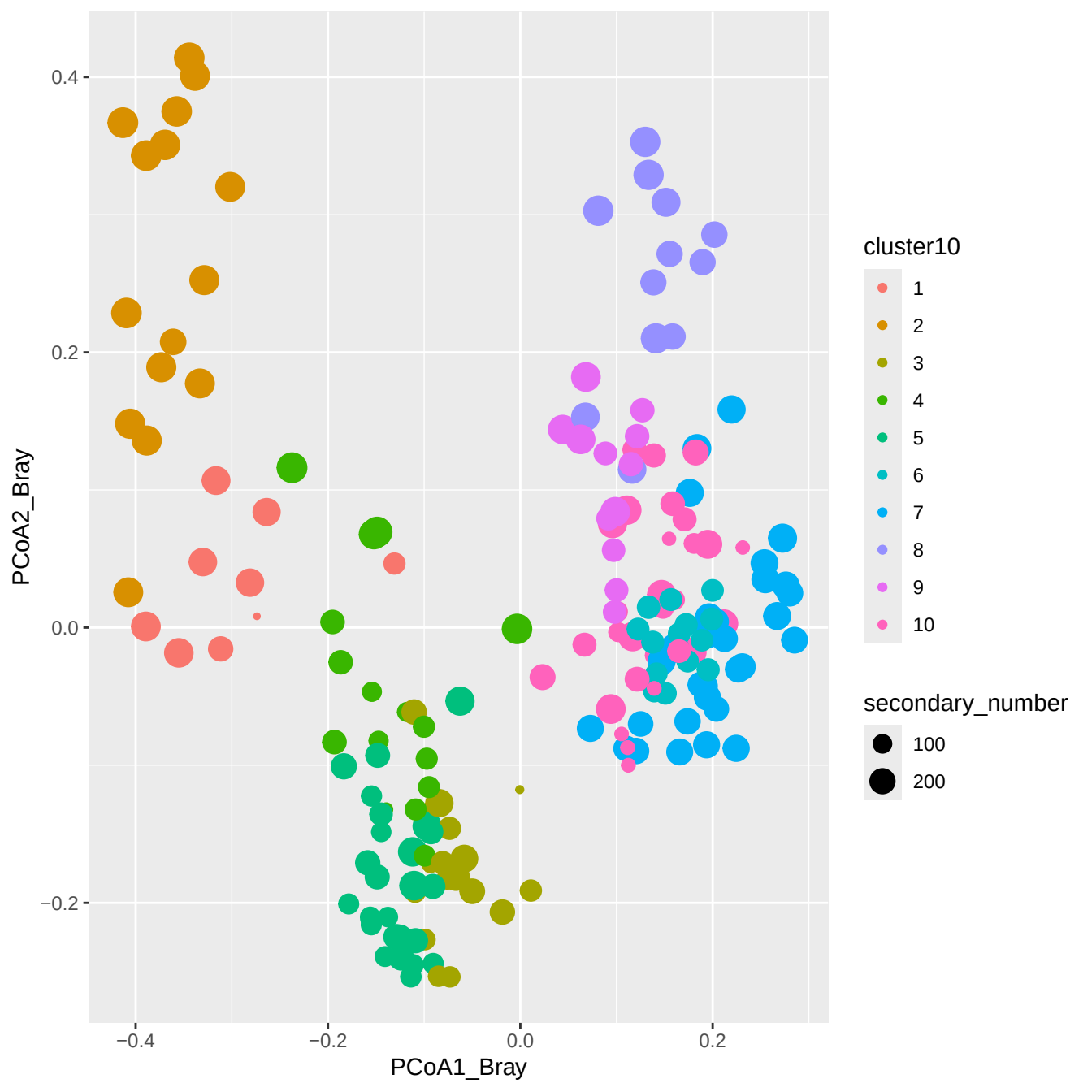
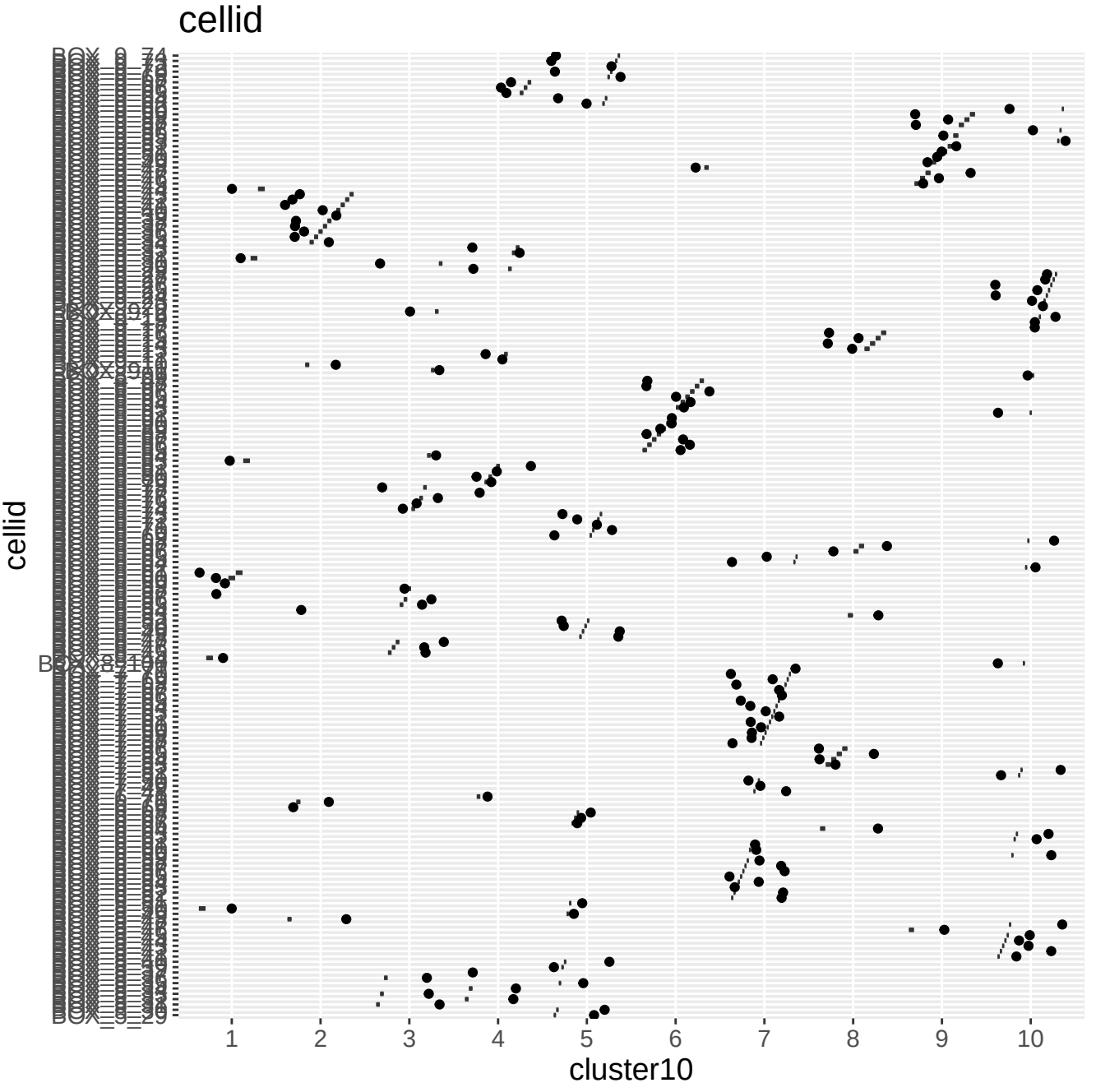
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
secondary_number
100
200


cellid
cellid
cluster10
1
2
3
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6
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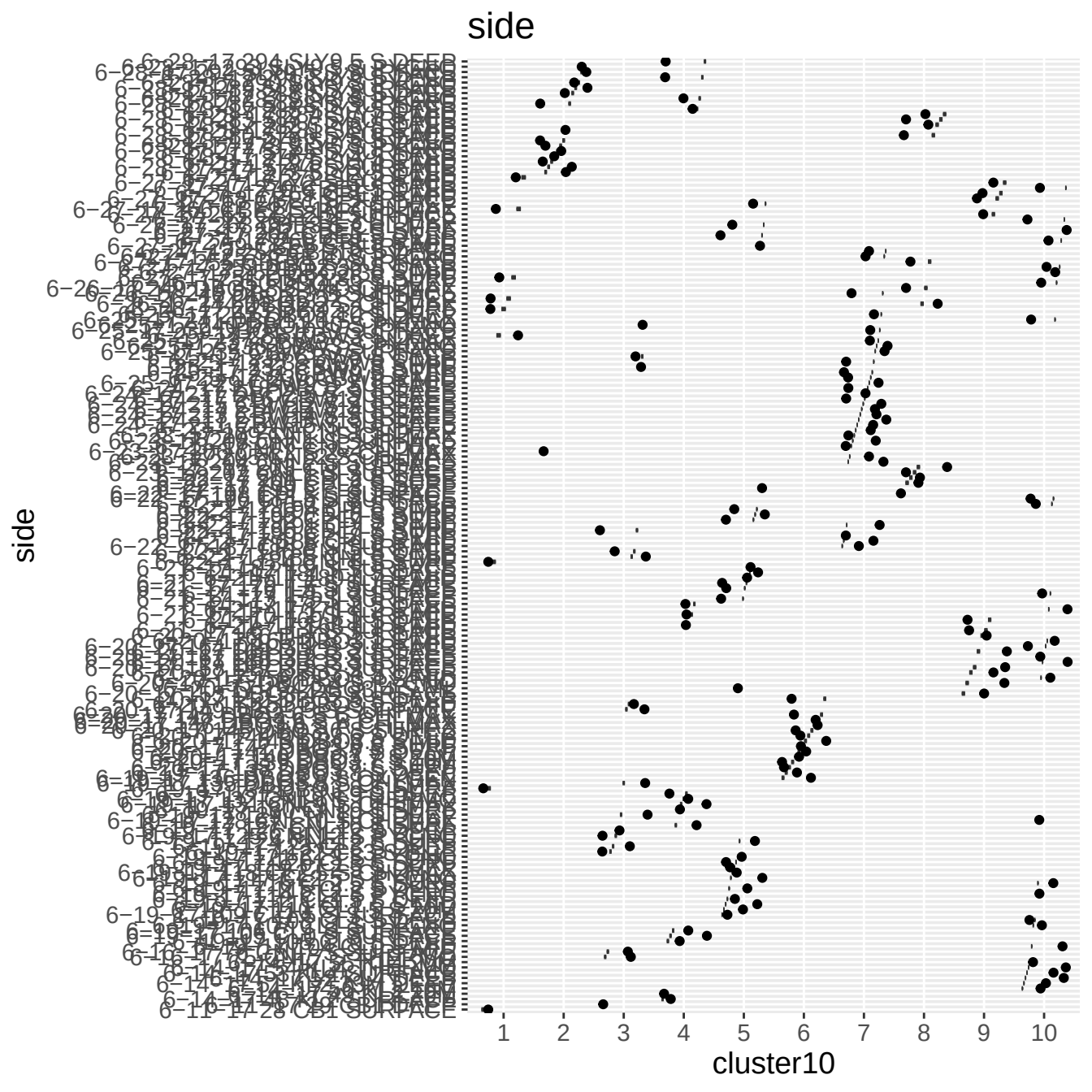
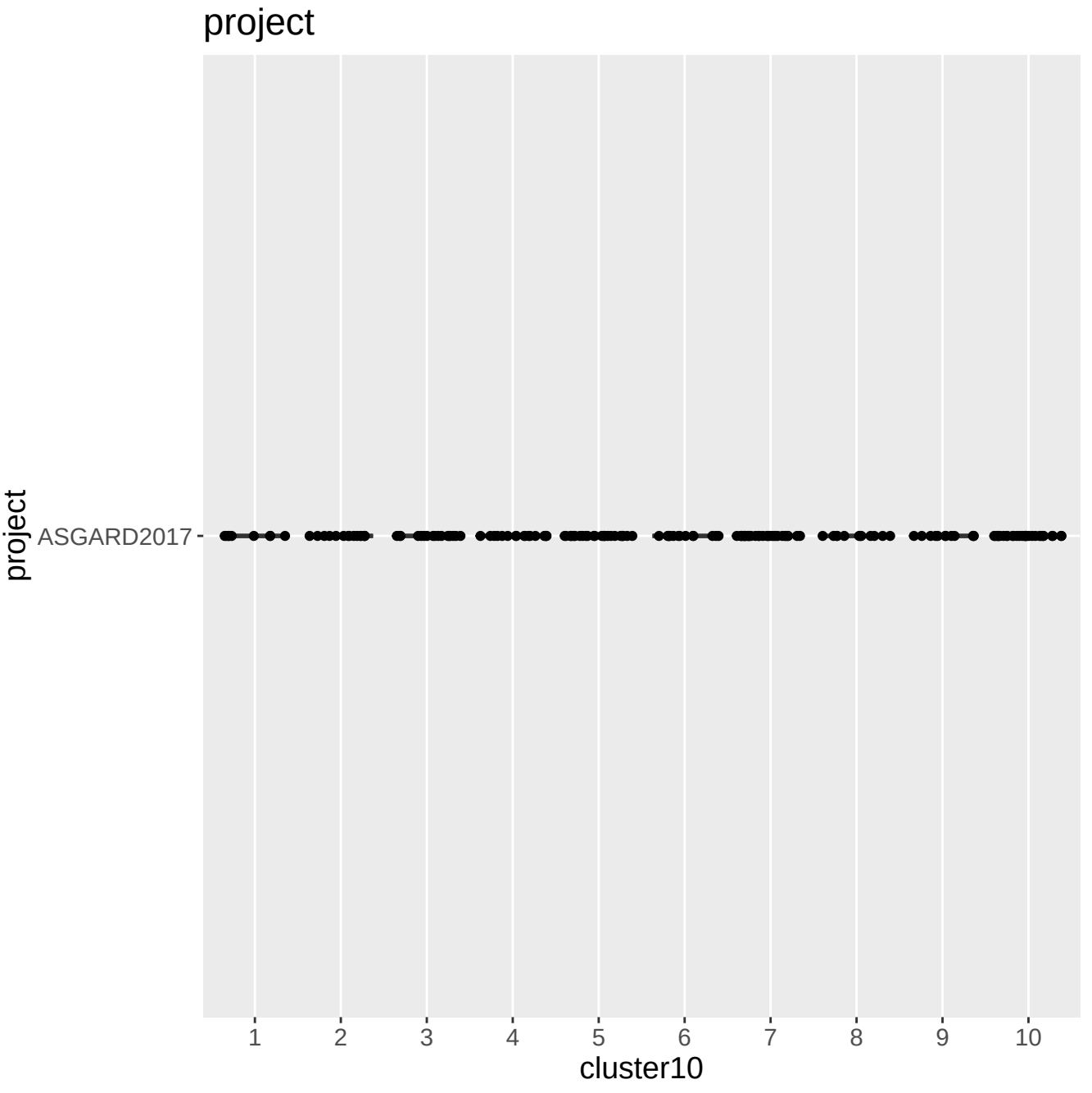
9
10

| | | |


side
side
1
2
3
4
5
6
7
8
9
10
cluster10

1 DBO3.5 S MID	●	6−21−17 172 IL2 S DEEP	●	6−22−17 195 CPL6 S DEEP	●	6−24−17 215	
2 DBO3.5 S DEEP	●	6−21−17 173 IL3 S SURFACE	●	6−22−17 196 CPL5 S SURFACE	●	6−24−17 216	
3 DBO3.4 S SURFACE	●	6−21−17 175 IL3 S DEEP	●	6−22−17 198 CPL3 S SURFACE	●	6−24−17 217	
4 DBO3.4 S ML	●	6−21−17 176 IL5 S SURFACE	●	6−22−17 199 CPL3 S DEEP	●	6−24−17 218	
5 DBO3.4 S PYCNO	●	6−21−17 179 IL7 S SURFACE	●	6−22−17 200 CPL2 S SURF	●	6−25−17 229	
6 DBO3.4 S MID	●	6−21−17 180 IL7 S MID	●	6−22−17 201 CPL2 S DEEP	●	6−25−17 230	
7 DBO3.4 S DEEP	●	6−21−17 181 IL7 S DEEP	●	6−23−17 202 CNL1 S SURFACE	●	6−25−17 231	
8 DBO3.3 S SURFACE	●	6−21−17 182 IL9 S SURFACE	●	6−23−17 203 CNL1 S DEEP	●	6−25−17 232	
0 DBO3.3 S DEEP	●	6−22−17 184 CNL5 S SURF	●	6−23−17 204 CNL2 S SURFACE	●	6−25−17 233	
1 DBO3.2 S SURFACE	●	6−22−17 185 CNL5 S MID	●	6−23−17 205 CNL2 S CHLMAX	●	6−25−17 234	
3 DBO3.2 S DEEP	●	6−22−17 186 CNL5 S DEEP	●	6−23−17 206 CNL2 S 2x CHLMAX	●	6−25−17 235	
4 DBO3.1 S SURFACE	●	6−22−17 187 CPL8 S SURFACE	●	6−23−17 207 CNL2 S DEEP	●	6−25−17 236	
5 DBO3.1 S MID	●	6−22−17 188 CPL8 S MID	●	6−23−17 208 CNL1 S SURFACE	●	6−25−17 237	
6 DBO3.1 S DEEP	●	6−22−17 189 CPL8 S DEEP	●	6−23−17 209 CNL1 S CHLMAX	●	6−25−17 238	
7 IL1 S SURFACE	●	6−22−17 190 CPL7 S SURF	●	6−23−17 210 CNL1 S DEEP	●	6−25−17 239	
8 IL1 S MID	●	6−22−17 191 CPL7 S MID	●	6−24−17 211 CBW15 S SURFACE	●	6−25−17 240	
9 IL1 S DEEP	●	6−22−17 192 CPL7 S DEEP	●	6−24−17 212 CBW15 S DEEP	●	6−25−17 241	
0 IL2 S SURFACE	●	6−22−17 193 CPL6 S SURF	●	6−24−17 213 CBW14 S SURFACE	●	6−25−17 242	


project
project
ASGARD2017
1
2
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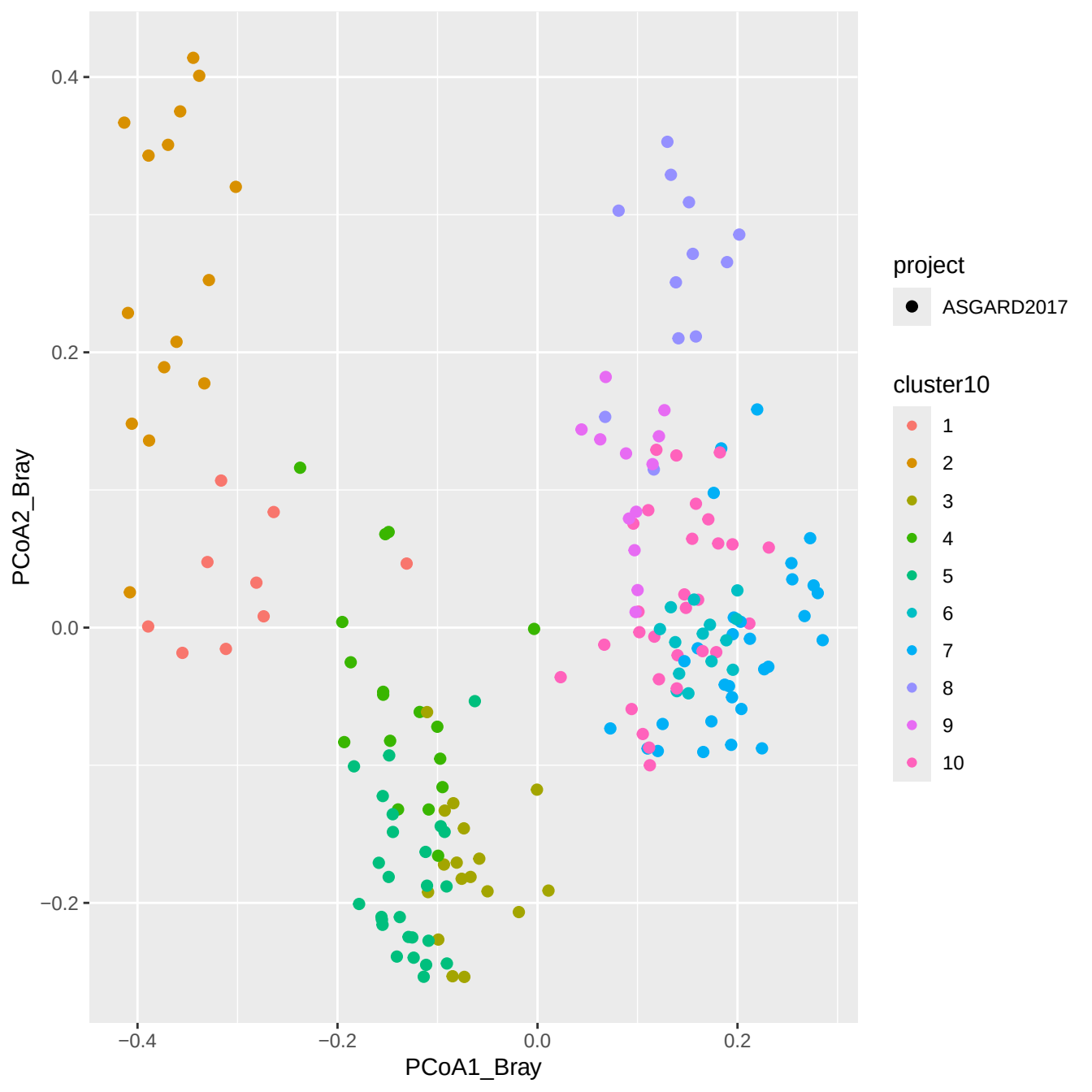
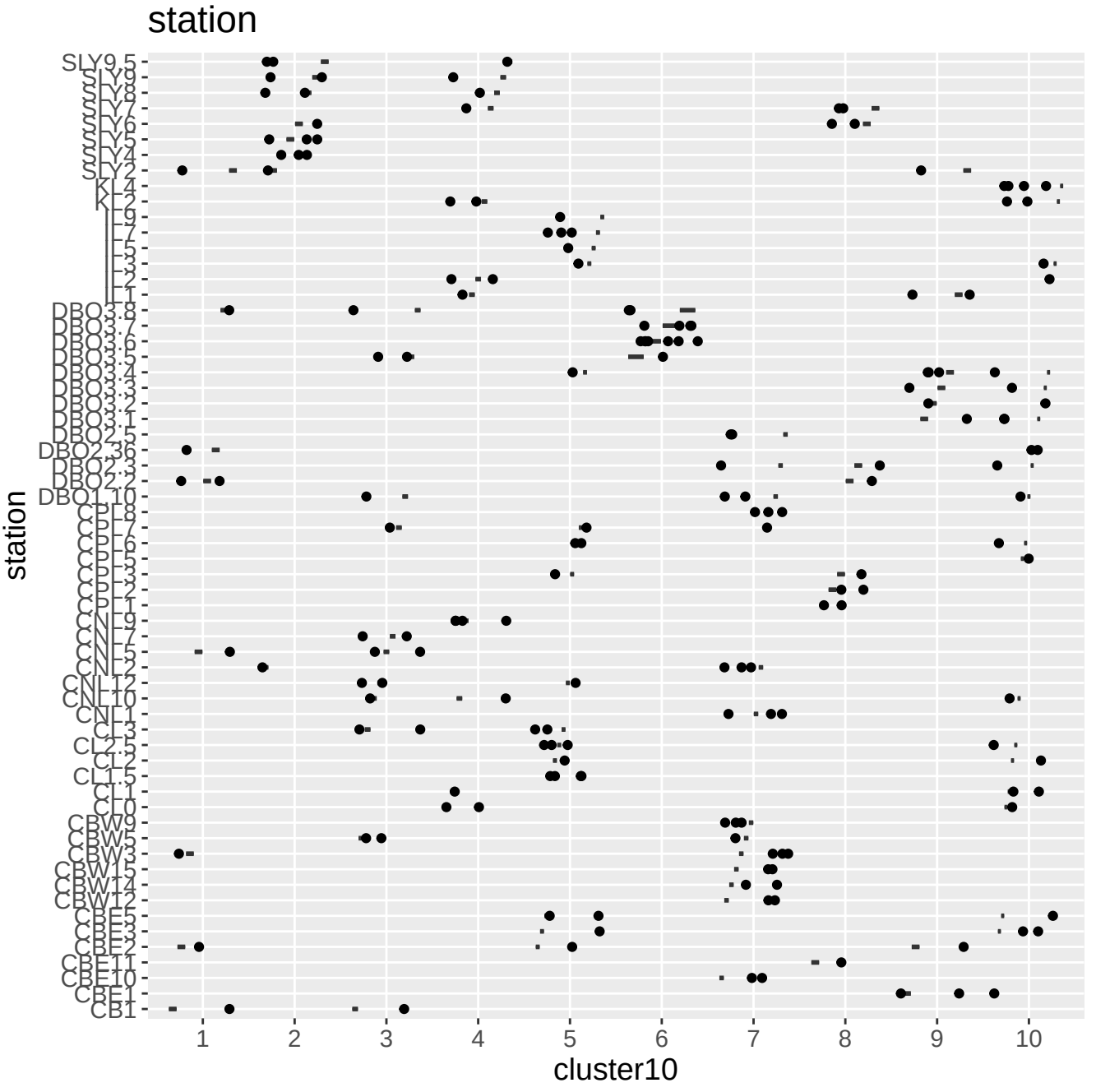
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
project
ASGARD2017
cluster10
1
2
3
4
5
6
7
8
9
10


station
SLY9.5
SLY9
SLY8
SLY7
SLY6
SLY5
SLY4
SLY2
KL4
KL2
L9
L7
L5
L3
L2
L1
DBO3.8
DBO3.7
DBO3.6
DBO3.5
DBO3.4
DBO3.3
DBO3.2
DBO3.1
DBO2.5
DBO2.36
DBO2.3
DBO2.2
DBO1.10
CPL8
CPL7
CPL6
CPL5
CPL3
CPL2
CPL1
CNL9
CNL7
CNL6
CNL5
CNL2
CNL12
CNL10
CNL1
CL3
CL2.5
CL2
CL1.5
CL1
CL0
CBW9
CBW5
CBW3
CBW15
CBW14
CBW12
CBE5
CBE3
CBE2
CBE11
CBE10
CBE1
CB1
station
1
2
3
4
5
6
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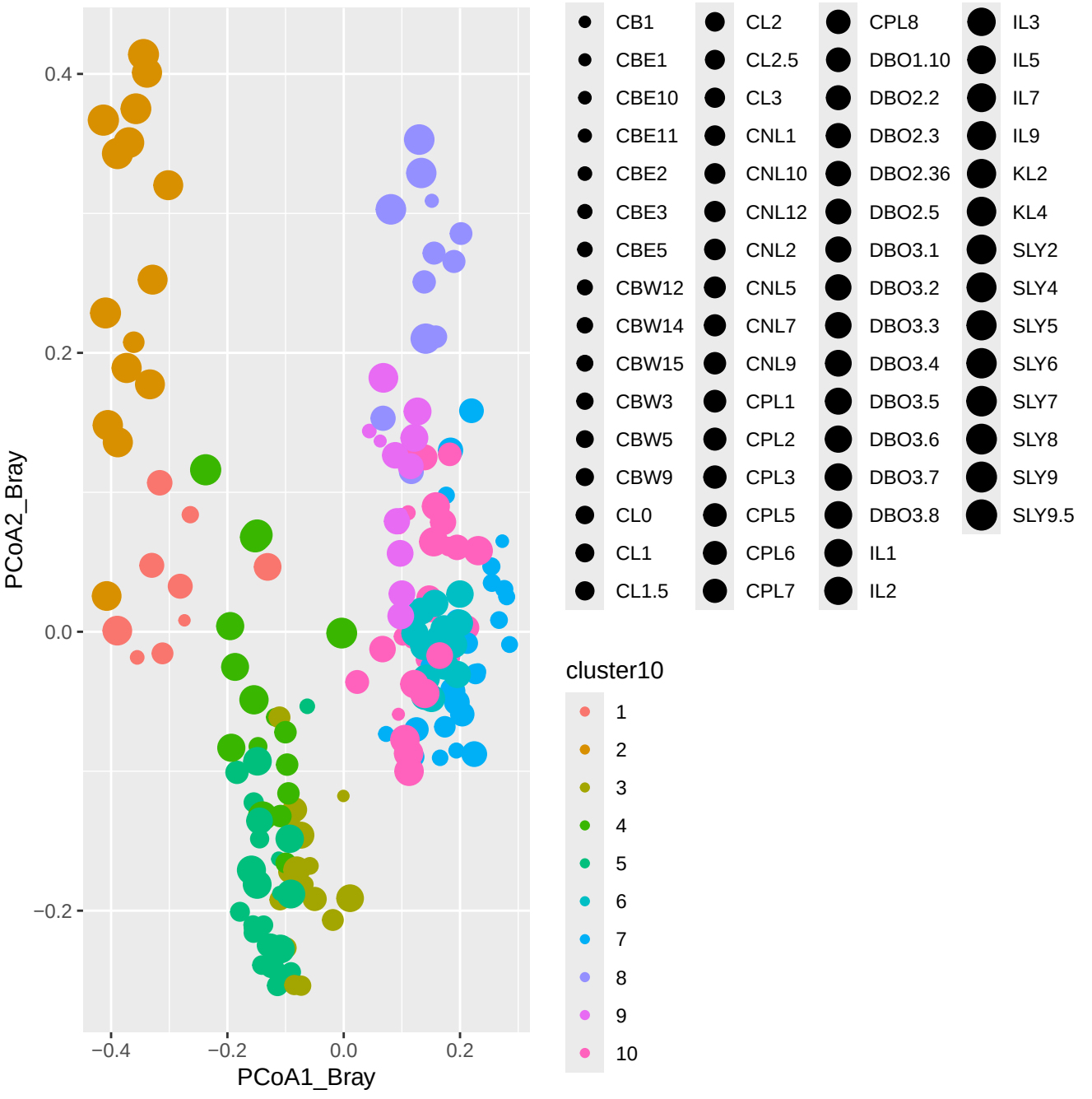
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
CB1
CBE1
CBE10
CBE11
CBE2
CBE3
CBE5
CBW12
CBW14
CBW15
CBW3
CBW5
CBW9
CL0
CL1
CL1.5
CL2
CL2.5
CL3
CNL1
CNL10
CNL12
CNL2
CNL5
CNL7
CNL9
CPL1
CPL2
CPL3
CPL5
CPL6
CPL7
CPL8
DBO1.10
DBO2.2
DBO2.3
DBO2.36
DBO2.5
DBO3.1
DBO3.2
DBO3.3
DBO3.4
DBO3.5
DBO3.6
DBO3.7
DBO3.8
IL1
IL2
IL3
IL5
IL7
IL9
KL2
KL4
SLY2
SLY4
SLY5
SLY6
SLY7
SLY8
SLY9
SLY9.5
cluster10
1
2
3
4
5
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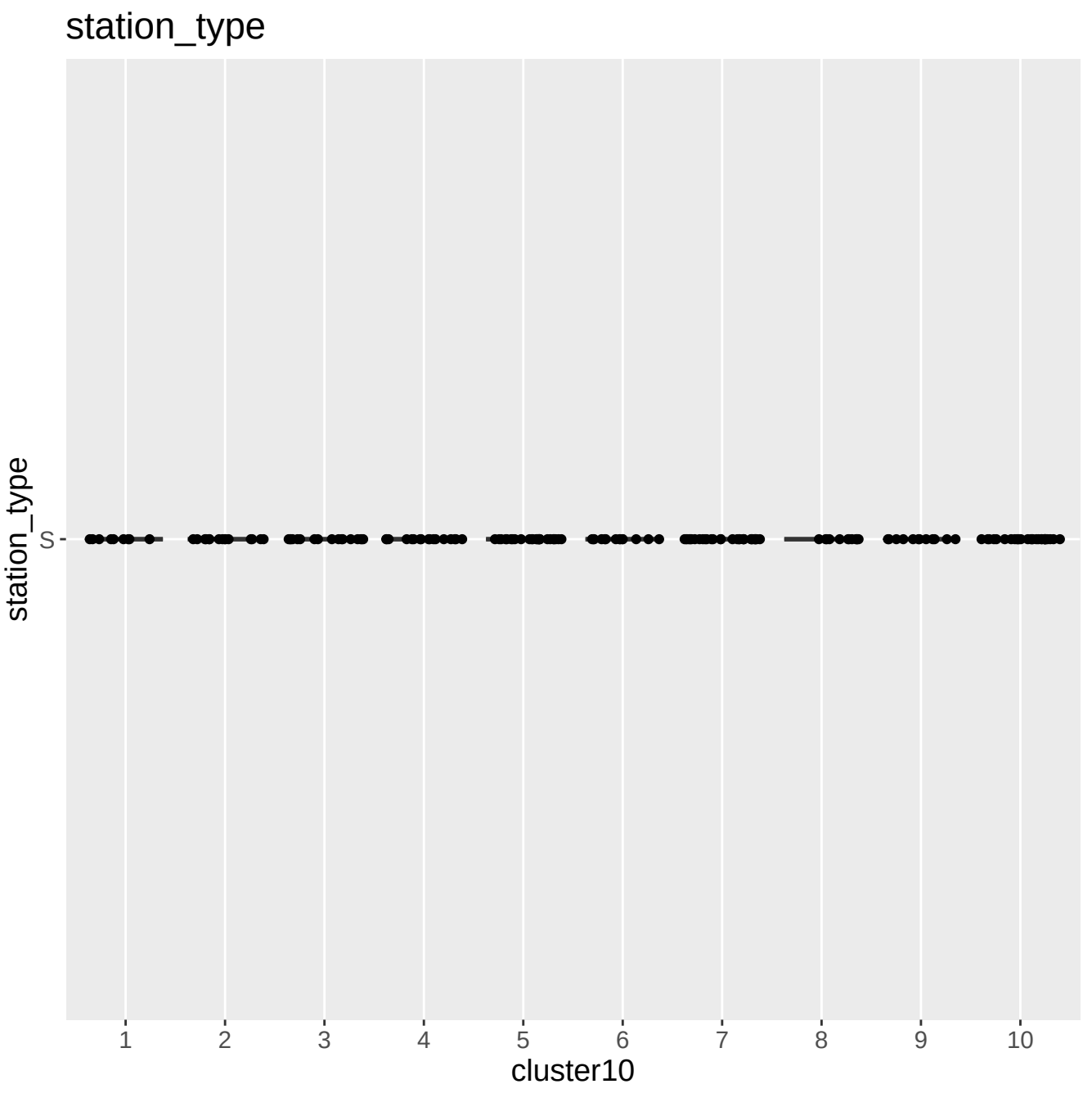
7
8
9
10


station_type
station_type
S
1
2
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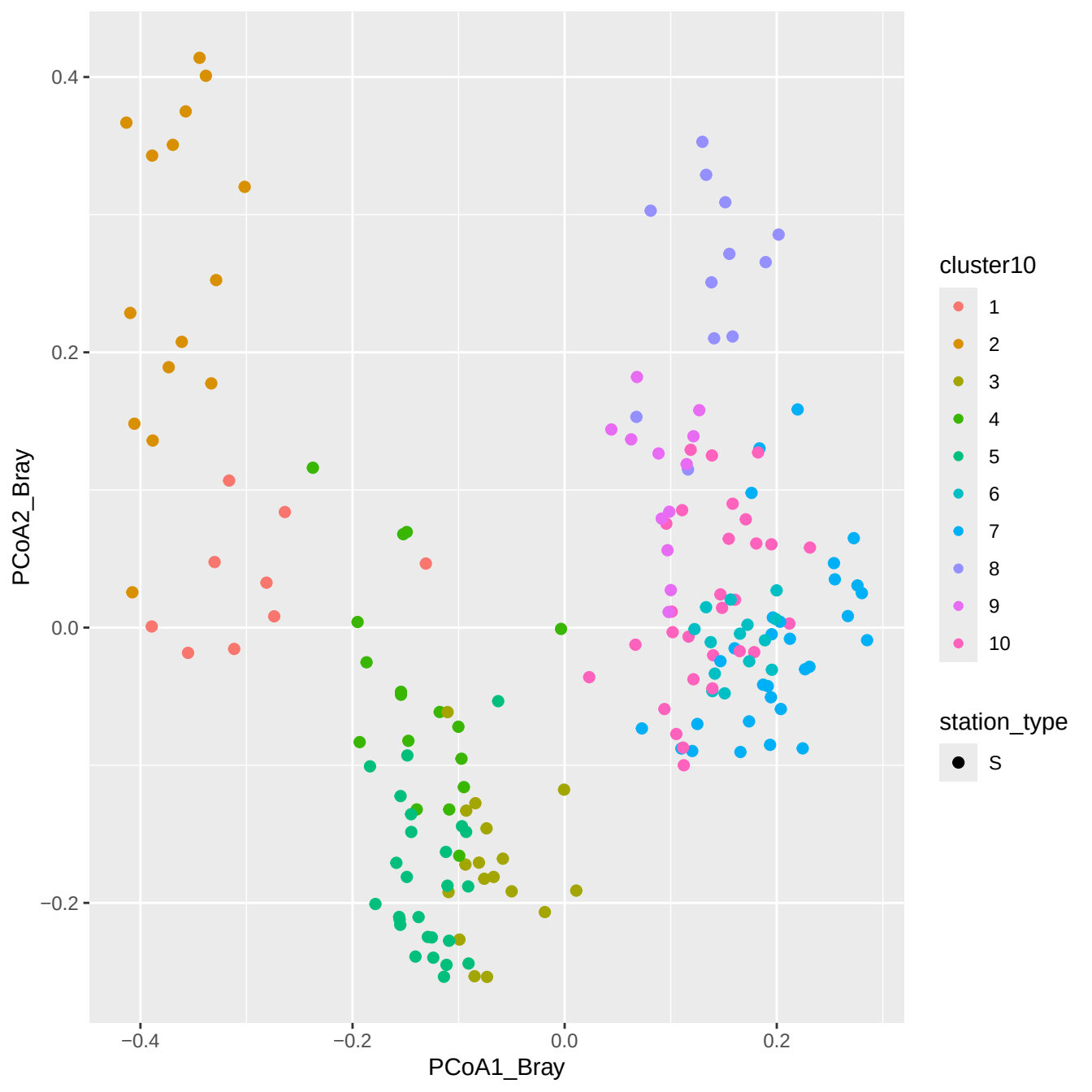
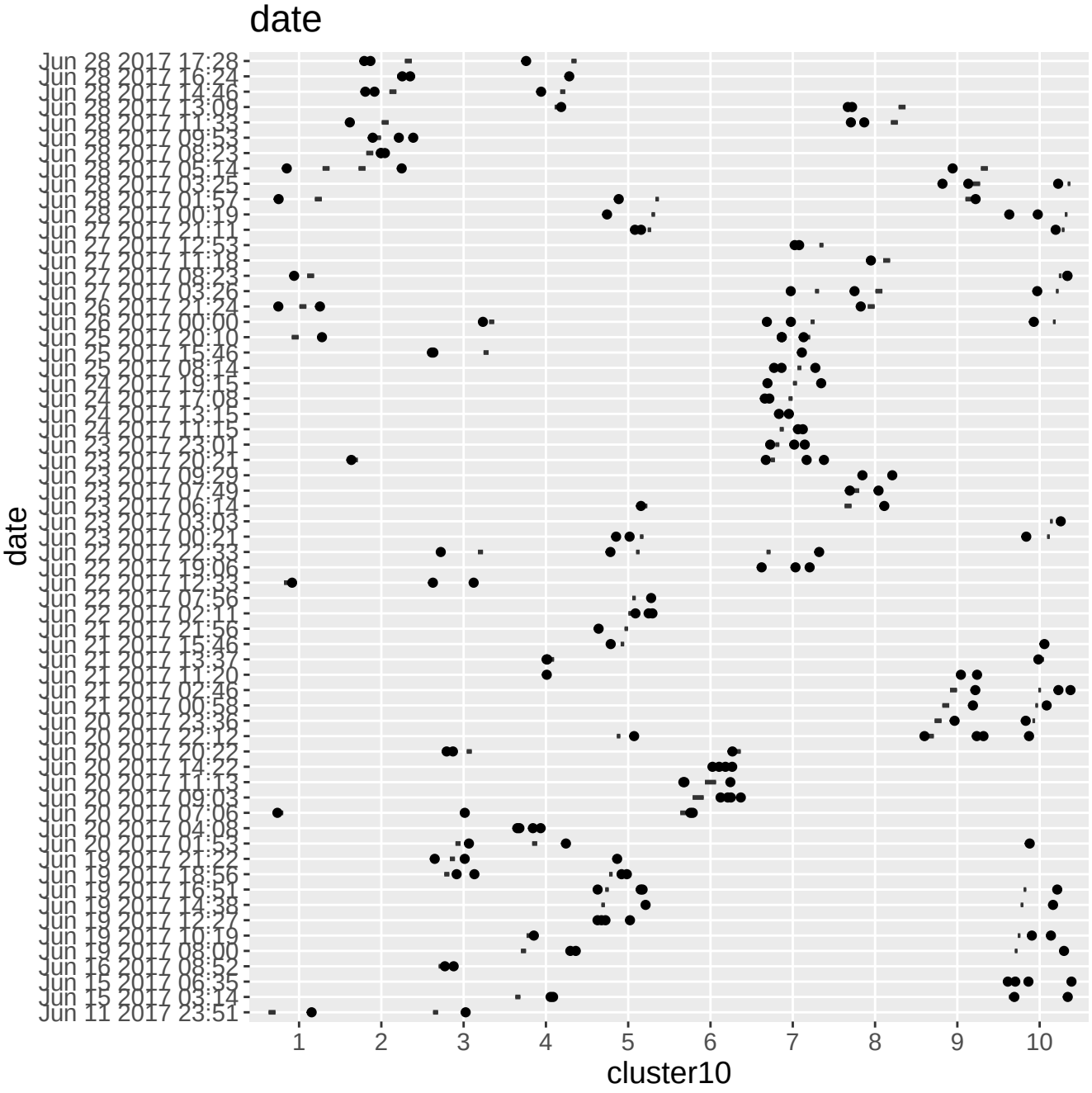
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
station_type
S


date
Jun 28 2017 17:28
Jun 28 2017 16:24
Jun 28 2017 14:46
Jun 28 2017 13:08
Jun 28 2017 11:33
Jun 28 2017 09:53
Jun 28 2017 08:23
Jun 28 2017 05:14
Jun 28 2017 03:25
Jun 28 2017 01:57
Jun 28 2017 00:19
Jun 27 2017 21:11
Jun 27 2017 12:53
Jun 27 2017 11:18
Jun 27 2017 08:23
Jun 27 2017 03:26
Jun 26 2017 21:24
Jun 26 2017 00:00
Jun 25 2017 20:10
Jun 25 2017 15:46
Jun 25 2017 08:14
Jun 24 2017 19:15
Jun 24 2017 17:08
Jun 24 2017 13:15
Jun 24 2017 11:15
Jun 23 2017 23:01
Jun 23 2017 20:21
Jun 23 2017 09:29
Jun 23 2017 07:49
Jun 23 2017 06:14
Jun 23 2017 03:03
Jun 23 2017 00:21
Jun 22 2017 22:33
Jun 22 2017 19:06
Jun 22 2017 12:33
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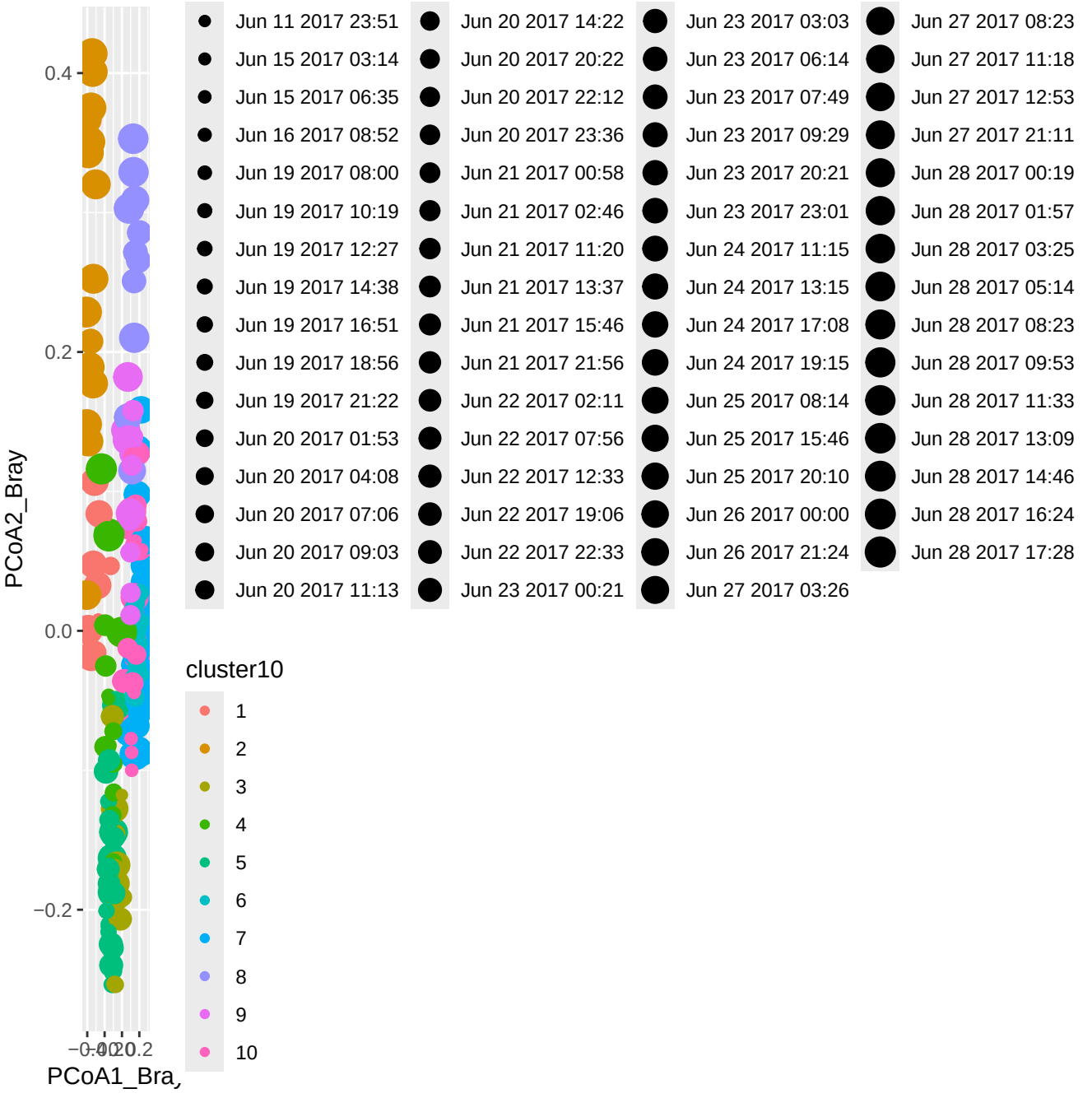
Jun 22 2017 02:11
Jun 21 2017 21:56
Jun 21 2017 15:46
Jun 21 2017 13:37
Jun 21 2017 11:20
Jun 21 2017 02:46
Jun 21 2017 00:58
Jun 20 2017 23:38
Jun 20 2017 22:12
Jun 20 2017 20:22
Jun 20 2017 14:22
Jun 20 2017 11:13
Jun 20 2017 09:03
Jun 20 2017 07:06
Jun 20 2017 04:08
Jun 20 2017 01:53
Jun 19 2017 21:32
Jun 19 2017 18:56
Jun 19 2017 16:51
Jun 19 2017 14:38
Jun 19 2017 12:27
Jun 19 2017 10:19
Jun 19 2017 08:00
Jun 16 2017 08:52
Jun 15 2017 06:35
Jun 15 2017 03:14
Jun 11 2017 23:51
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
Jun 11 2017 23:51
Jun 15 2017 03:14
Jun 15 2017 06:35
Jun 16 2017 08:52
Jun 19 2017 08:00
Jun 19 2017 10:19
Jun 19 2017 12:27
Jun 19 2017 14:38
Jun 19 2017 16:51
Jun 19 2017 18:56
Jun 19 2017 21:22
Jun 20 2017 01:53
Jun 20 2017 04:08
Jun 20 2017 07:06
Jun 20 2017 09:03
Jun 20 2017 11:13
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Jun 21 2017 11:20
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Jun 21 2017 15:46
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Jun 22 2017 02:11
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Jun 22 2017 12:33
Jun 22 2017 19:06
Jun 22 2017 22:33
Jun 23 2017 00:21
Jun 23 2017 03:03
Jun 23 2017 06:14
Jun 23 2017 07:49
Jun 23 2017 09:29
Jun 23 2017 20:21
Jun 23 2017 23:01
Jun 24 2017 11:15
Jun 24 2017 13:15
Jun 24 2017 17:08
Jun 24 2017 19:15
Jun 25 2017 08:14
Jun 25 2017 15:46
Jun 25 2017 20:10
Jun 26 2017 00:00
Jun 26 2017 21:24
Jun 27 2017 03:26
Jun 27 2017 08:23
Jun 27 2017 11:18
Jun 27 2017 12:53
Jun 27 2017 21:11
Jun 28 2017 00:19
Jun 28 2017 01:57
Jun 28 2017 03:25
Jun 28 2017 05:14
Jun 28 2017 08:23
Jun 28 2017 09:53
Jun 28 2017 11:33
Jun 28 2017 13:09
Jun 28 2017 14:46
Jun 28 2017 16:24
Jun 28 2017 17:28
cluster10
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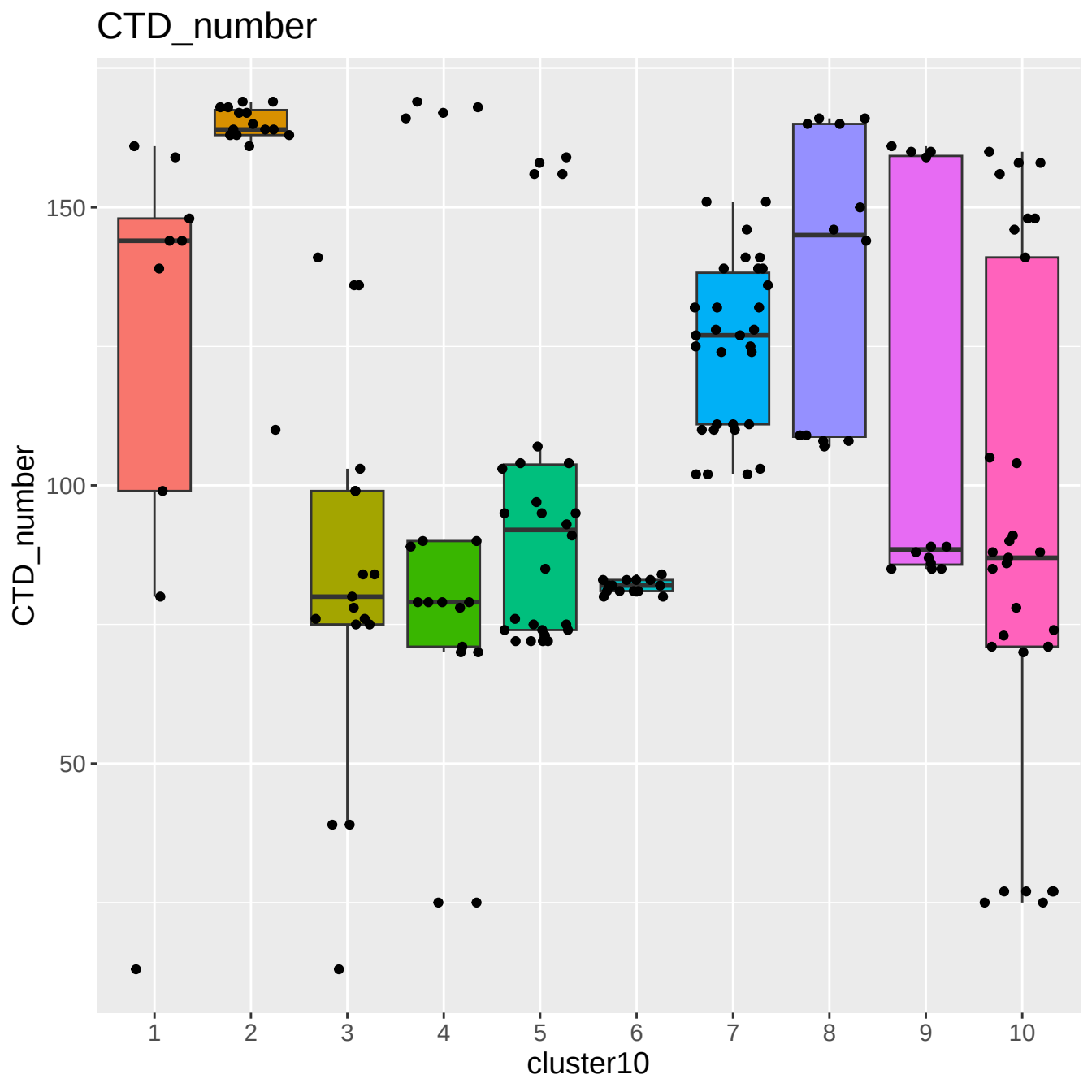
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CTD_number
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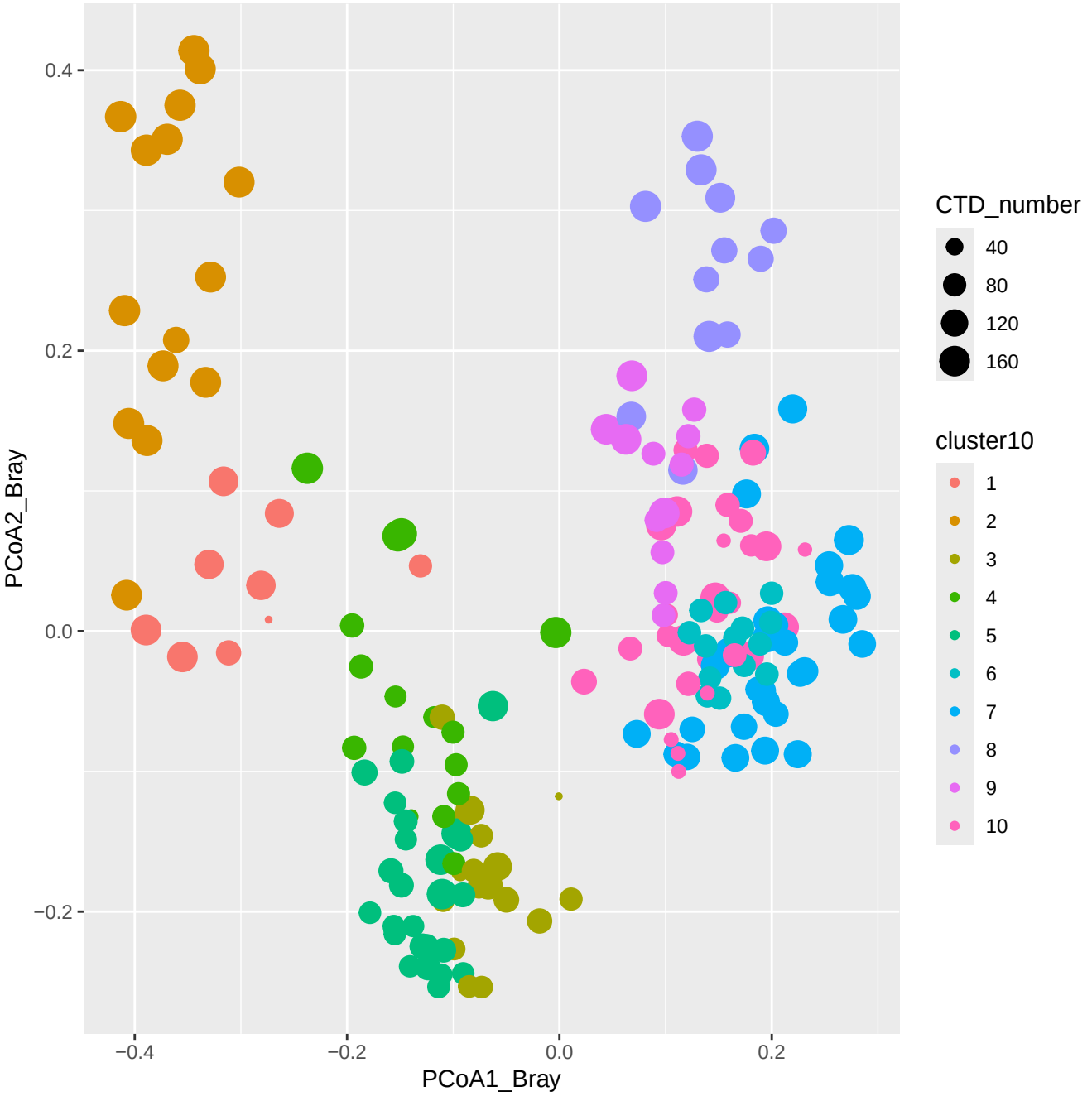
cluster10


0.4
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0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
CTD_number
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80
120
160
cluster10
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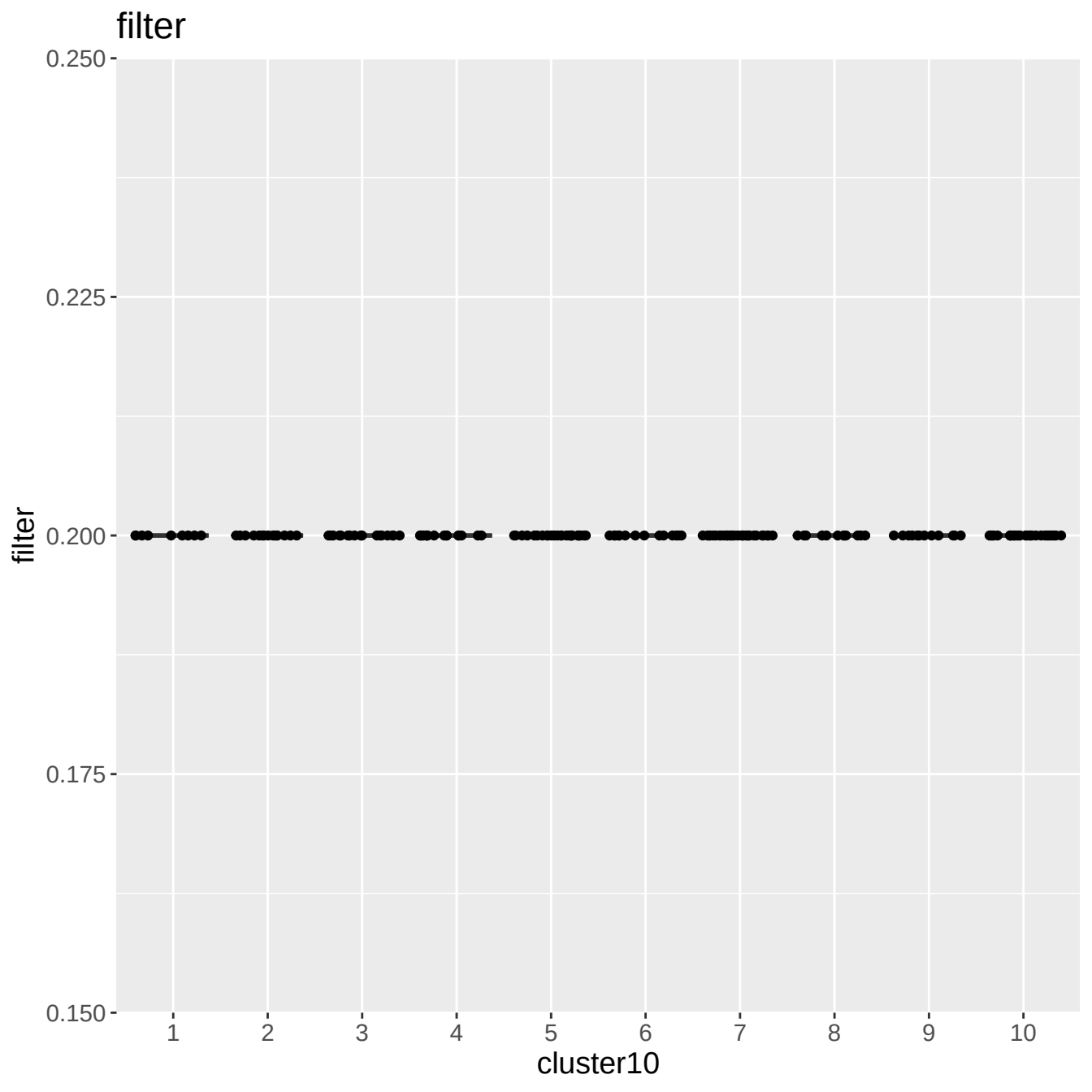
filter
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0.175
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filter
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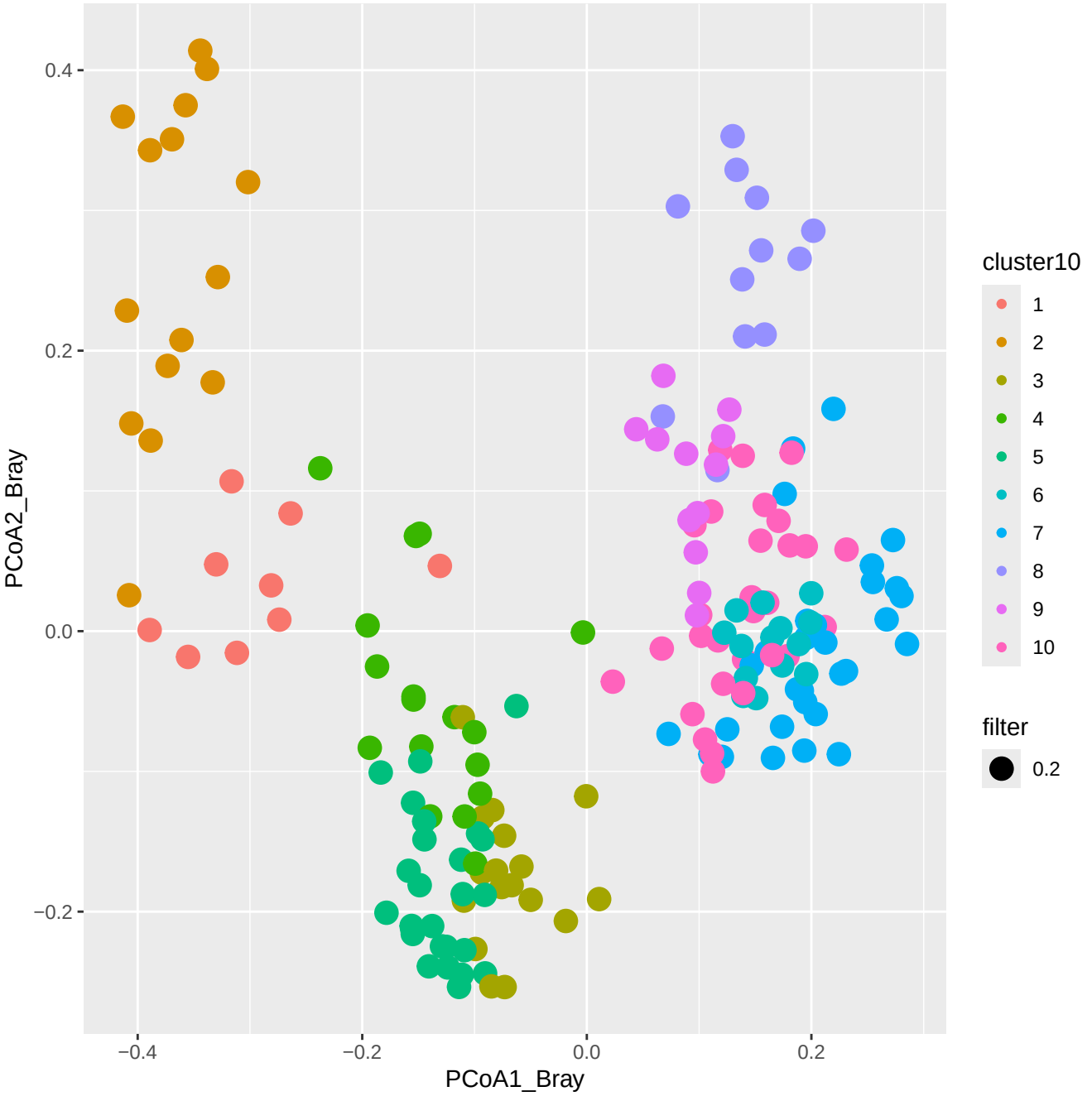
cluster10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PCoA2_Bray
−0.4
−0.2
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PCoA1_Bray
cluster10
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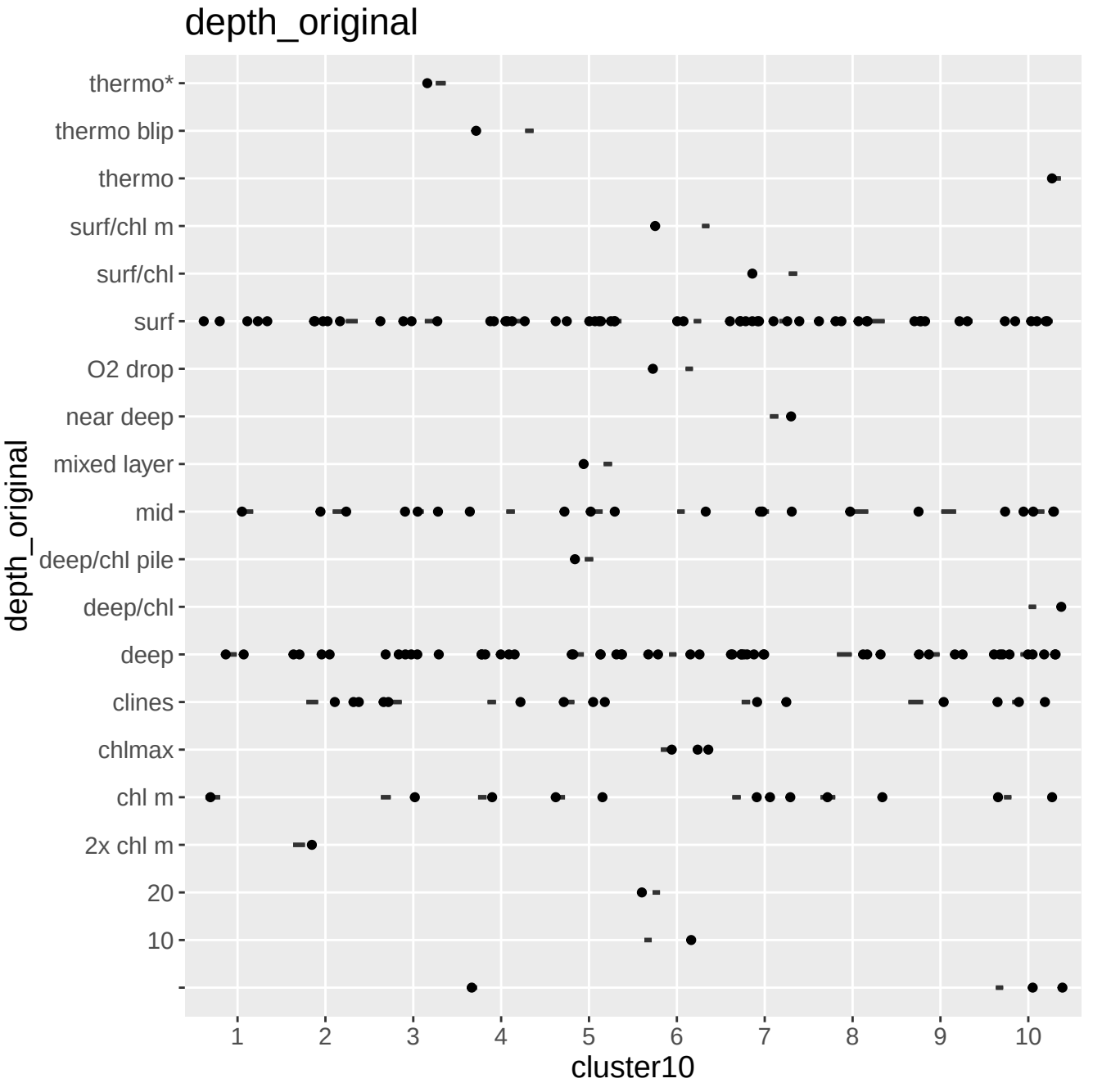
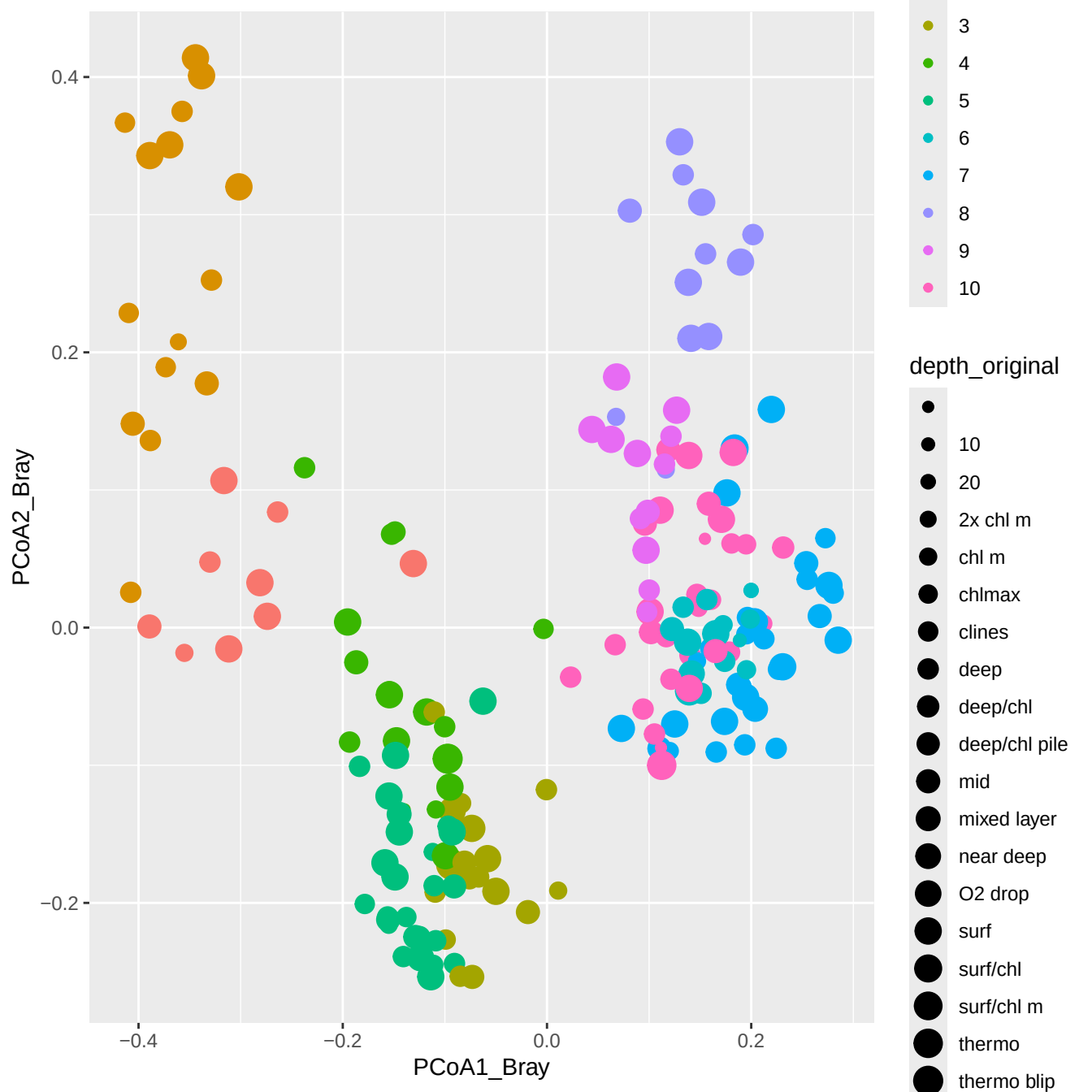
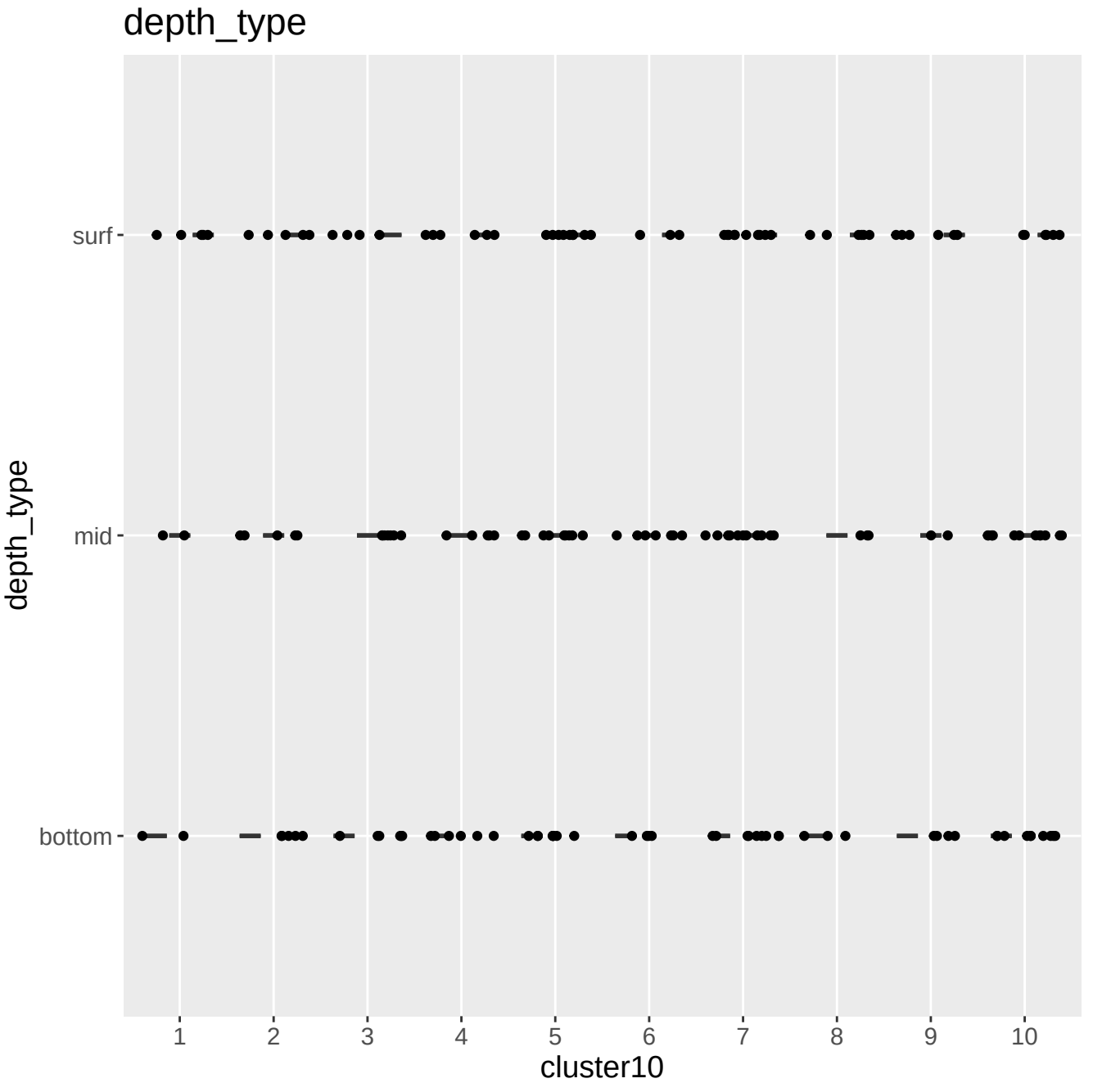
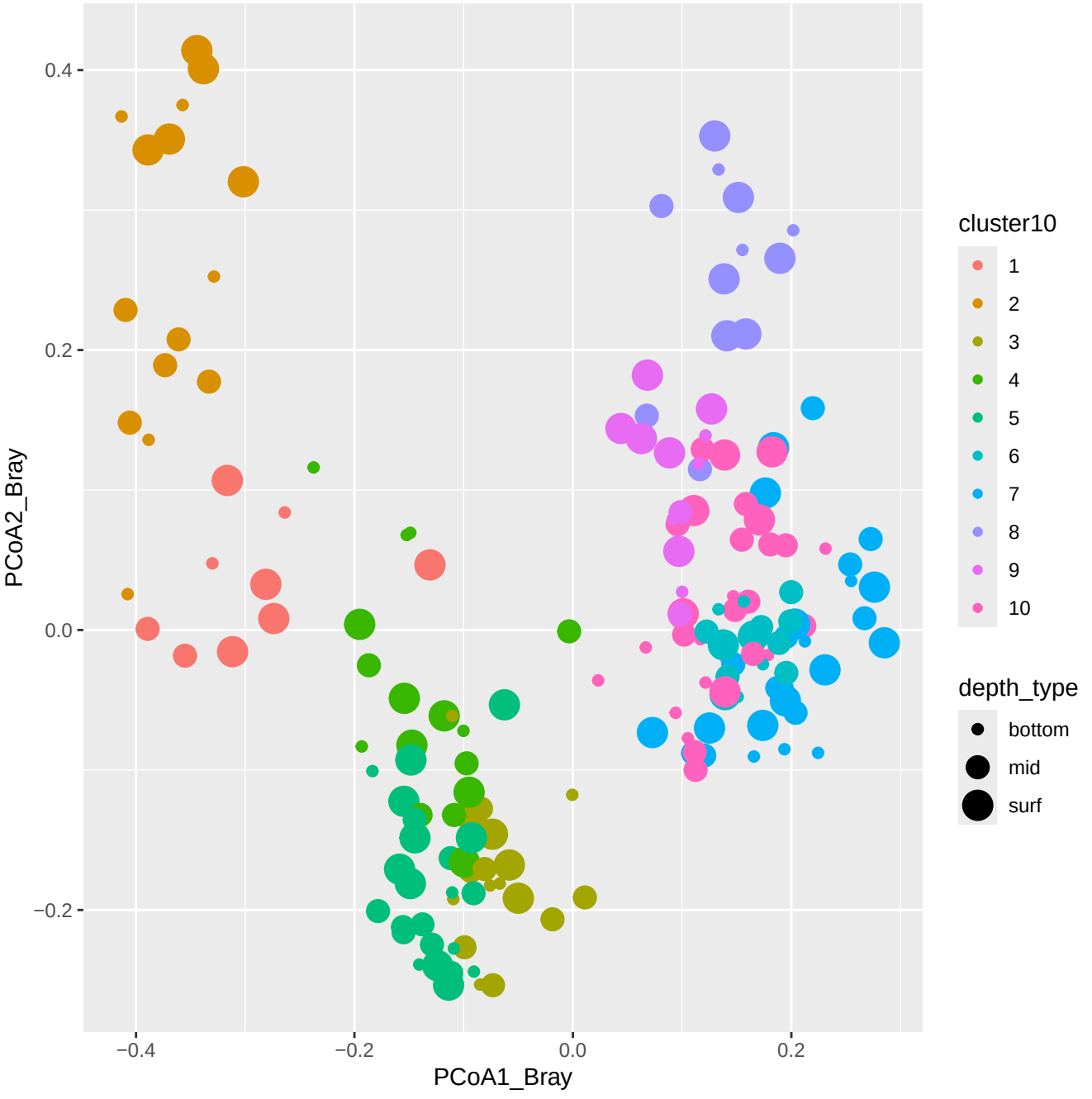
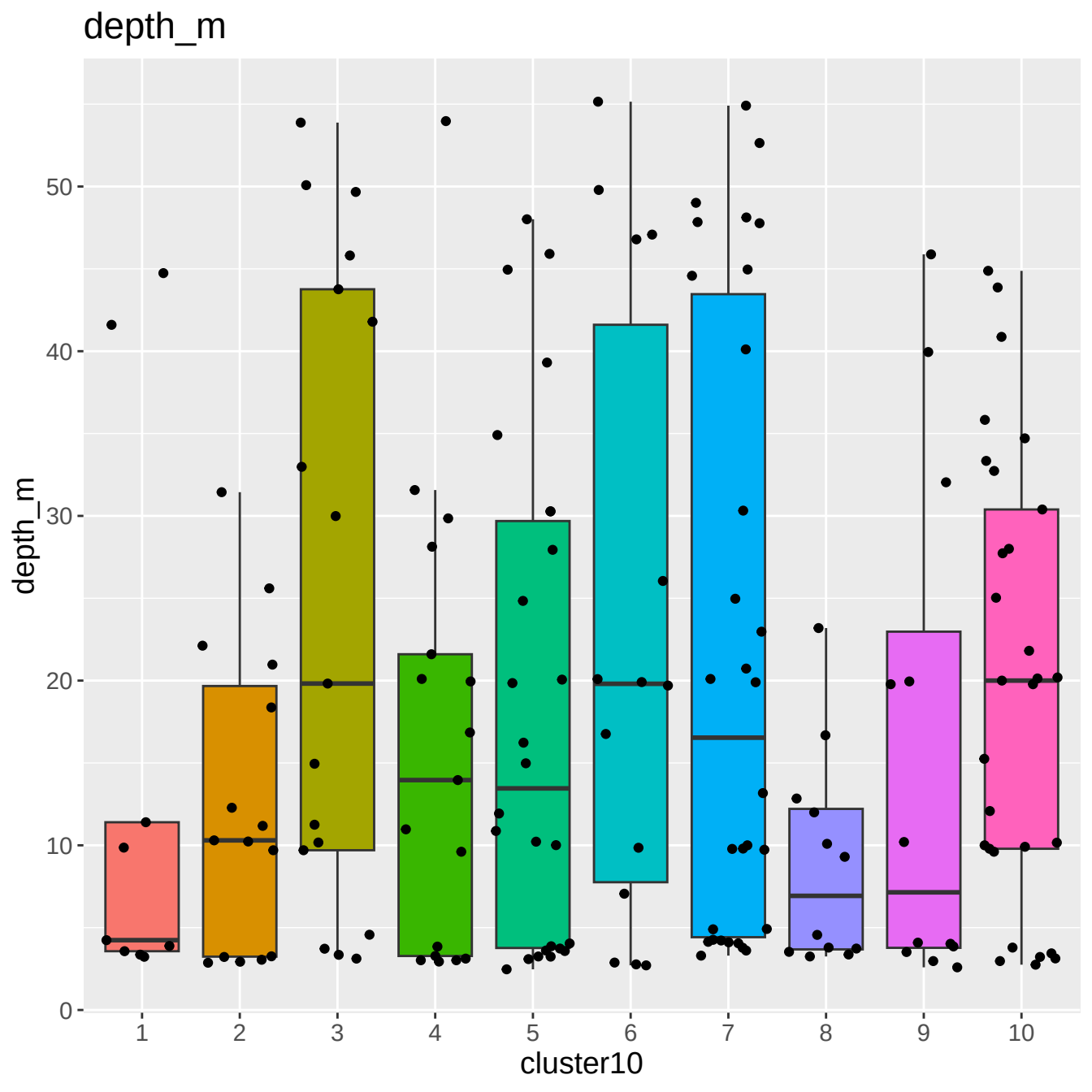
filter
0.2


depth_original
thermo*
thermo blip
thermo
surf/chl m
surf/chl
surf
O2 drop
near deep
mixed layer
mid
deep/chl pile
deep/chl
deep
clines
chlmax
chl m
2x chl m
20
10
-
depth_original
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
3
4
5
6
7
8
9
10
depth_original
10
20
2x chl m
chl m
chlmax
clines
deep
deep/chl
deep/chl pile
mid
mixed layer
near deep
O2 drop
surf
surf/chl
surf/chl m
thermo
thermo blip


depth_type
surf
mid
bottom
depth_type
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
depth_type
bottom
mid
surf


depth_m
depth_m
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10
20
30
40
50
1
2
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4
5
6
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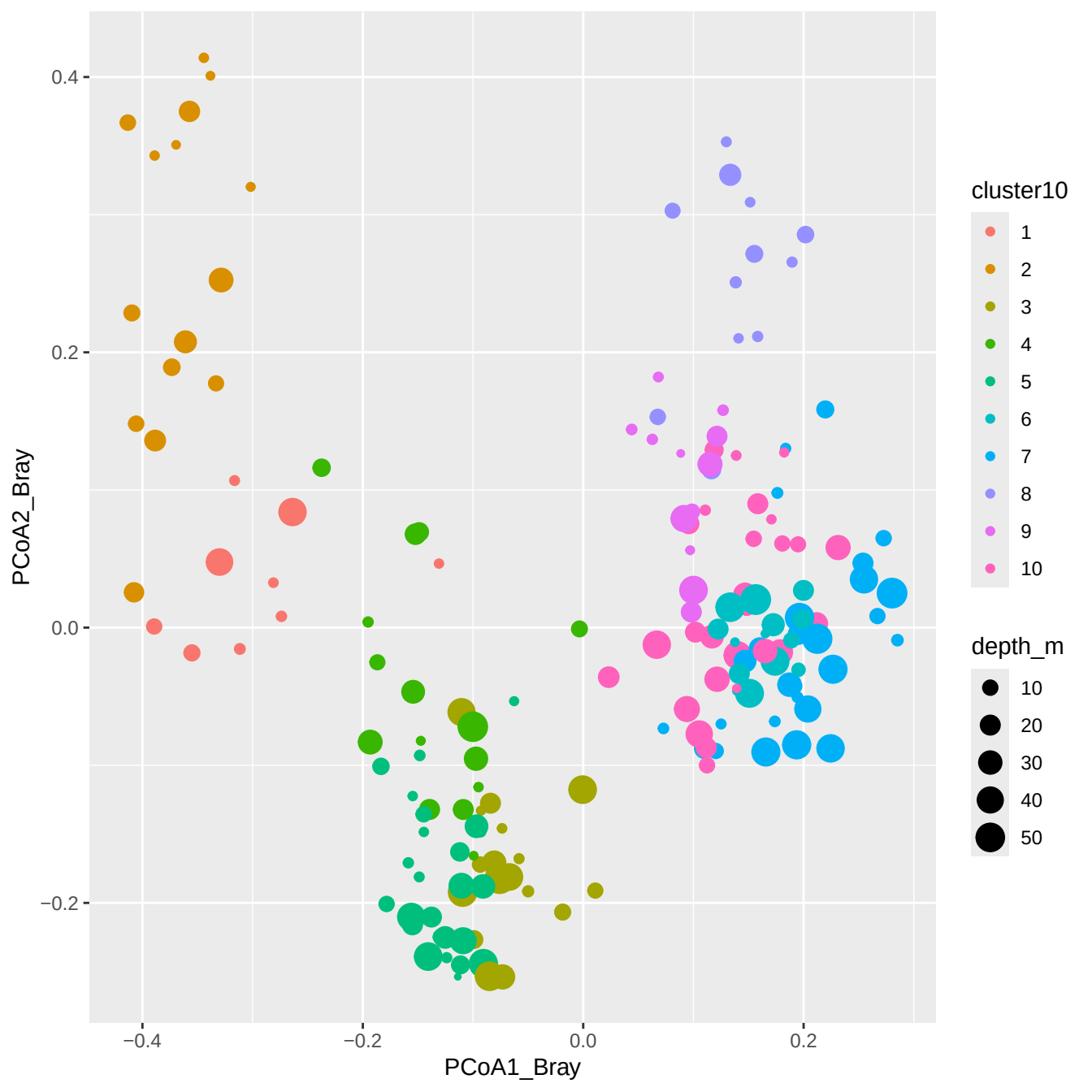
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
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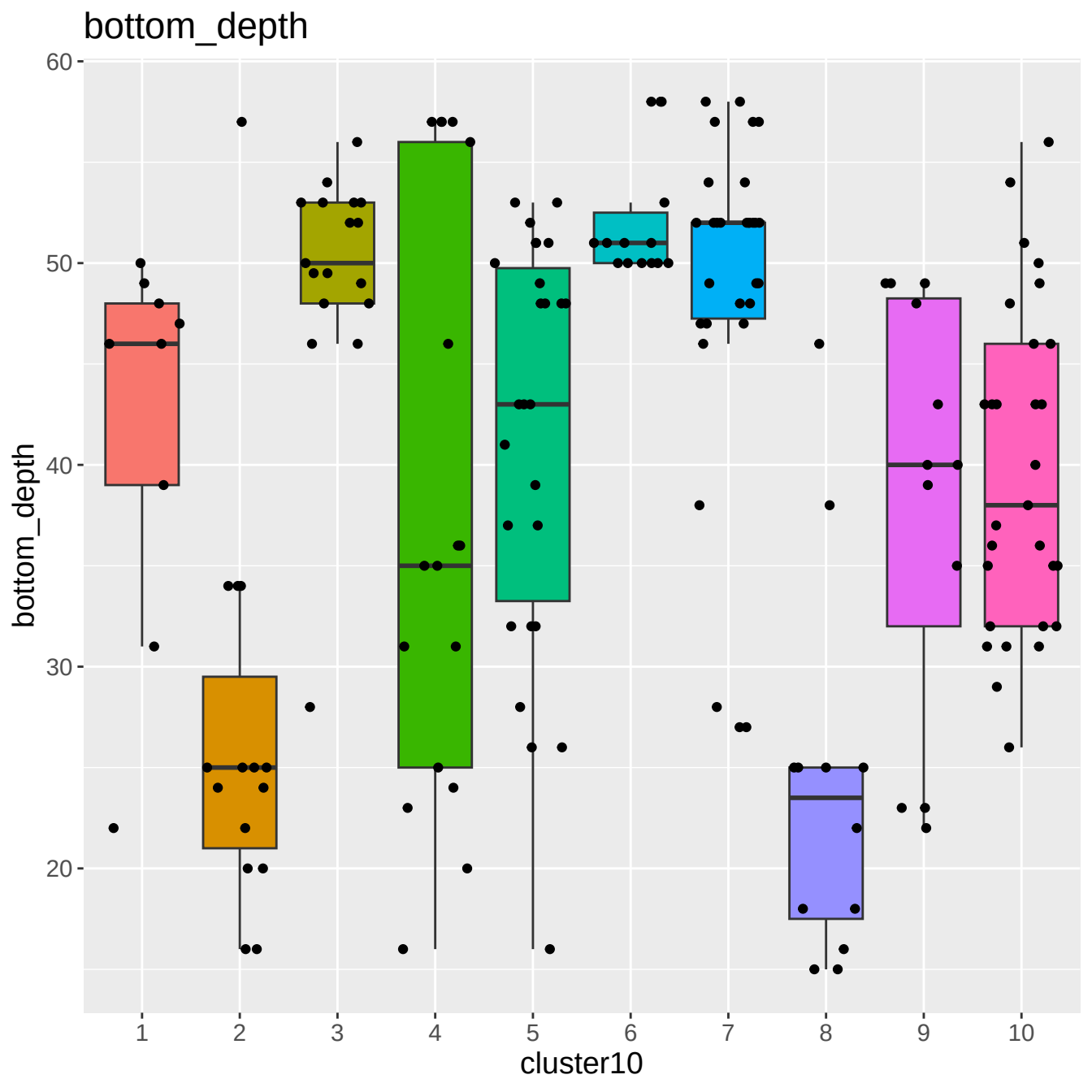
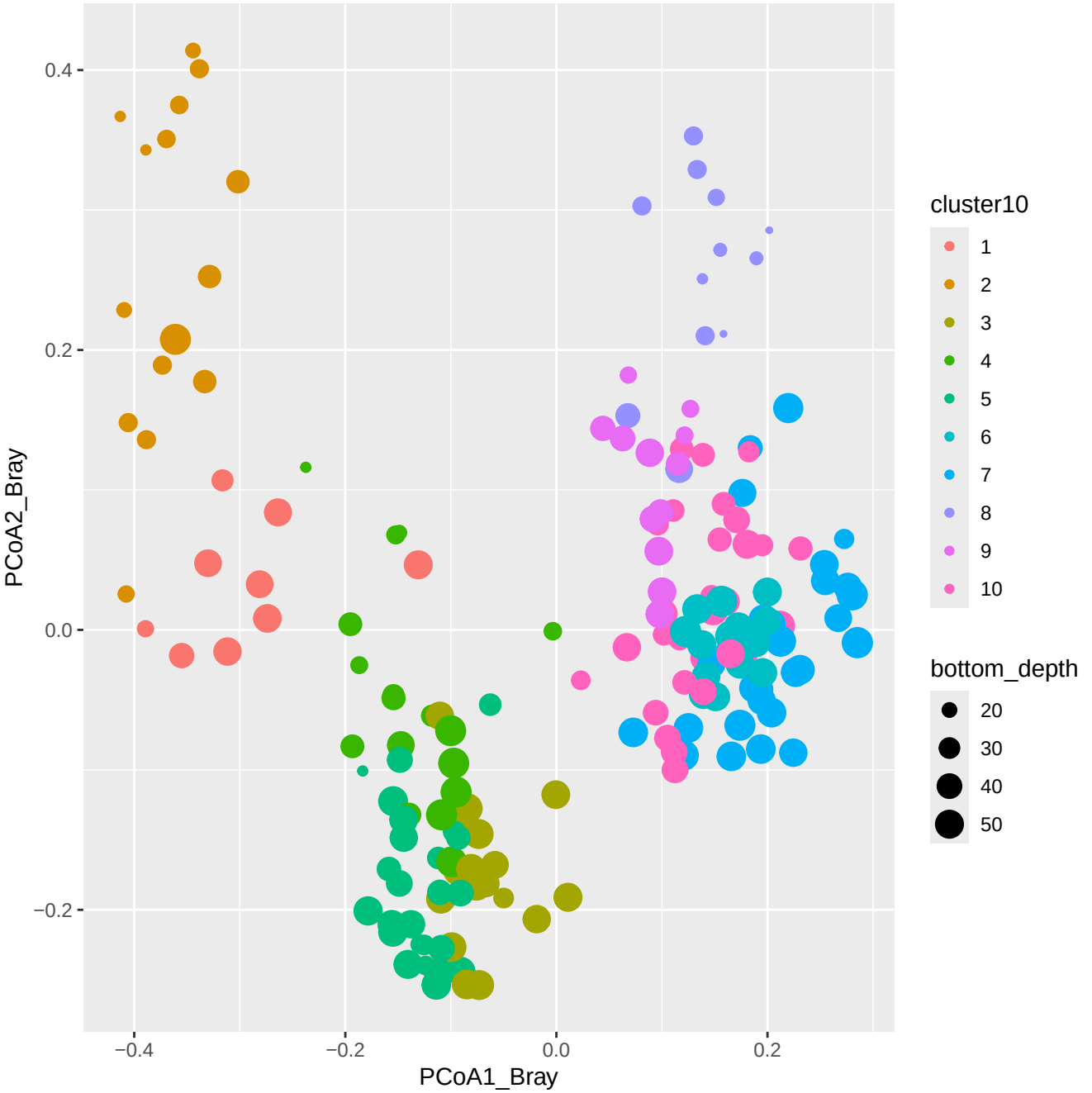
2
3
4
5
6
7
8
9
10
depth_m
10
20
30
40
50


bottom_depth
60
50
40
30
20
bottom_depth
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
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4
5
6
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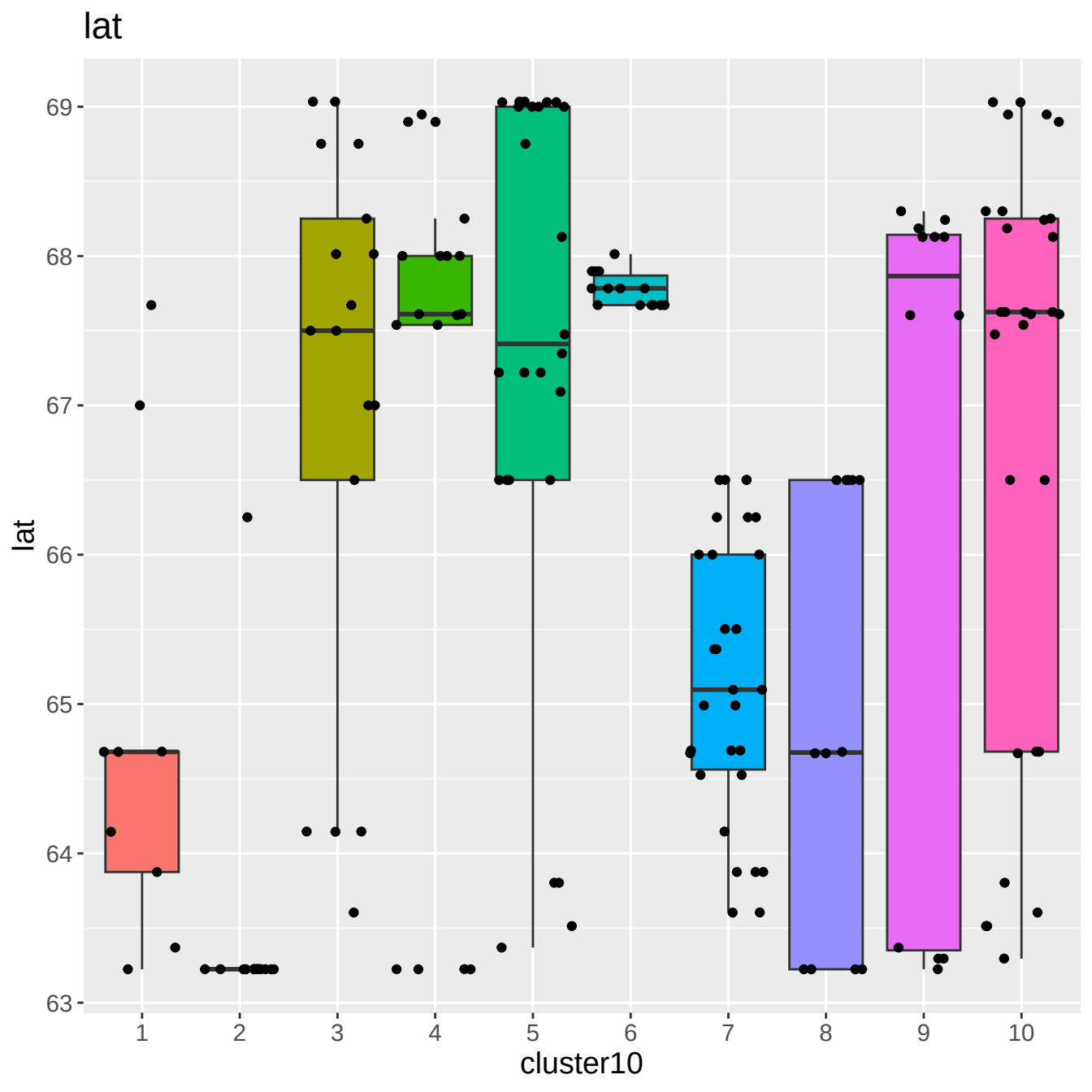
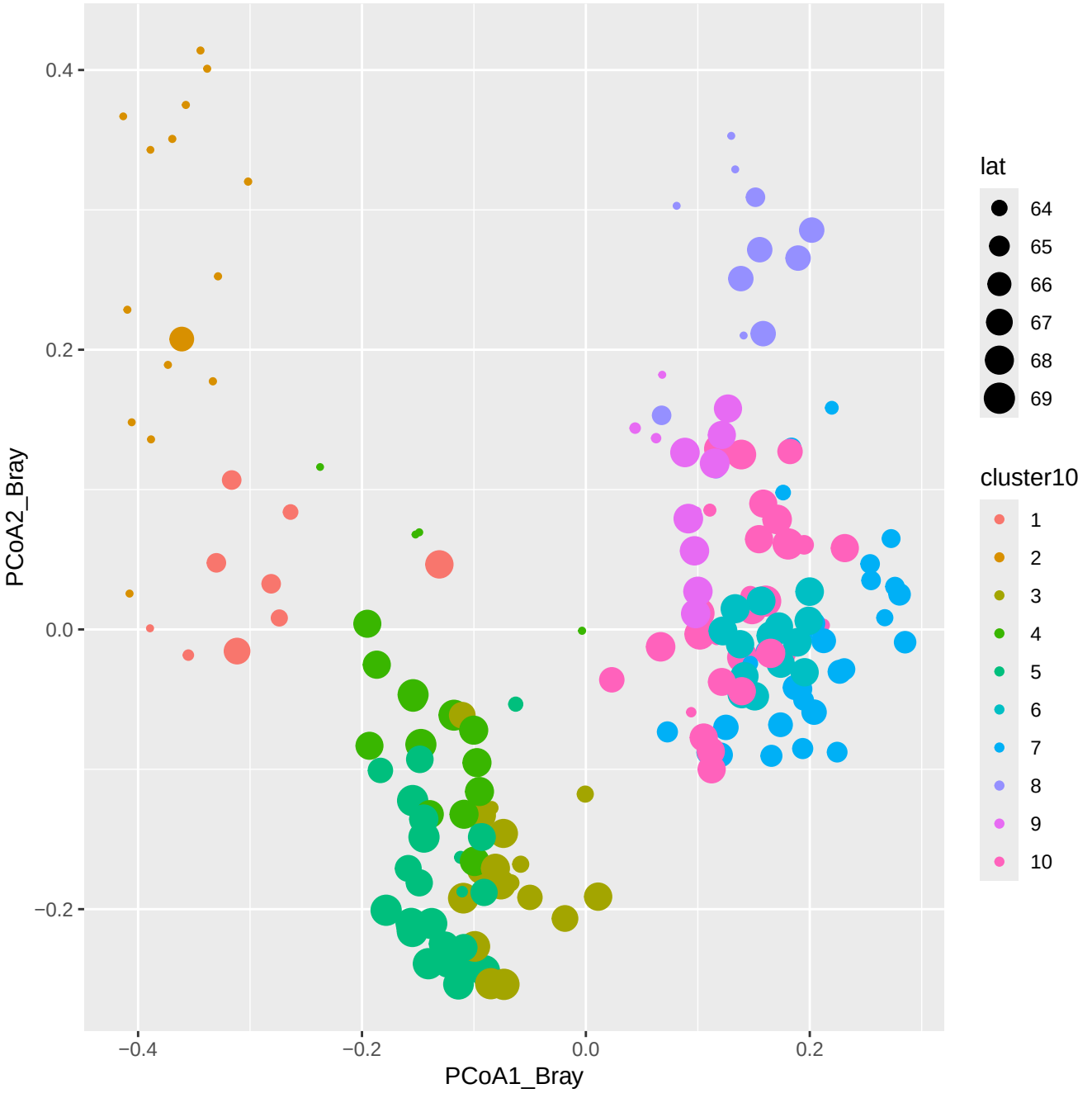
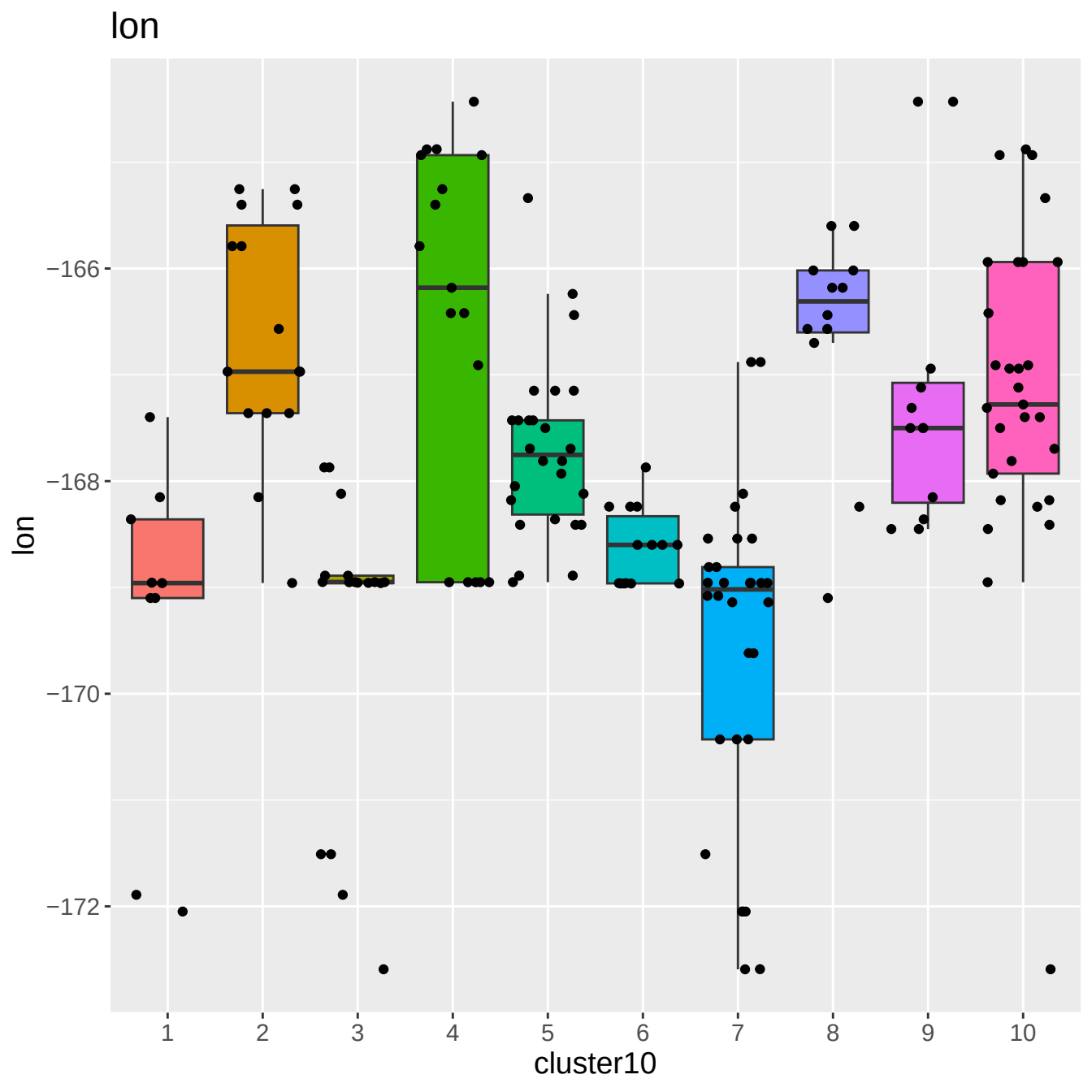
8
9
10
bottom_depth
20
30
40
50


lat
69
68
67
66
65
64
63
lat
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
lat
64
65
66
67
68
69
cluster10
1
2
3
4
5
6
7
8
9
10


lon
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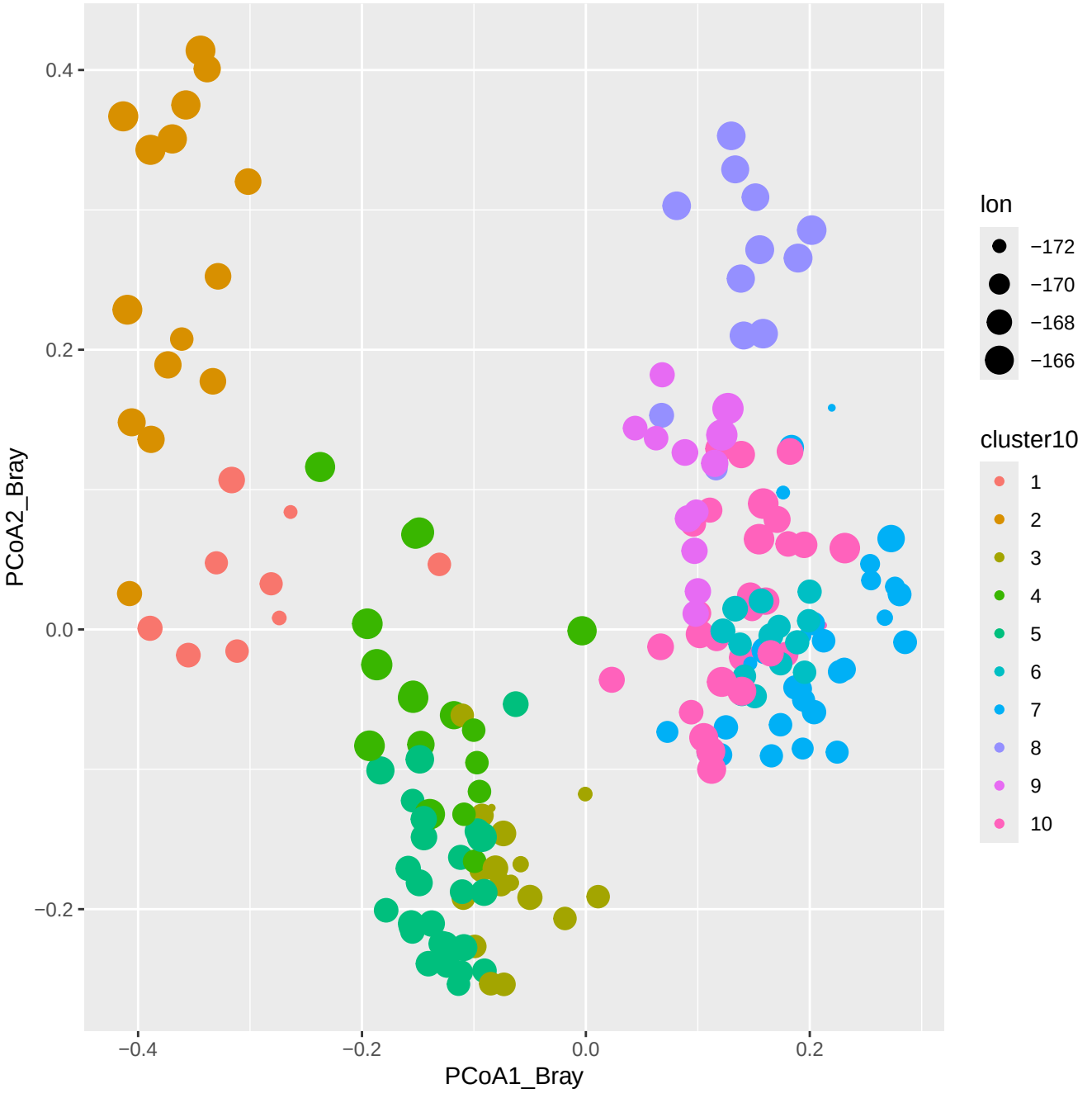
−168
−170
−172
lon
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
lon
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−170
−168
−166
cluster10
1
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3
4
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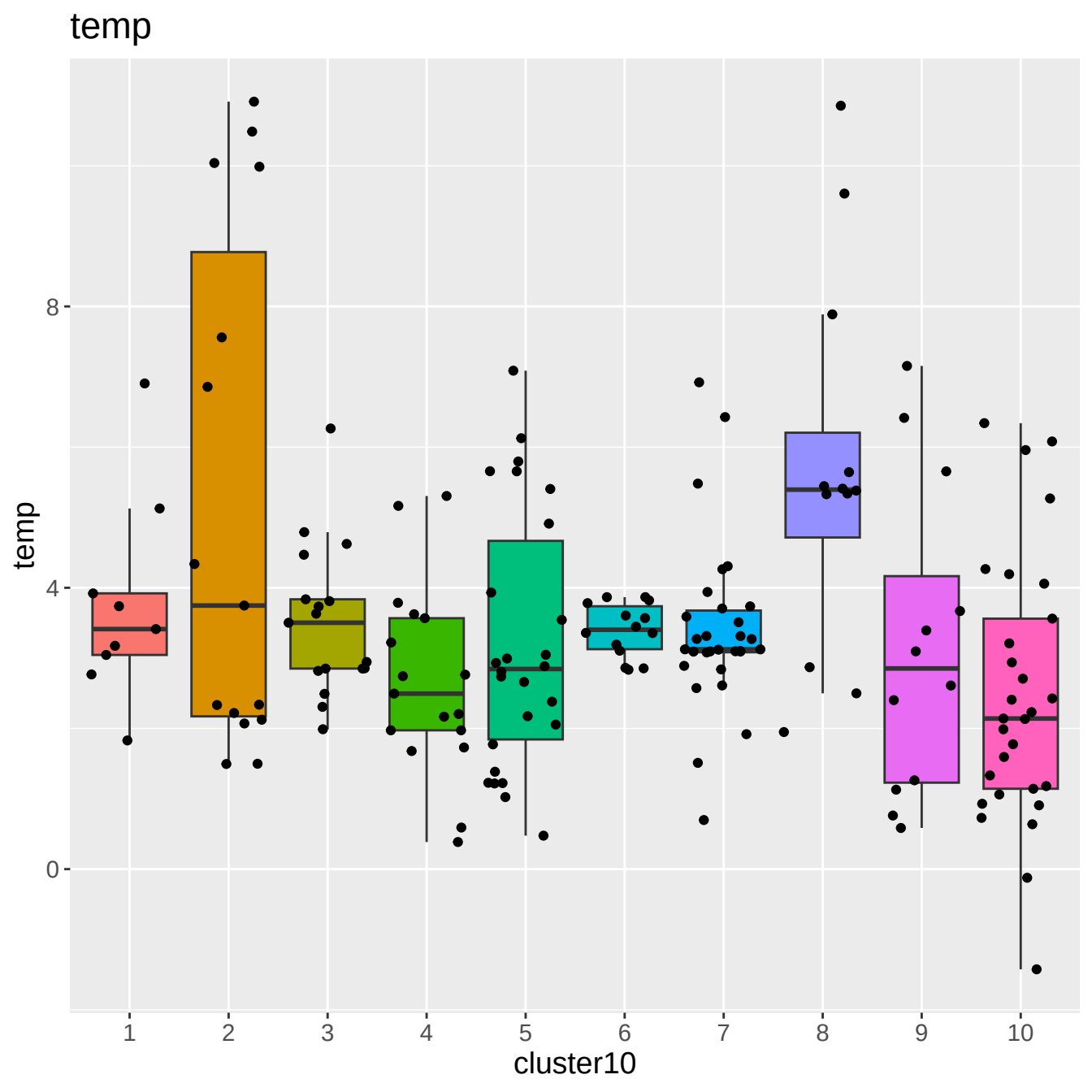
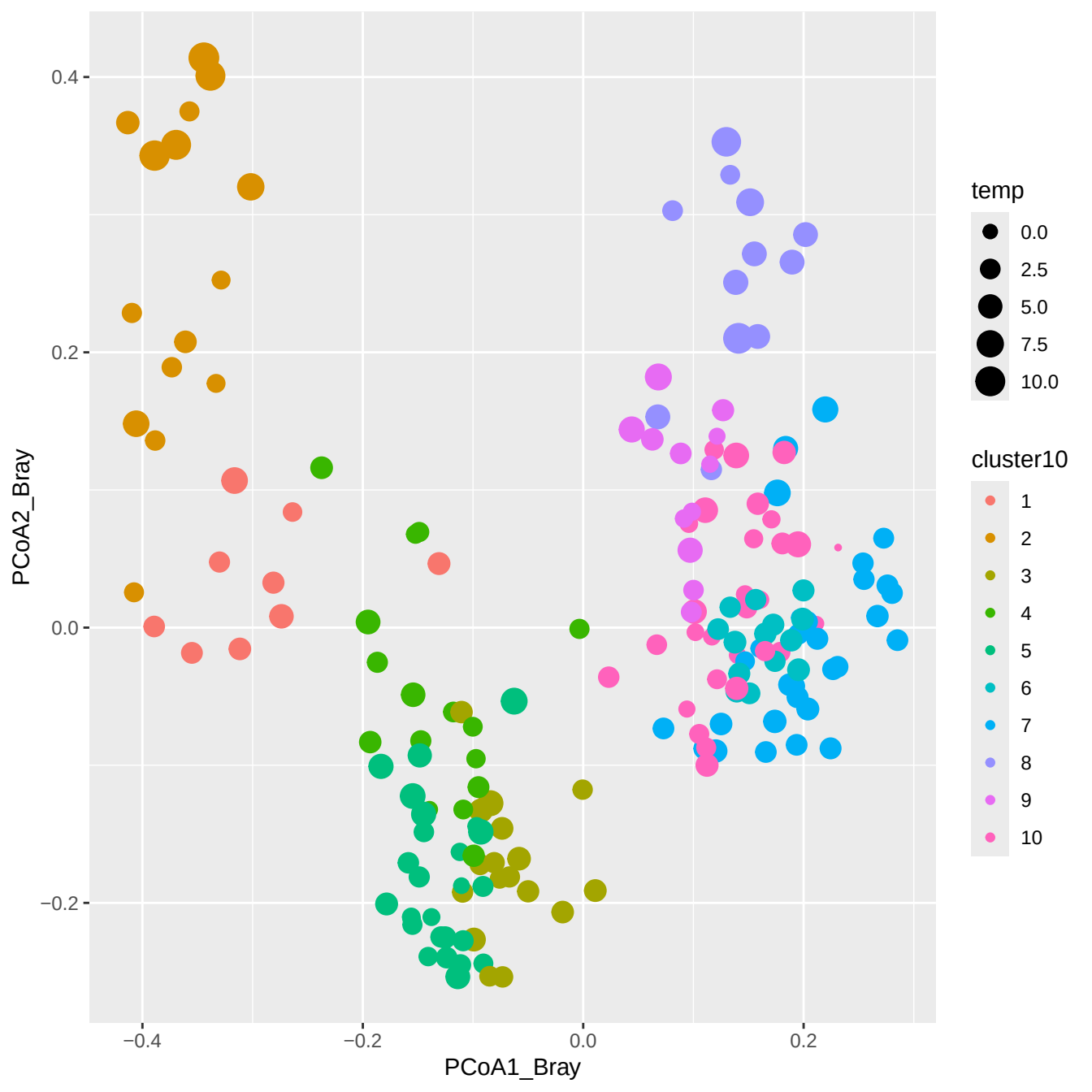
7
8
9
10


temp
temp
8
4
0
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
temp
0.0
2.5
5.0
7.5
10.0
cluster10
1
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3
4
5
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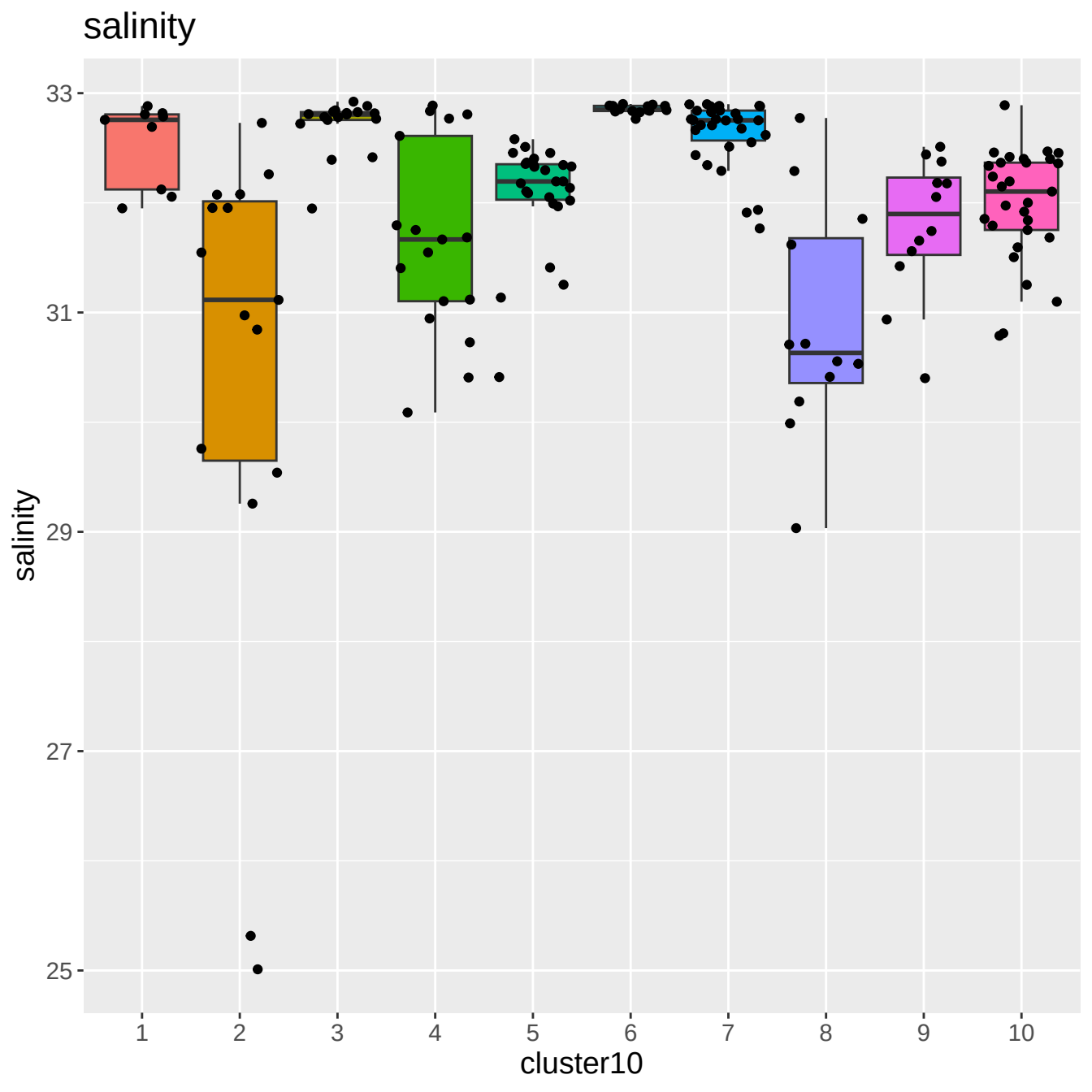
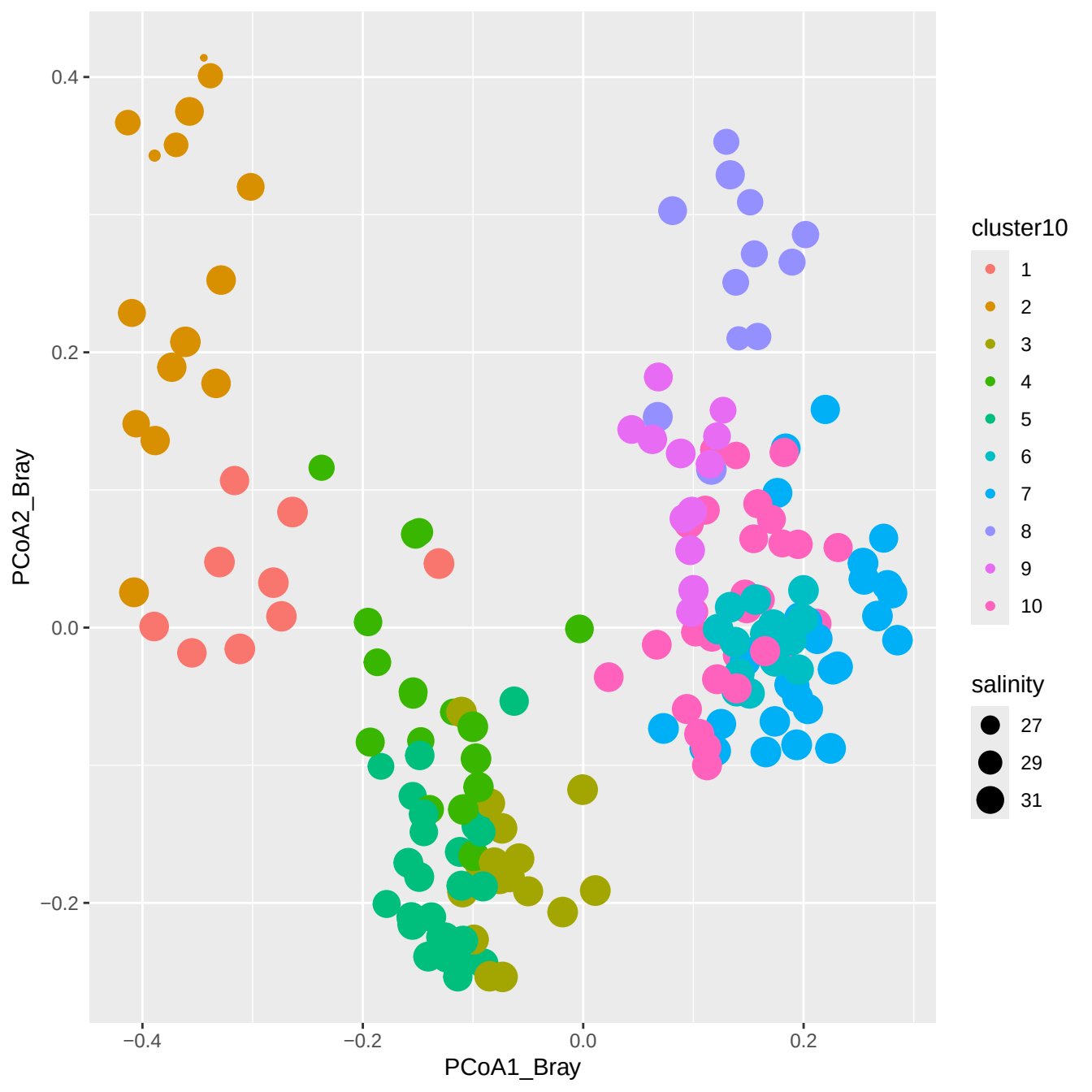
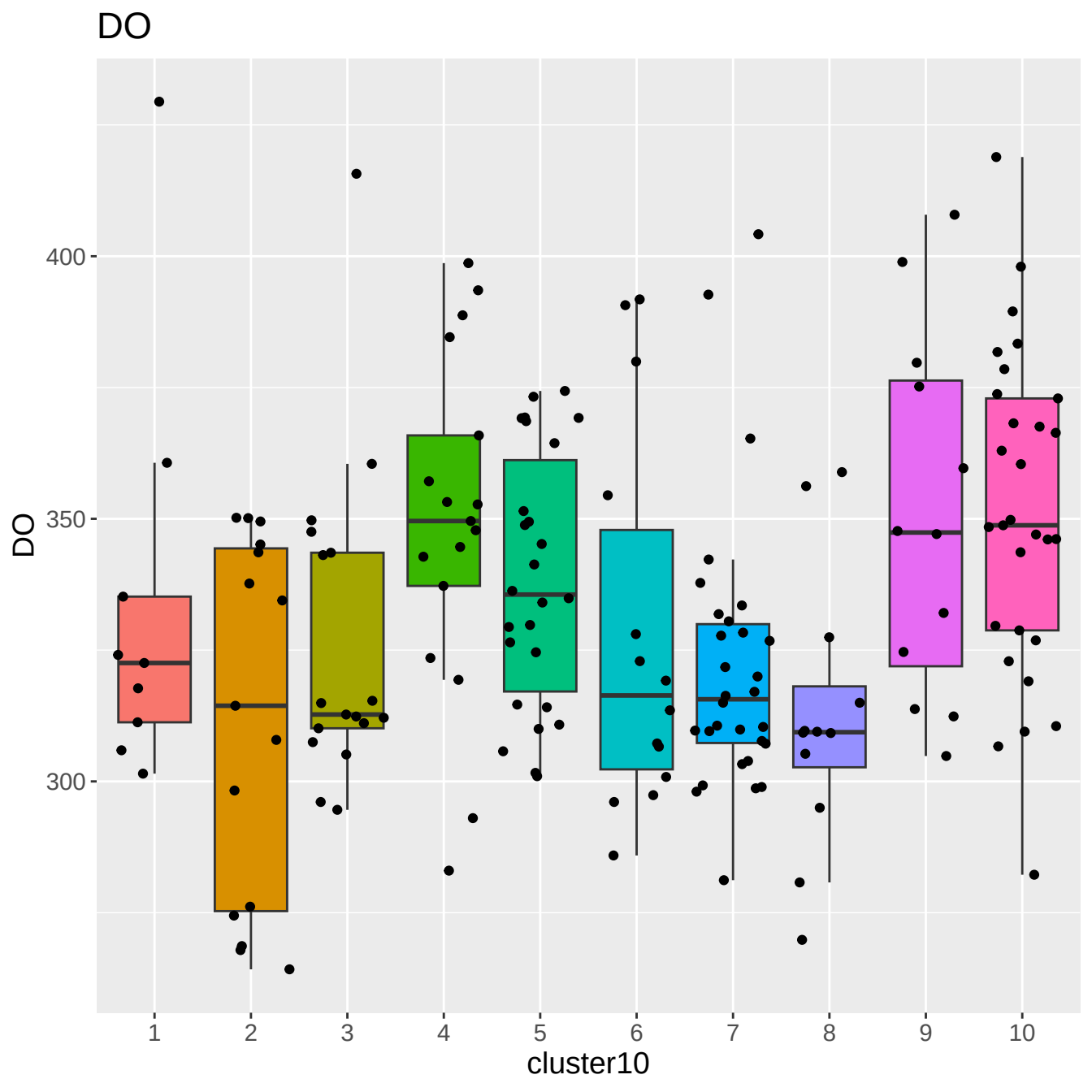
7
8
9
10


salinity
33
31
29
27
25
salinity
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
salinity
27
29
31


DO
400
350
300
DO
1
2
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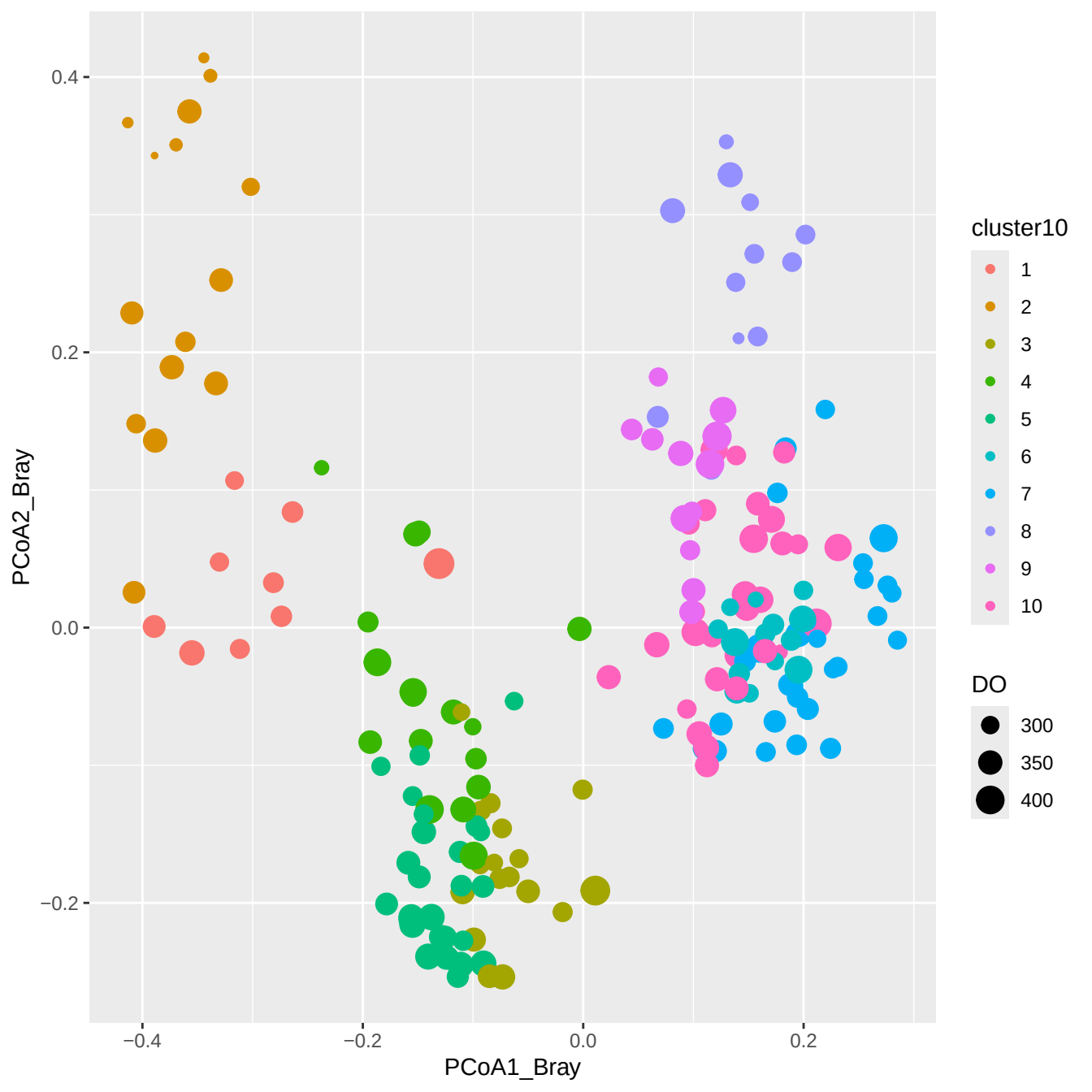
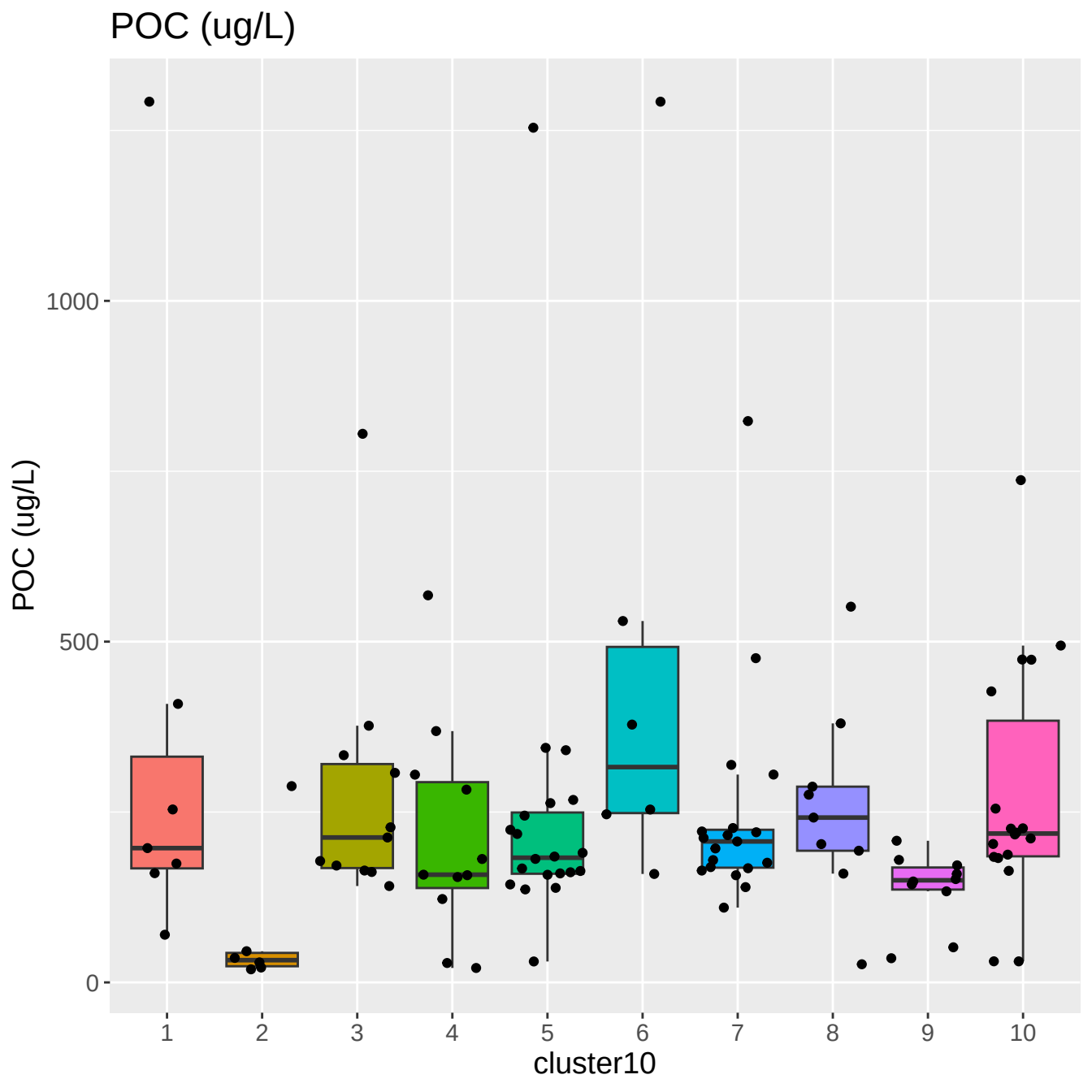
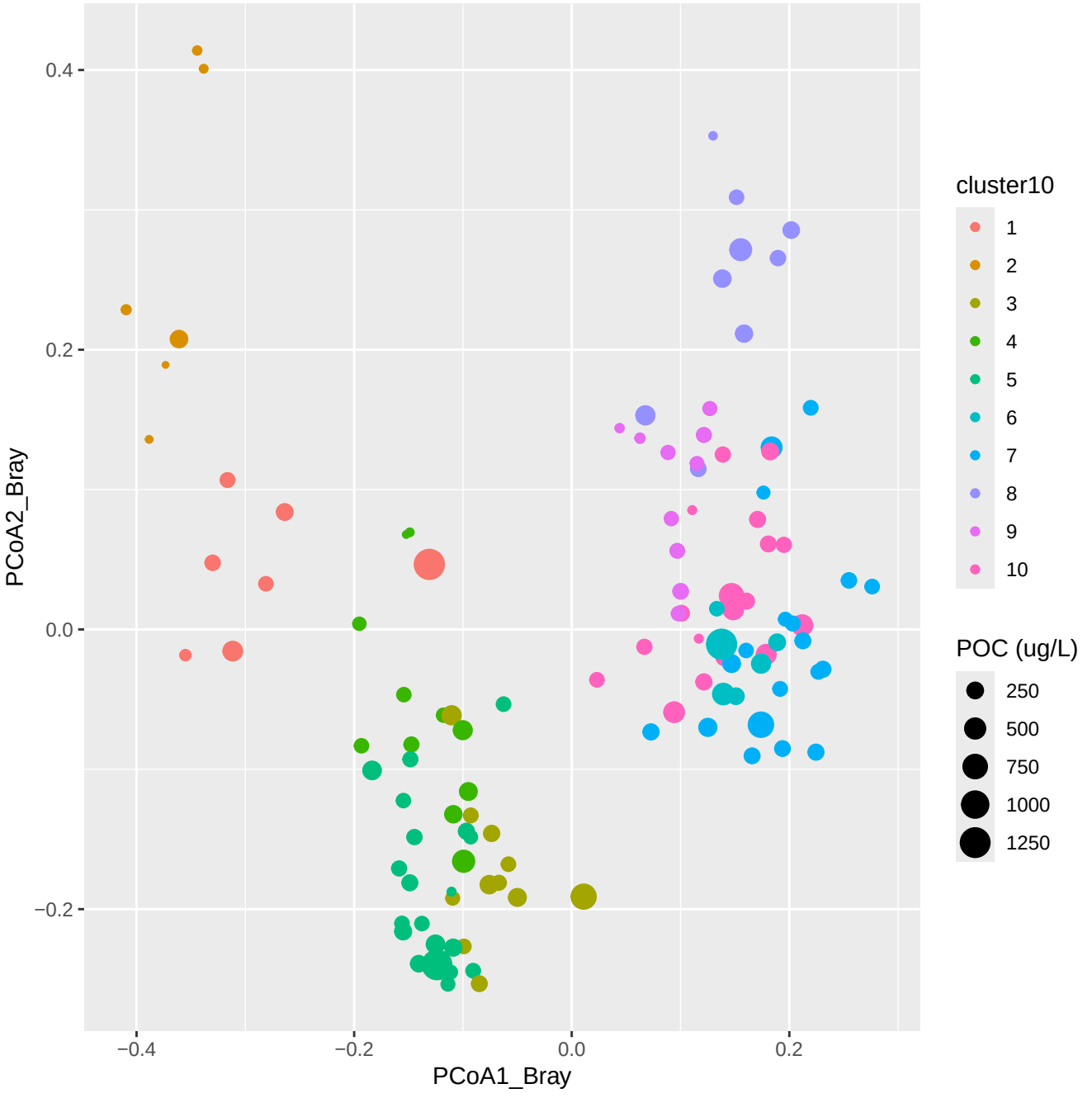
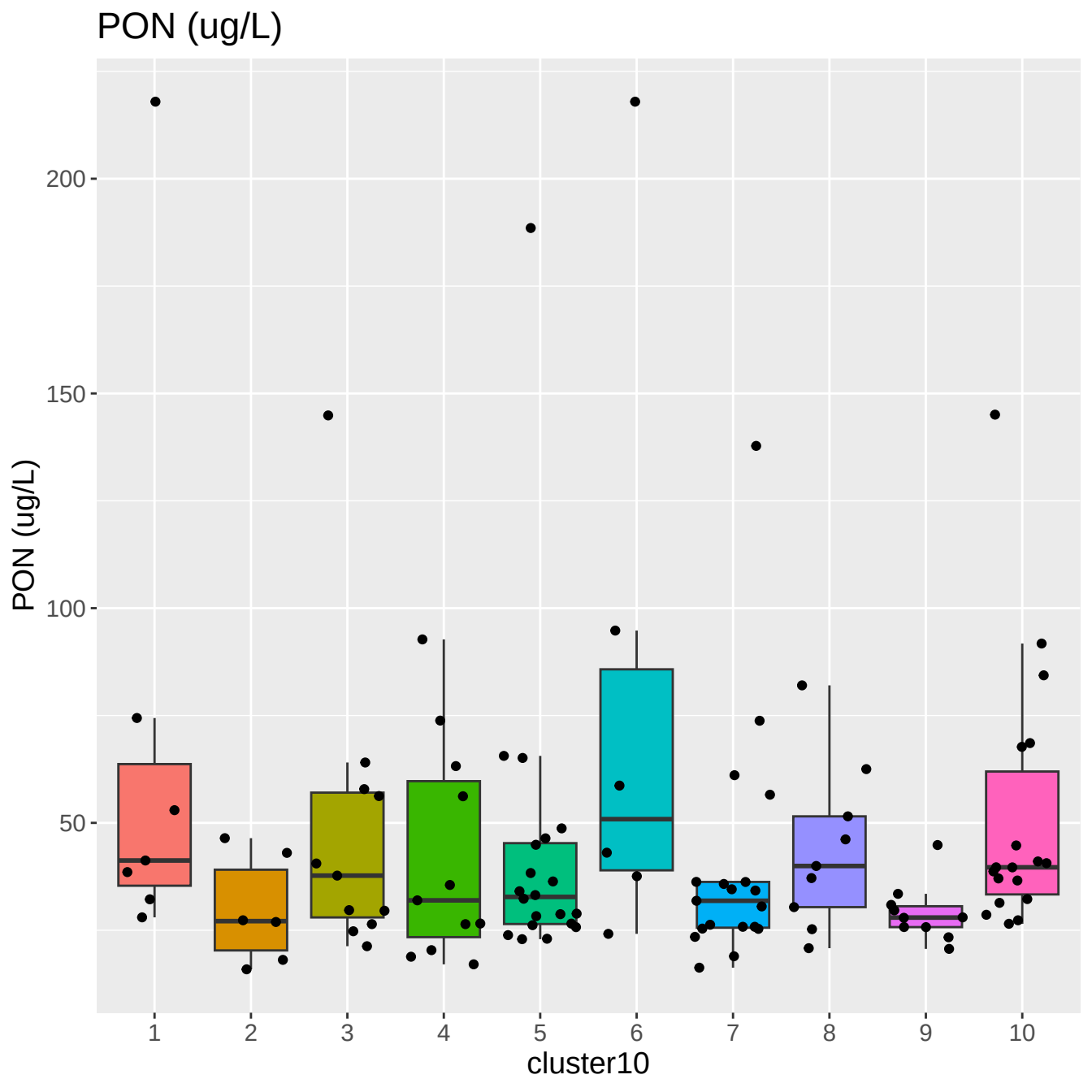
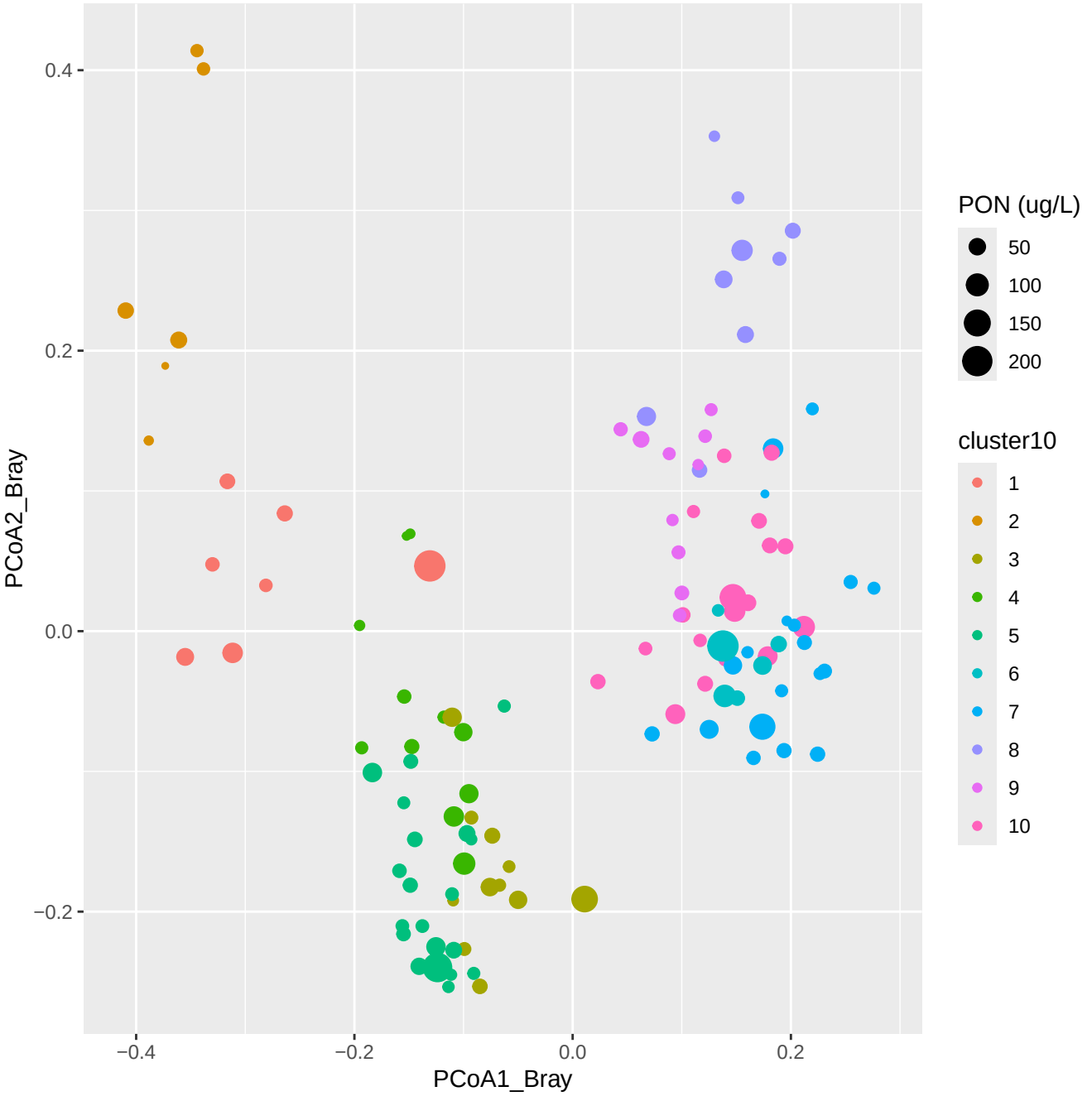
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
DO
300
350
400


POC (ug/L)
POC (ug/L)
1000
500
0
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
POC (ug/L)
250
500
750
1000
1250


PON (ug/L)
PON (ug/L)
200
150
100
50
1
2
3
4
5
6
7
8
9
10
cluster10


PON (ug/L)
50
100
150
200
cluster10
1
2
3
4
5
6
7
8
9
10
PCoA2_Bray
PCoA1_Bray
0.4
0.2
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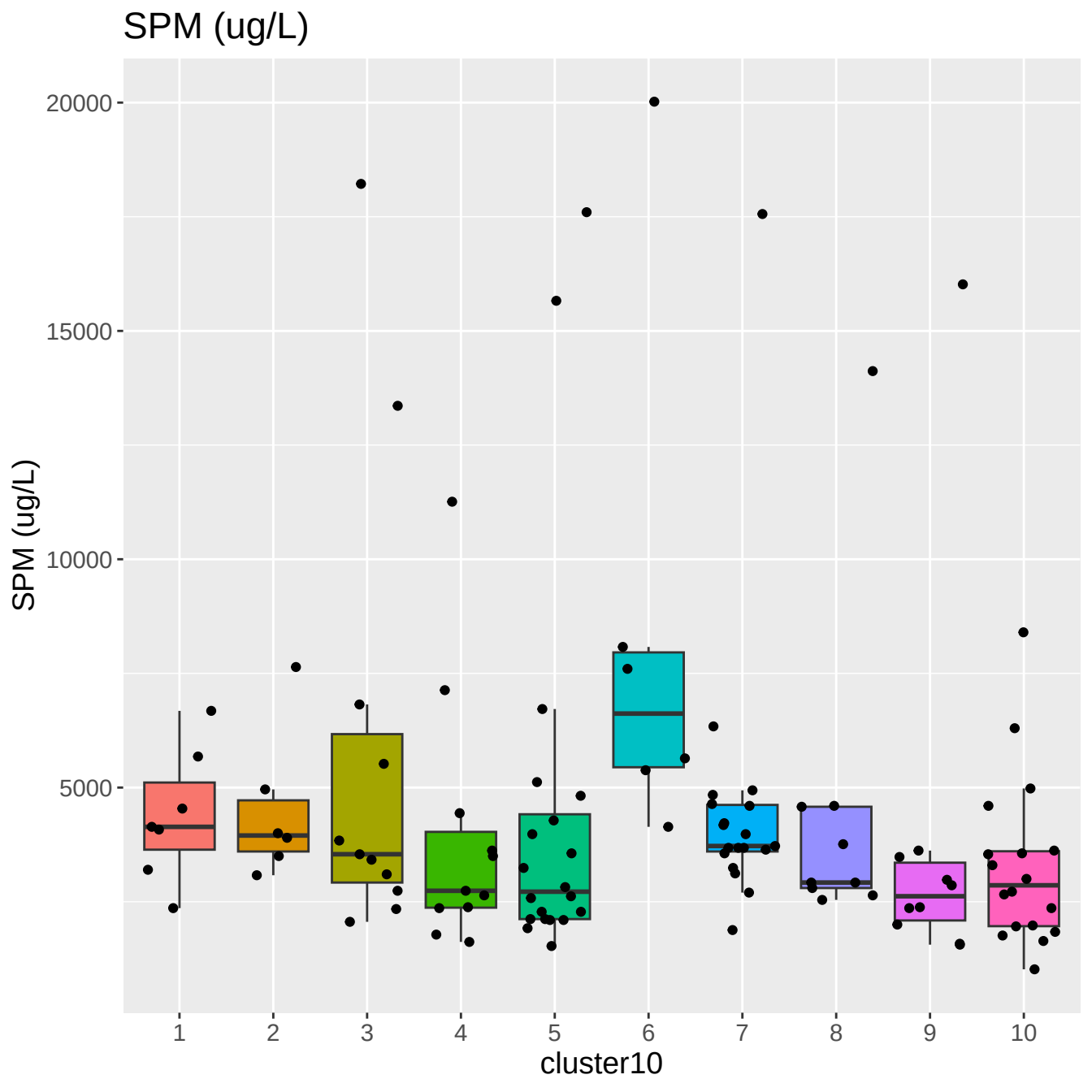
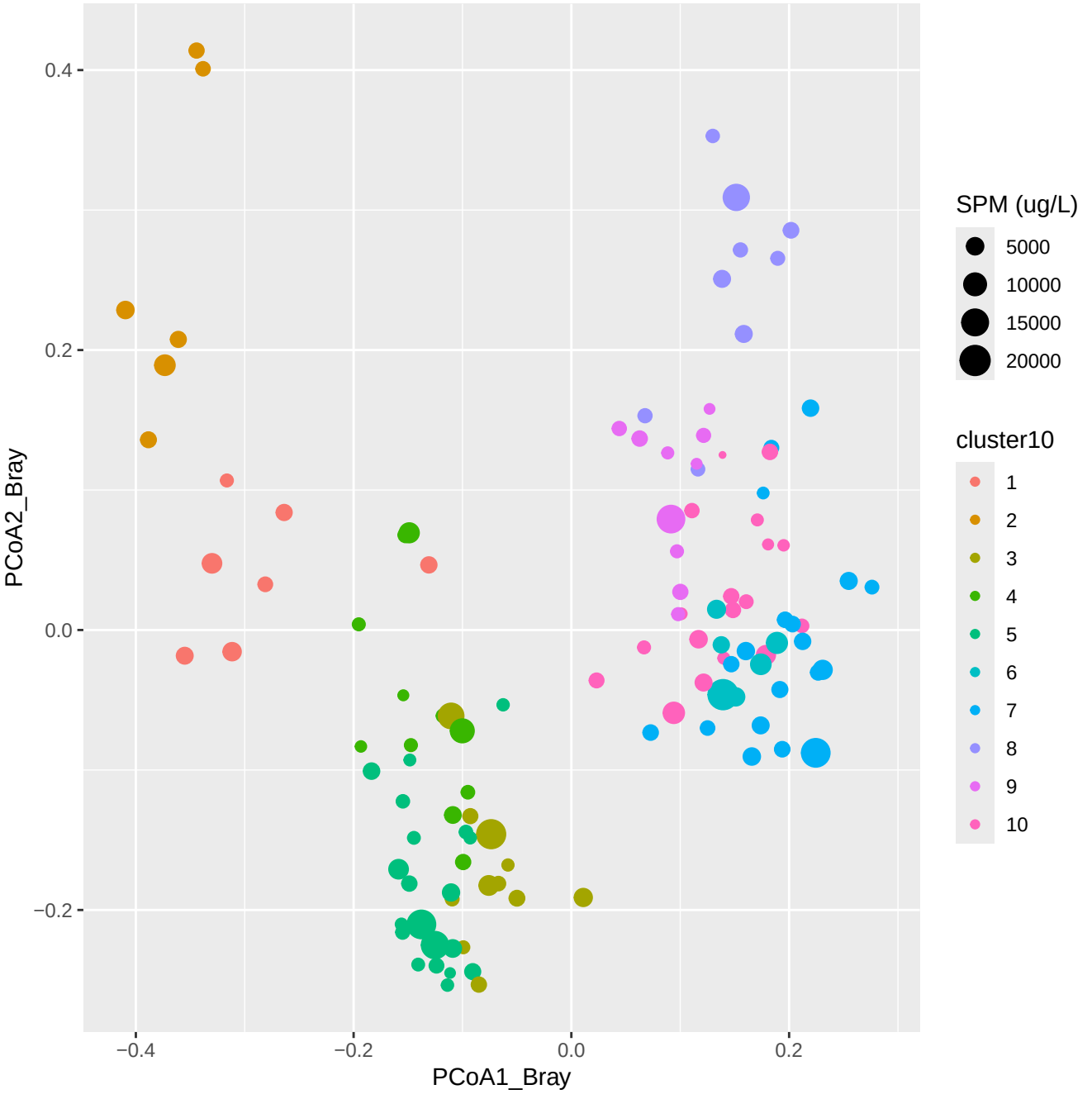
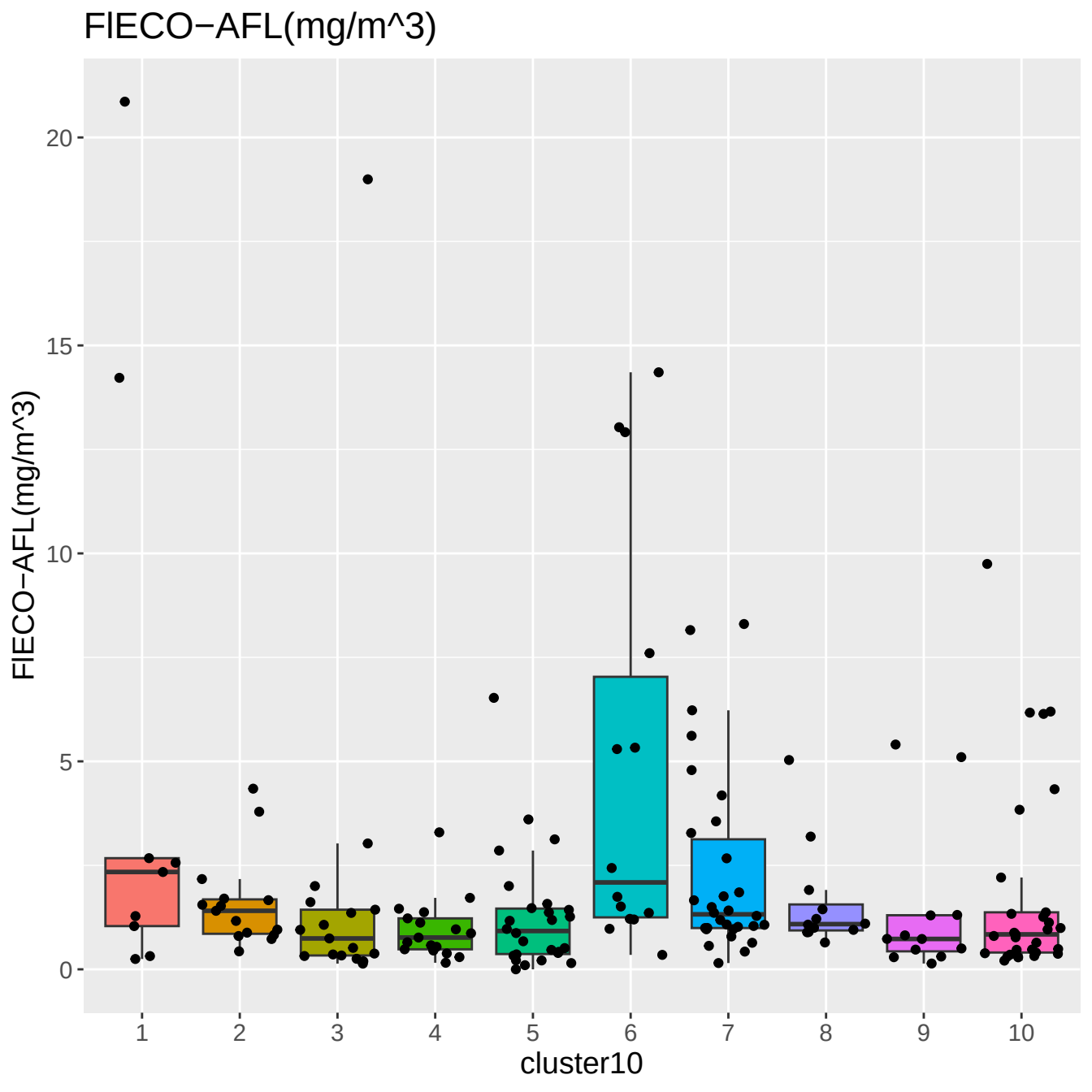
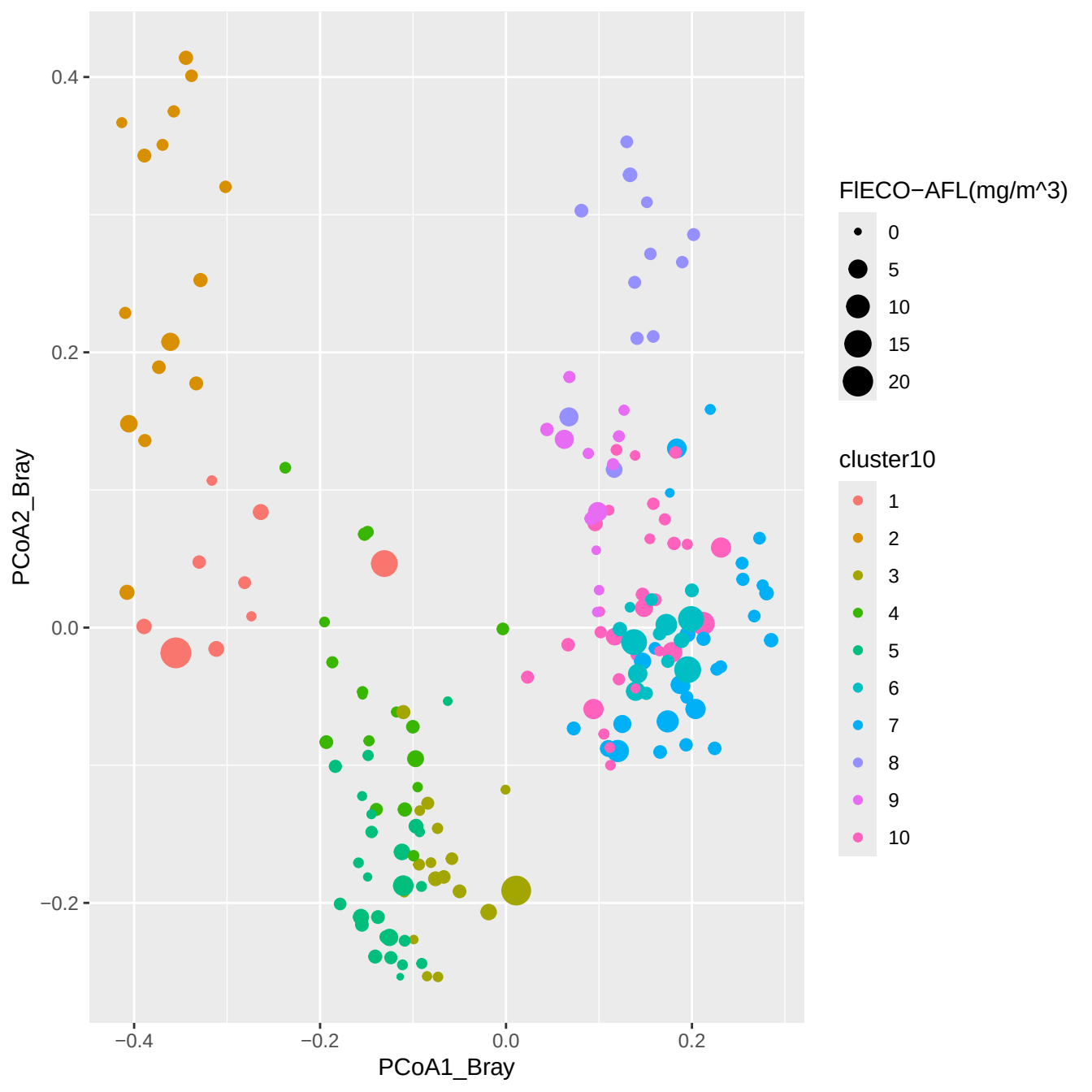
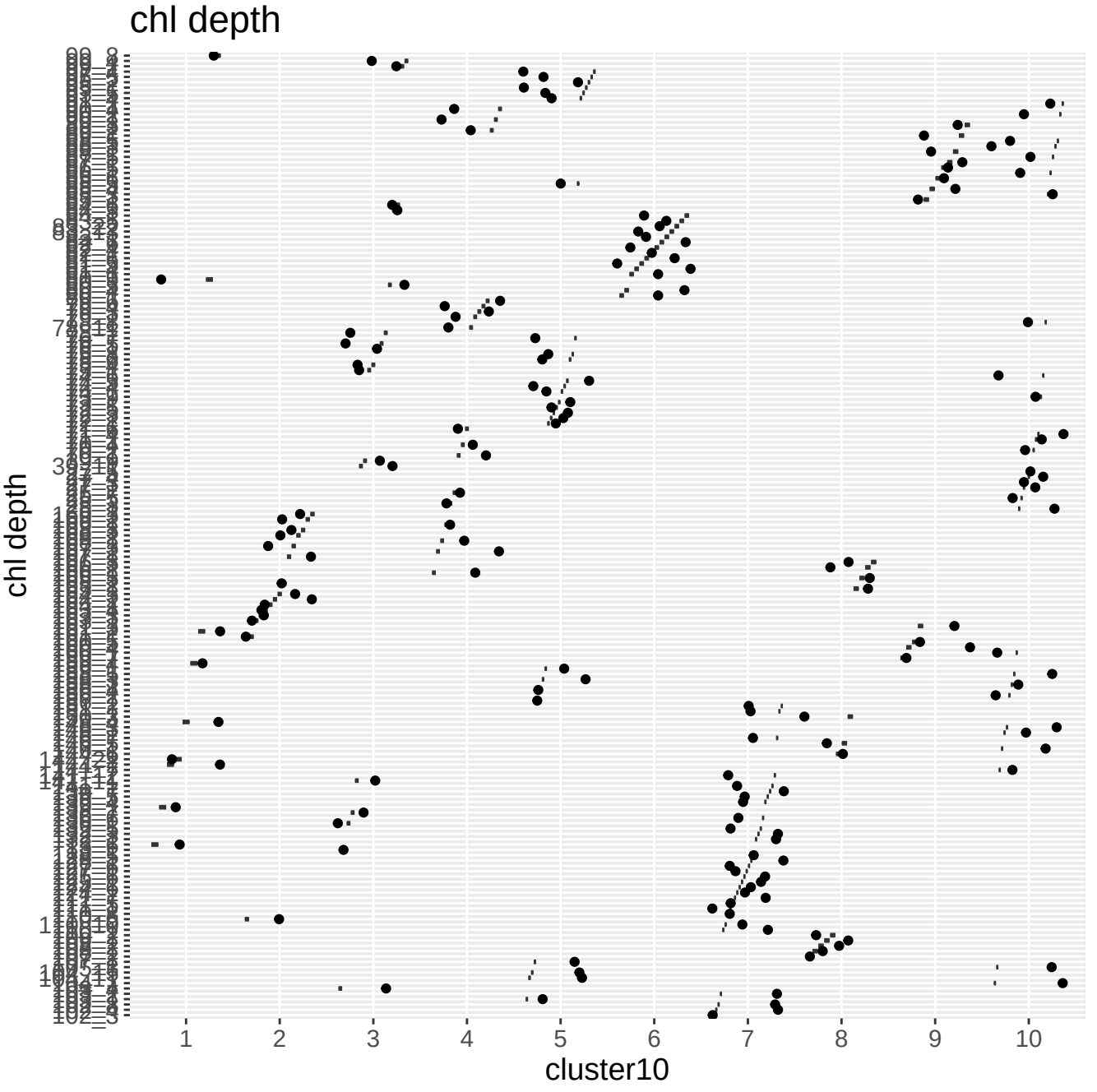
−0.2
−0.4
−0.2
0.0
0.2


SPM (ug/L)
20000
15000
10000
5000
SPM (ug/L)
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
SPM (ug/L)
5000
10000
15000
20000
cluster10
1
2
3
4
5
6
7
8
9
10


FIECO−AFL(mg/m^3)
FIECO−AFL(mg/m^3)
20
15
10
5
0
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
FlECO−AFL(mg/m^3)
0
5
10
15
20
cluster10
1
2
3
4
5
6
7
8
9
10


chl depth
chl depth
cluster10
1
2
3
4
5
6
7
8
9
10

- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10

chl depth

- 102_3
- 102_4
- 102_8
- 103_1
- 103_2
- 103_4
- 104_1
- 104_11
- 104_15
- 105_3
- 107_1
- 107_2
- 108_1
- 108_2
- 109_1
- 109_2
- 110_1
- 110_10
- 110_8
- 111_1
- 111_5
- 111_7
- 124_1
- 124_6
- 125_1
- 125_6
- 127_1
- 127_6
- 128_1
- 128_5
- 13_1
- 13_6
- 132_1
- 132_3
- 132_5
- 136_1
- 136_7
- 139_1
- 139_4
- 139_5
- 139_7
- 141_1
- 141_11
- 141_12
- 141_7
- 144_1
- 144_22
- 144_6
- 146_1
- 146_5
- 146_7
- 148_1
- 148_3
- 148_4
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- 156_1
- 156_2
- 156_4
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- 168_3
- 169_1
- 169_2
- 25_1
- 25_3
- 25_5
- 25_7
- 27_1
- 27_3
- 27_4
- 27_5
- 39_11
- 39_9
- 70_1
- 70_2
- 70_4
- 71_1
- 71_4
- 71_6
- 72_1
- 72_2
- 72_5
- 73_1
- 73_9
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- 74_4
- 74_5
- 74_6
- 75_1
- 75_4
- 75_6
- 75_8
- 76_1
- 76_5
- 76_7
- 78_1
- 78_11
- 78_8
- 79_1
- 79_4
- 79_6
- 80_1
- 80_4
- 80_5
- 80_9
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- 81_4
- 81_5
- 81_6
- 82_1
- 82_4
- 82_6
- 83_1
- 83_13
- 83_22
- 83_5
- 84_1
- 84_6
- 85_1
- 85_3
- 85_4
- 85_5
- 85_6
- 86_1
- 86_5
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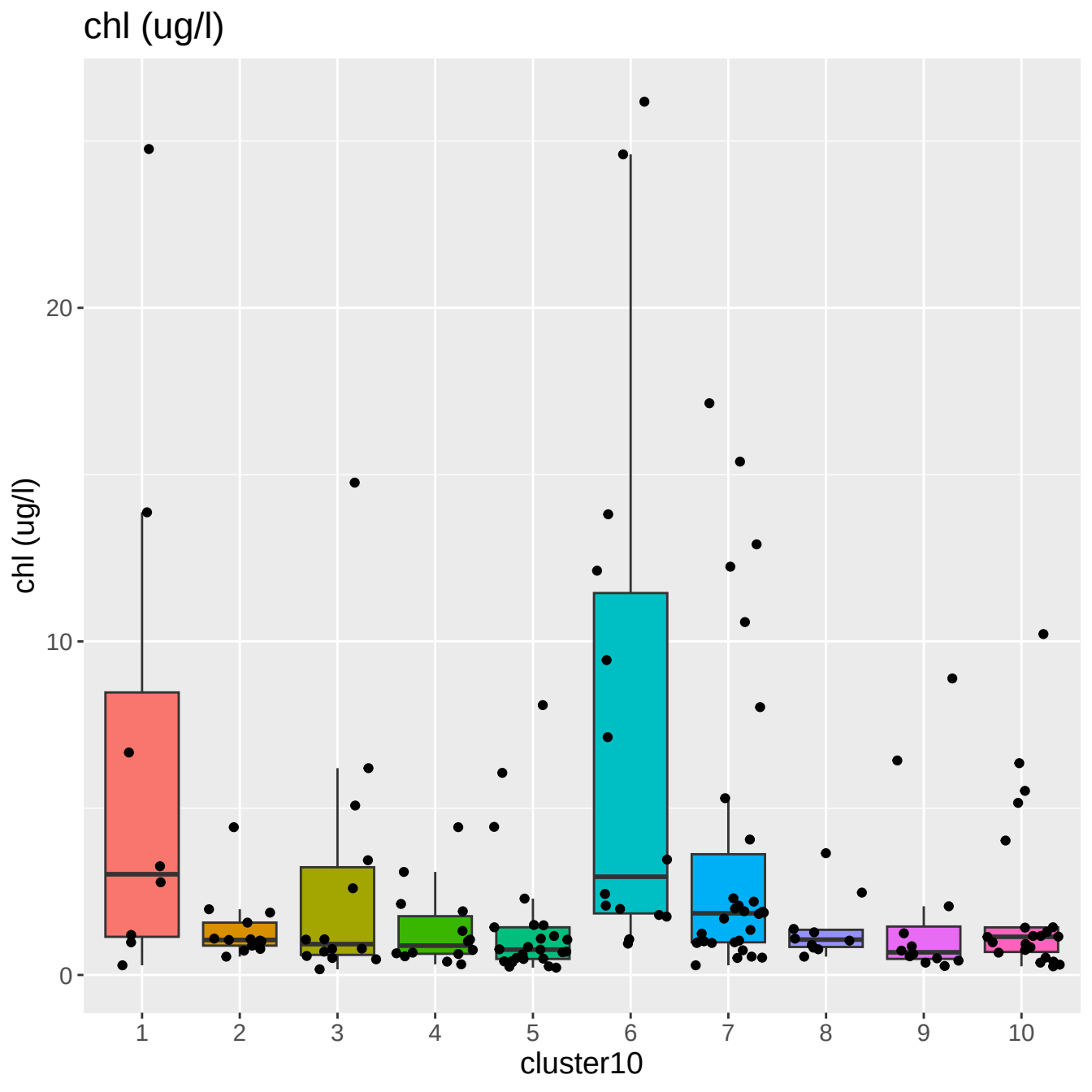
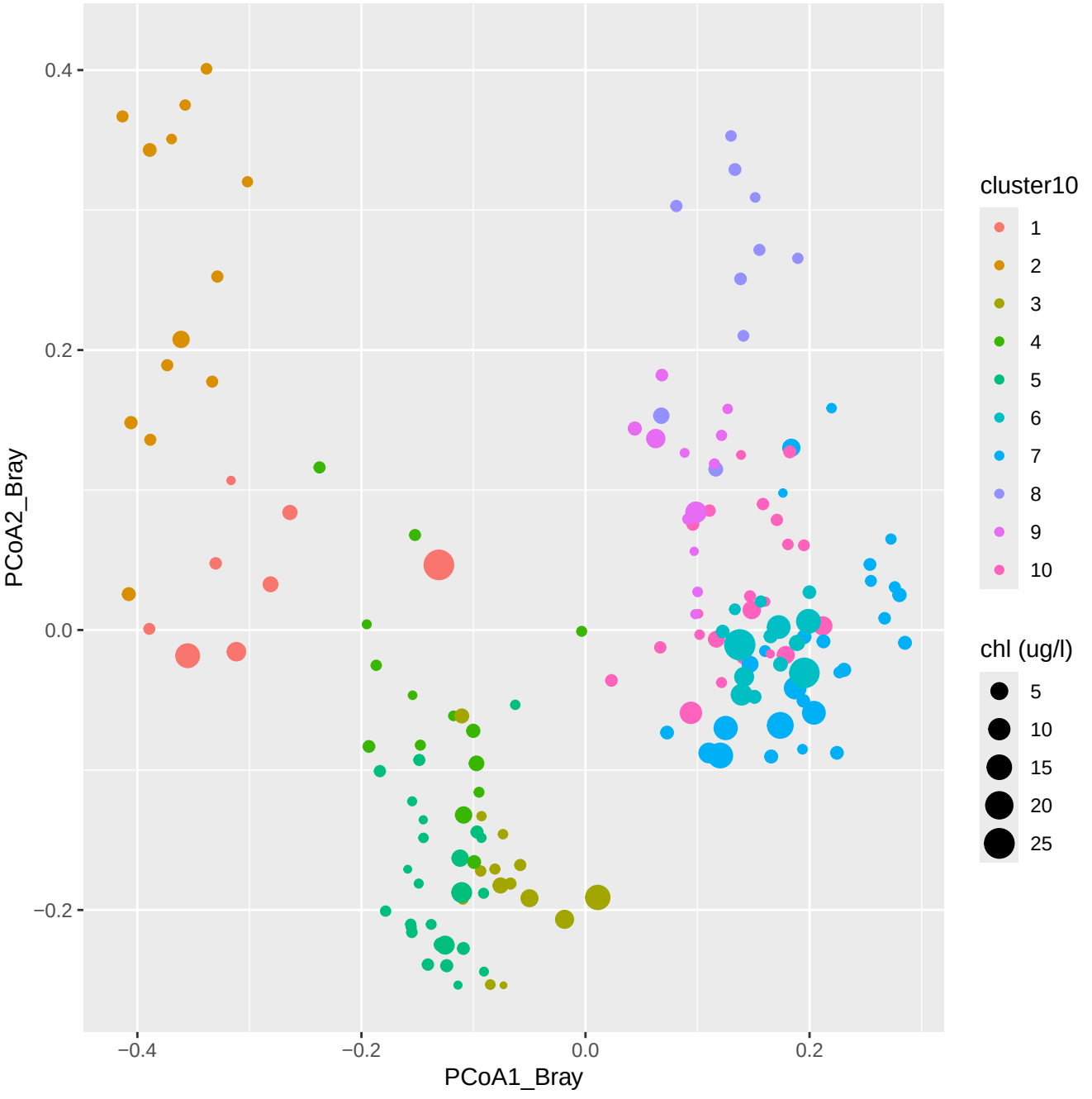
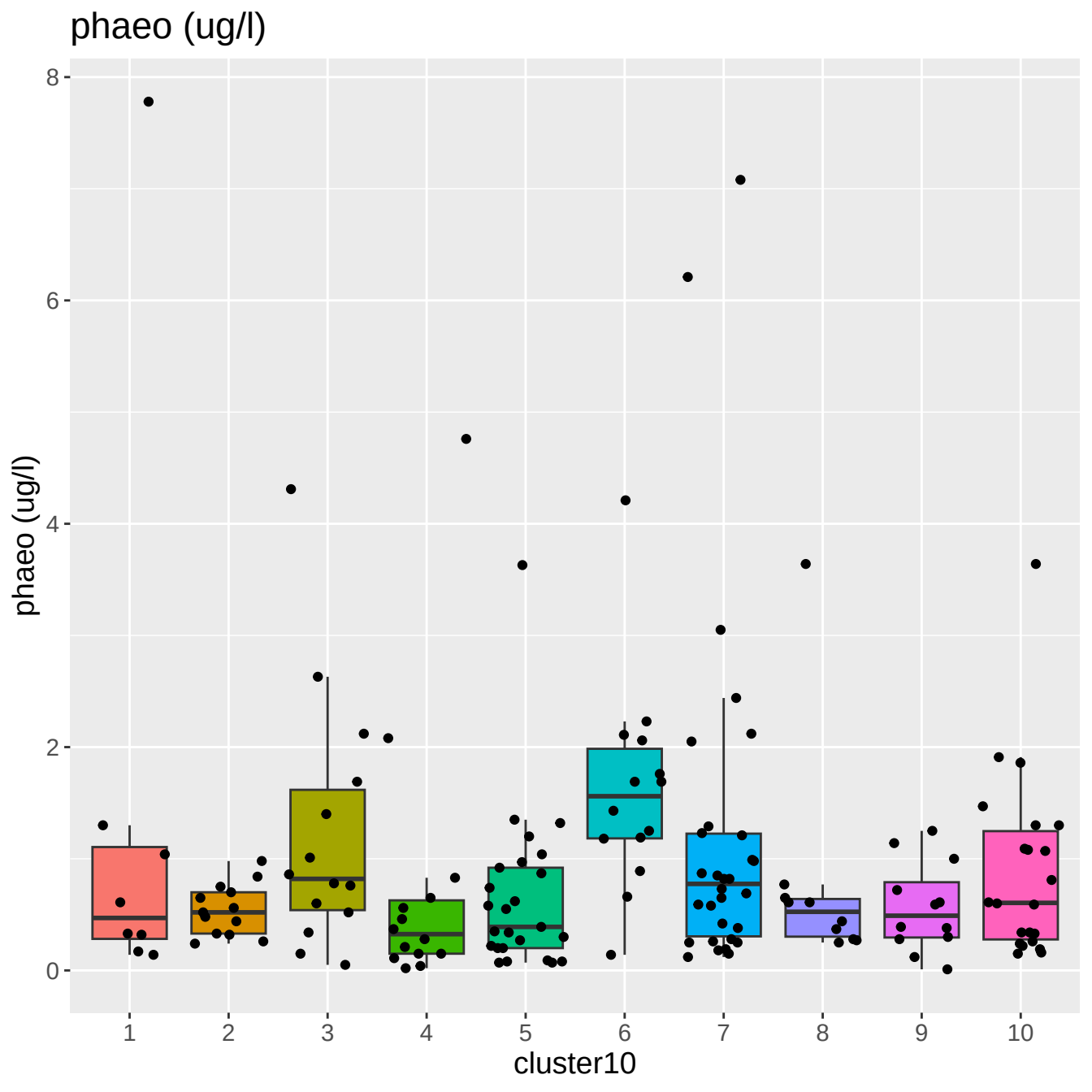
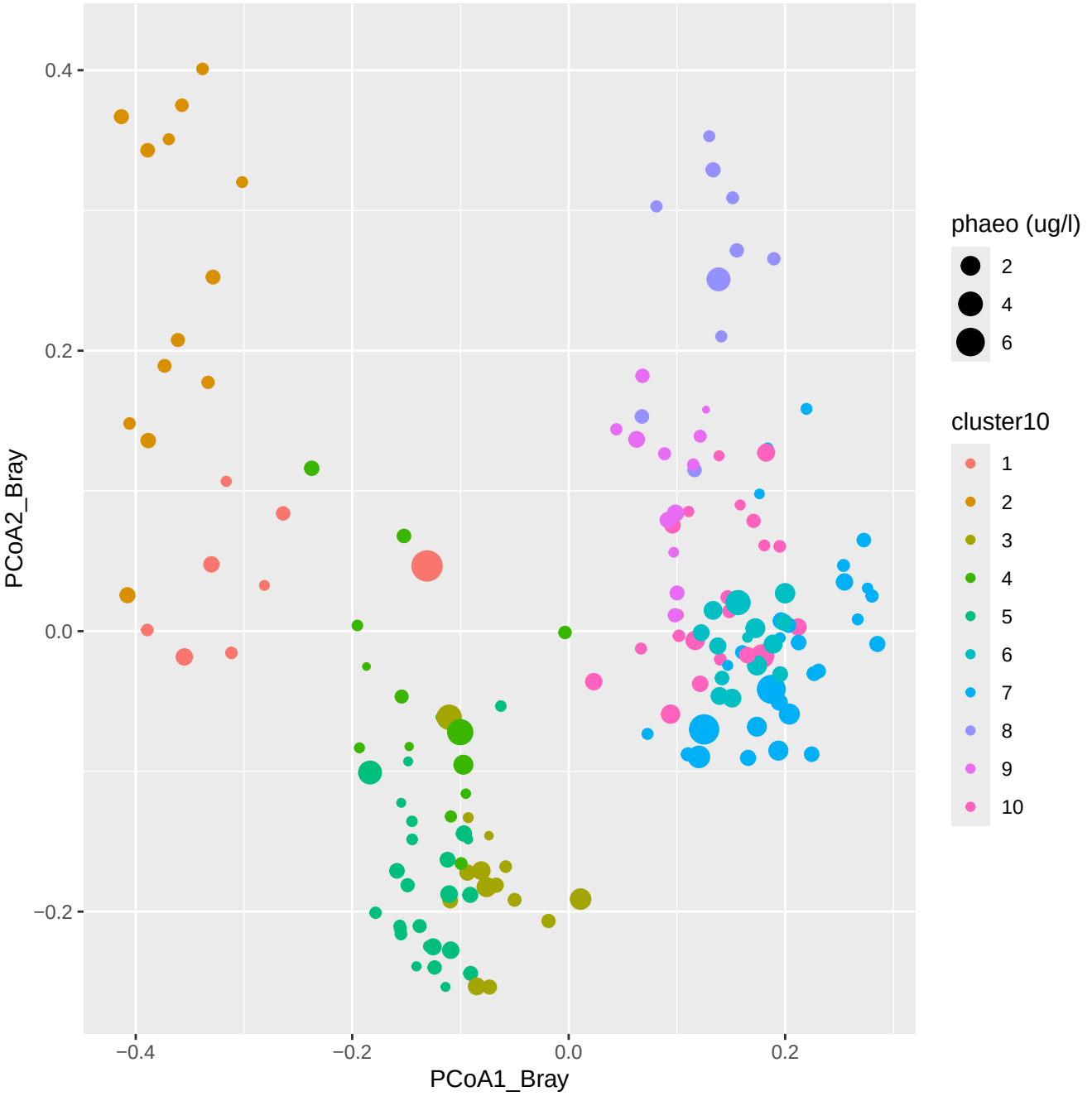
- 90_2


chl (ug/l)
chl (ug/l)
20
10
0
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
chl (ug/l)
5
10
15
20
25


phaeo (ug/l)
phaeo (ug/l)
8
6
4
2
0
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
phaeo (ug/l)
2
4
6
cluster10
1
2
3
4
5
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7
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9
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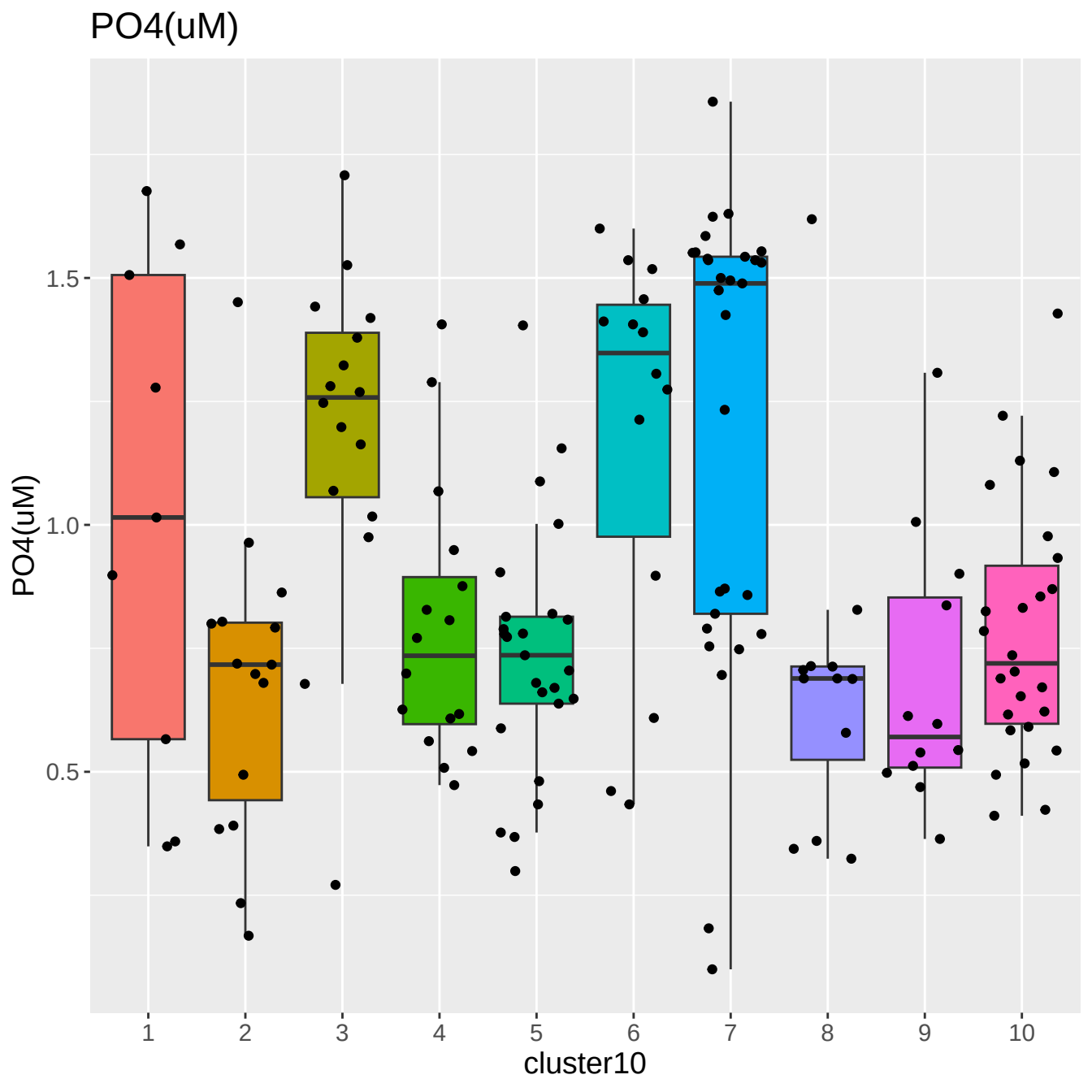
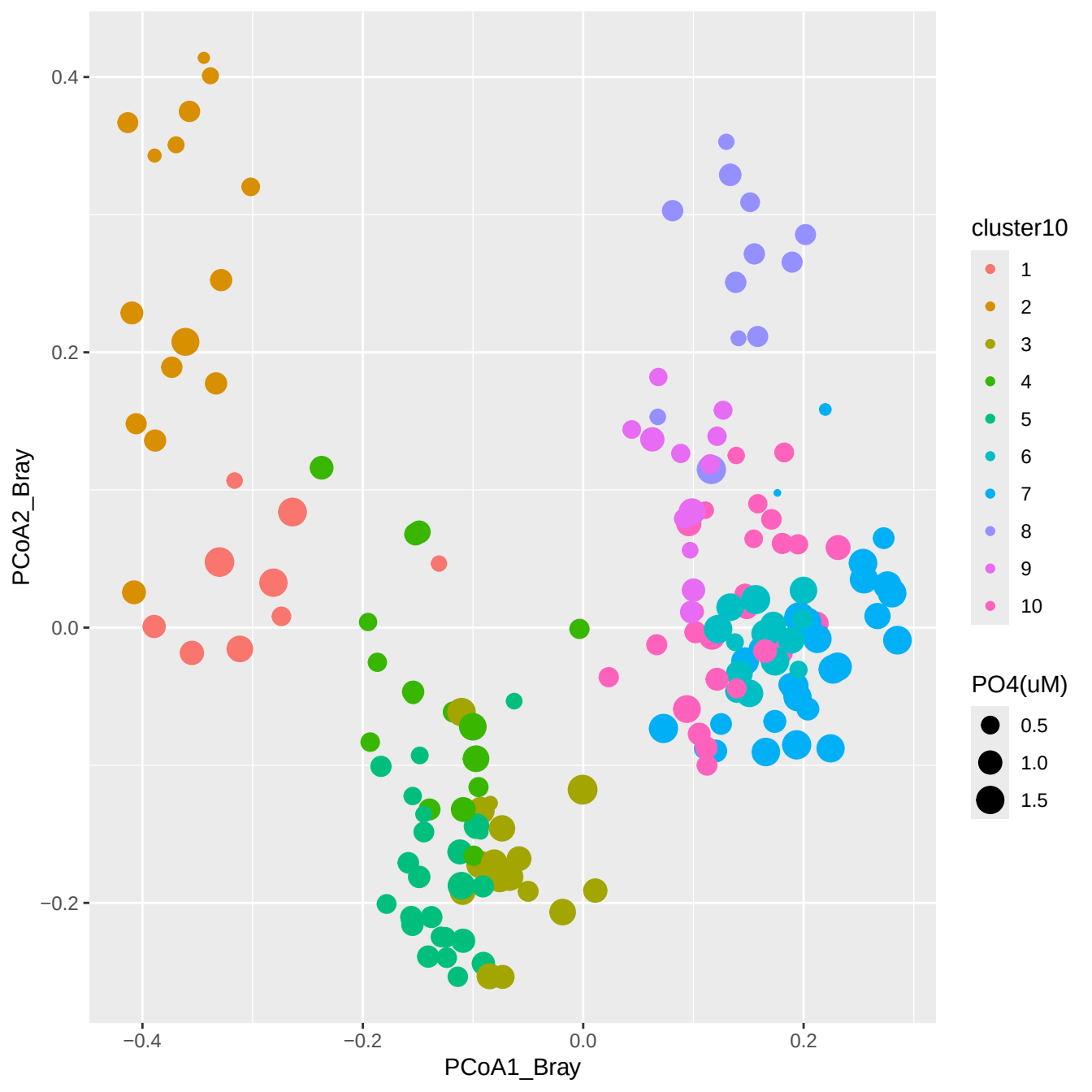
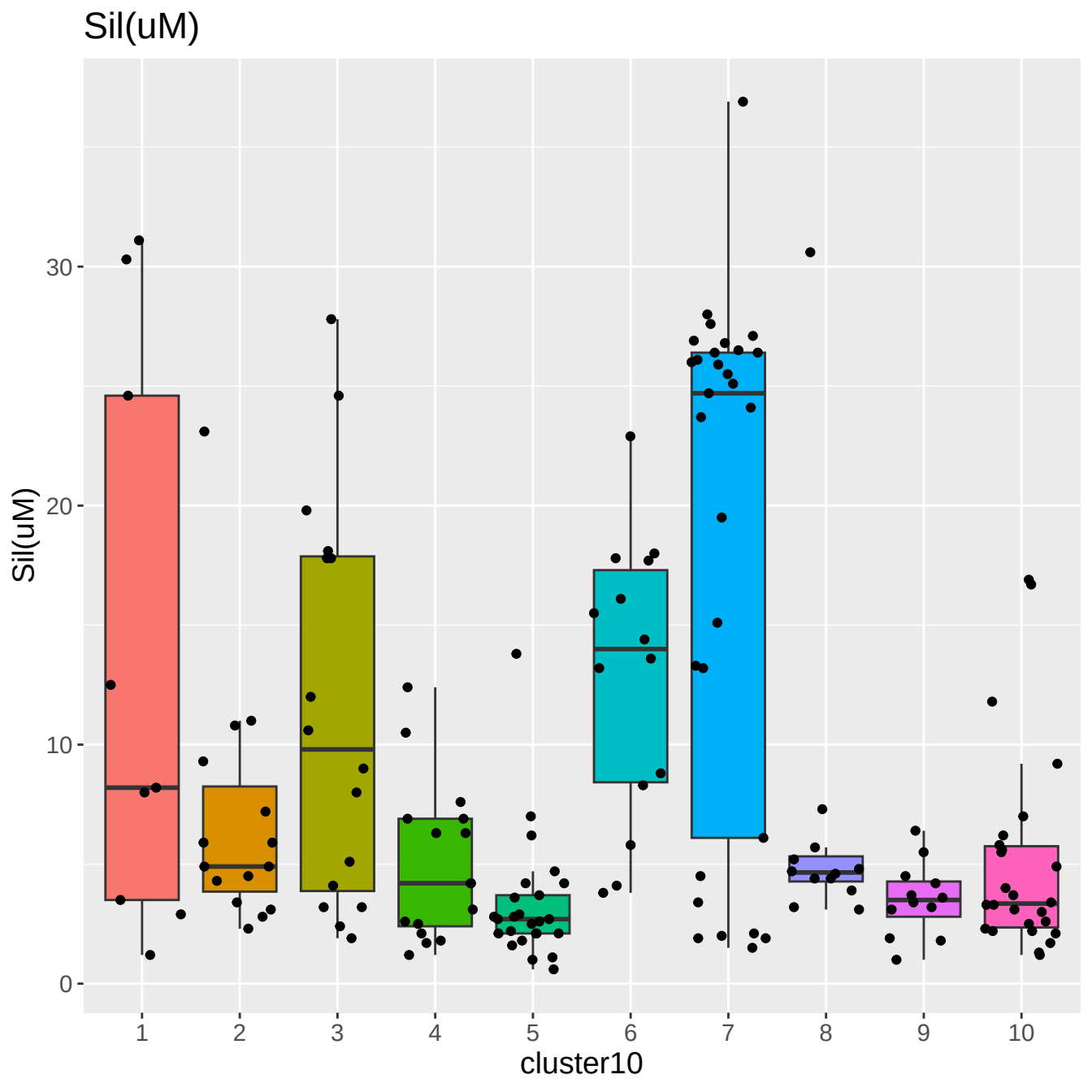
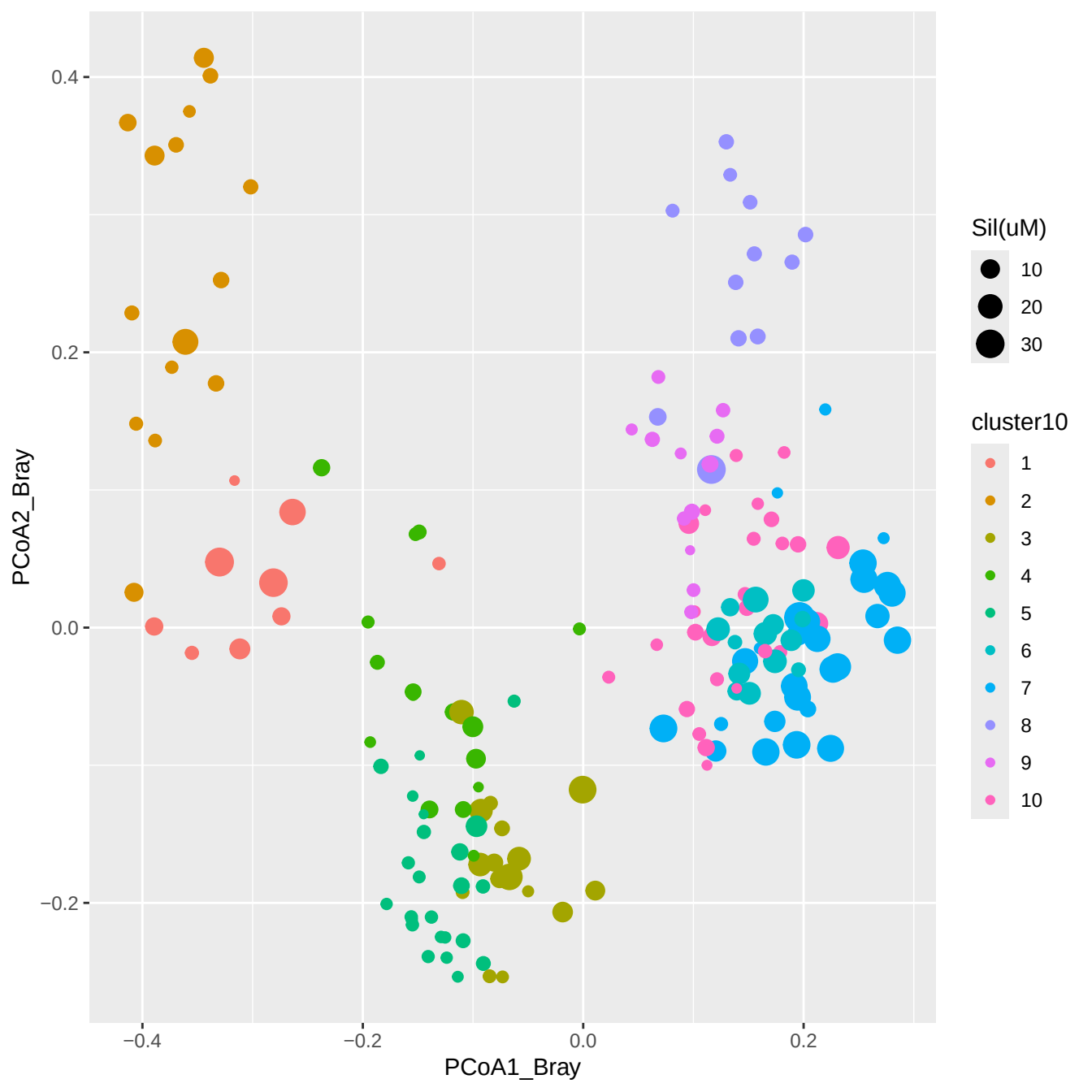
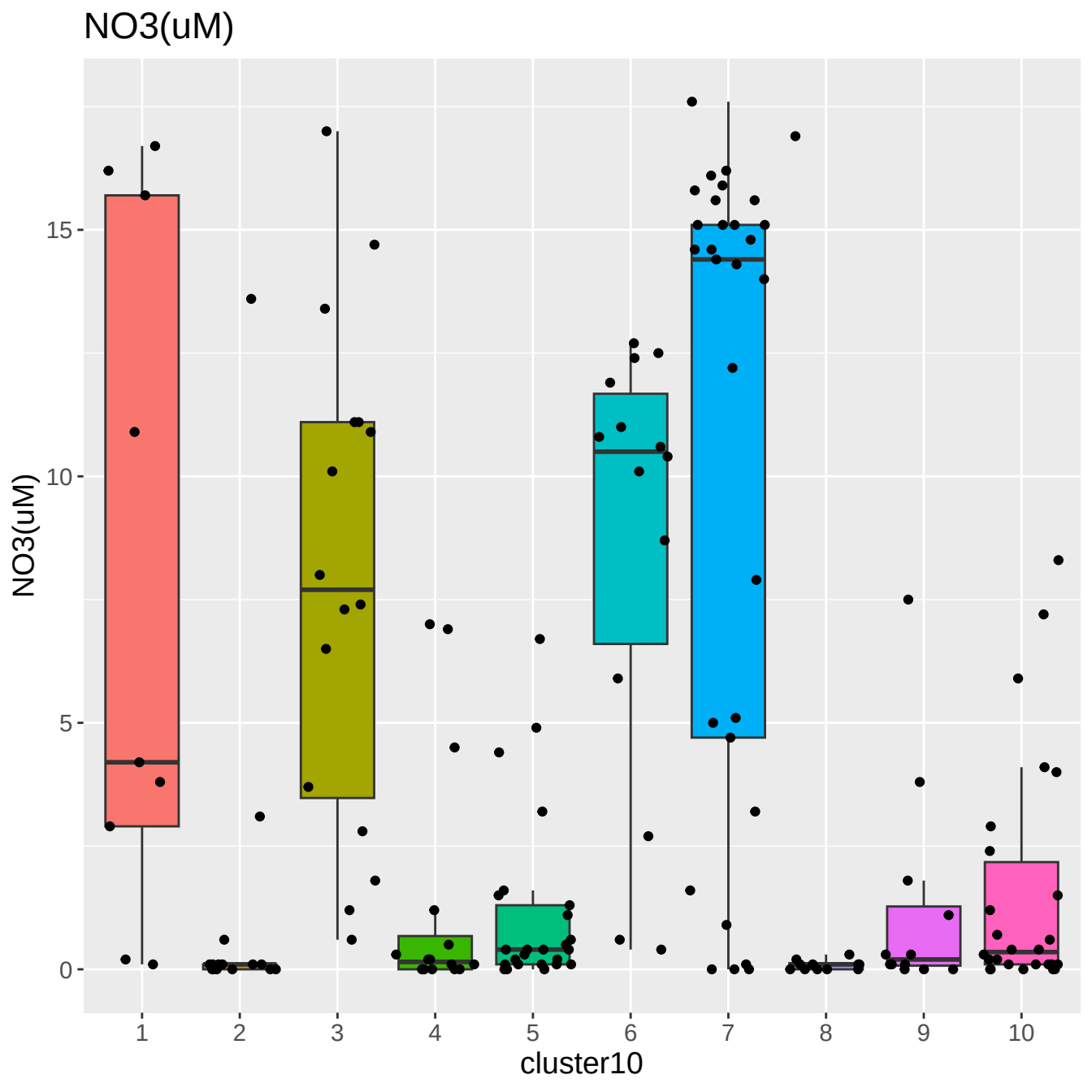
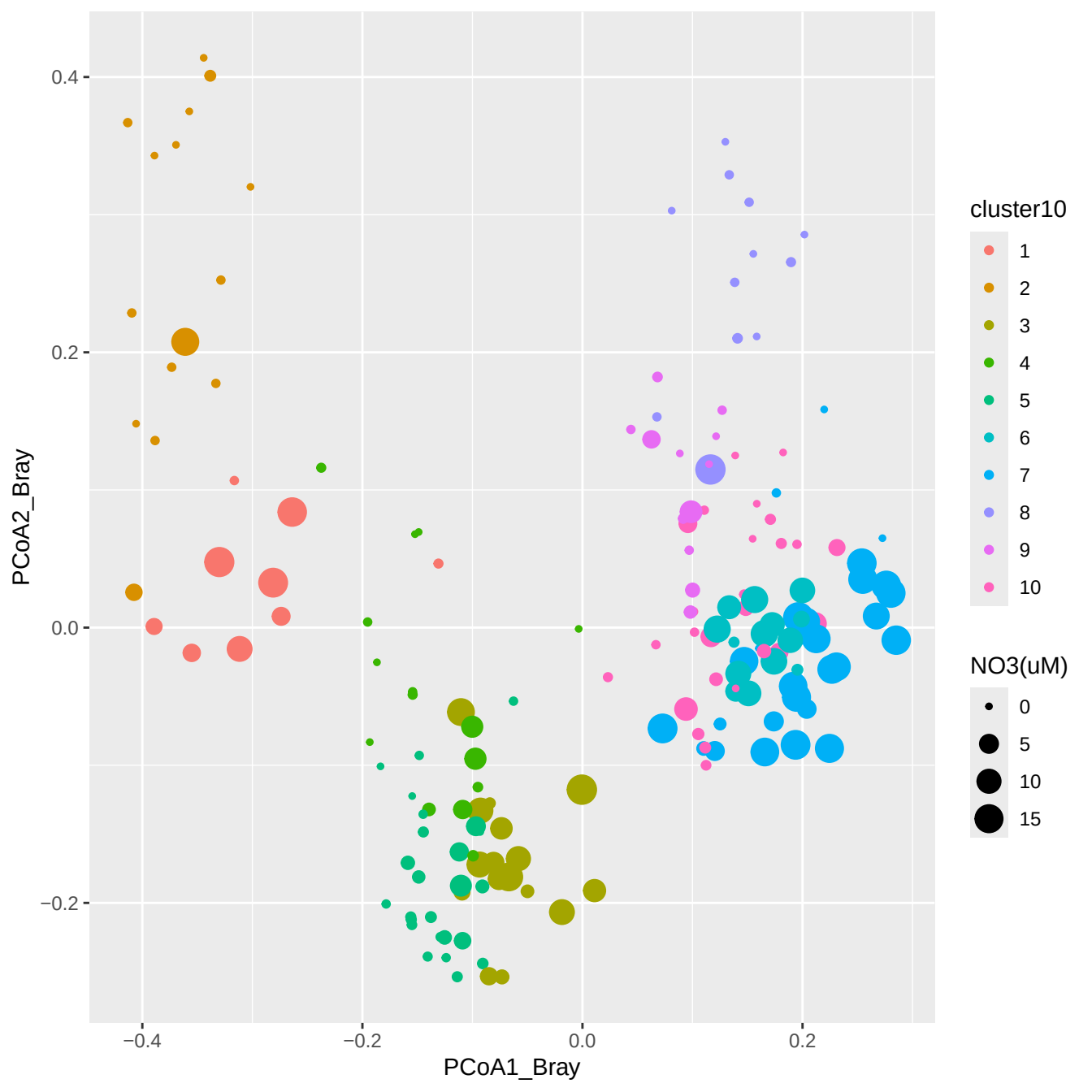

PO4(uM)
PO4(uM)
1.5
1.0
0.5
1
2
3
4
5
6
7
8
9
10
cluster10

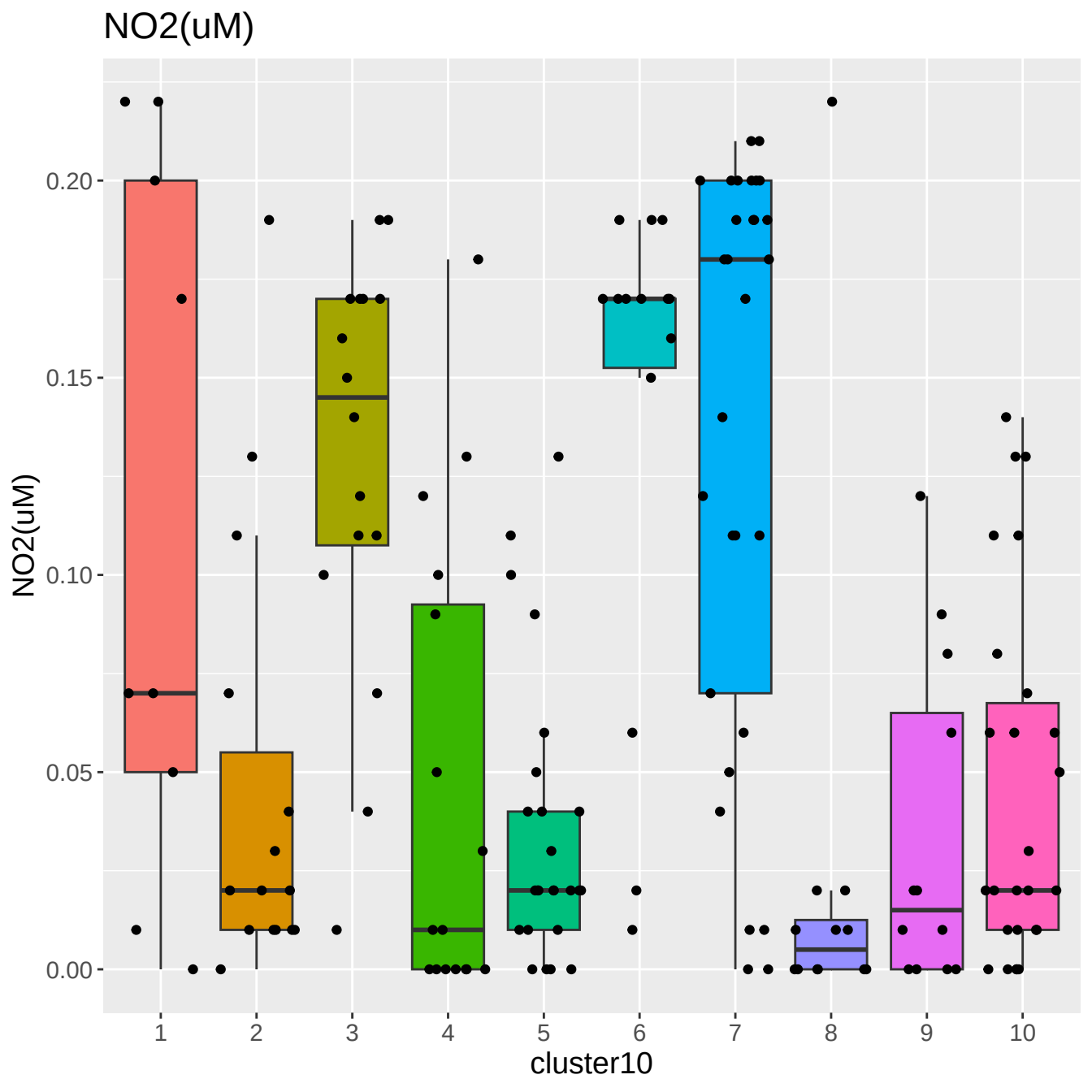

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
PO4(uM)
0.5
1.0
1.5

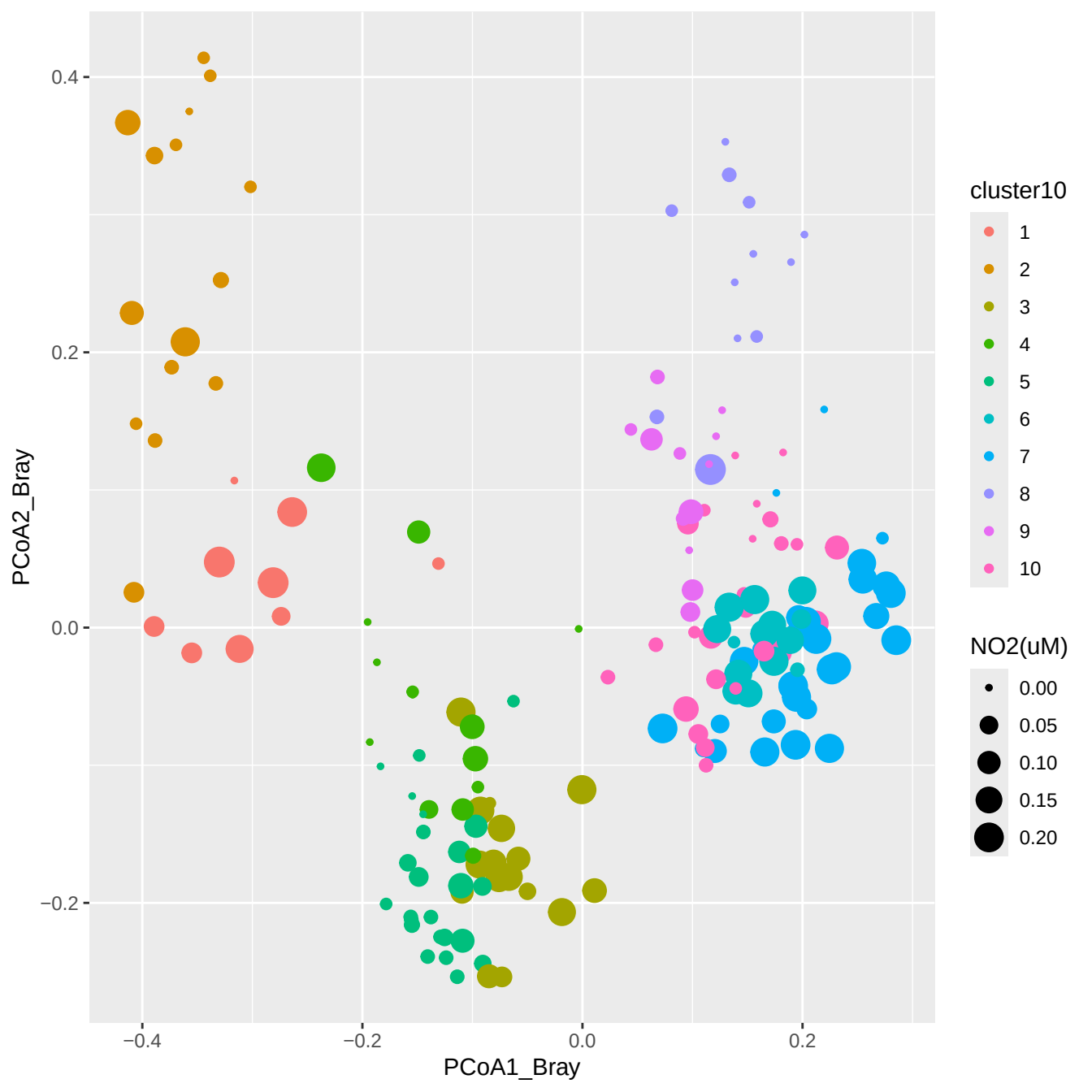

Sil(uM)
Sil(uM)
30
20
10
0
1
2
3
4
5
6
7
8
9
10
cluster10

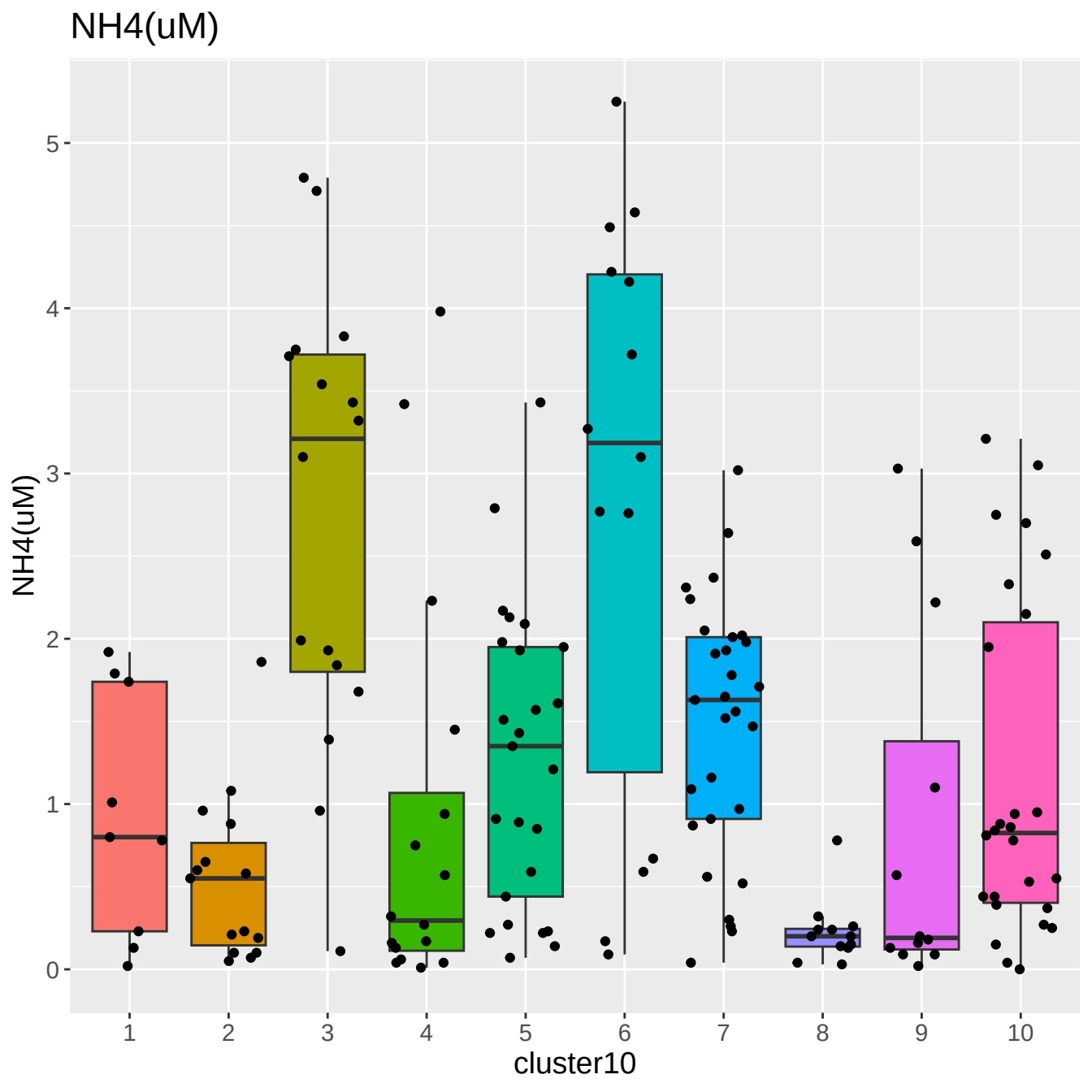

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
Sil(uM)
10
20
30
cluster10
1
2
3
4
5
6
7
8
9
10

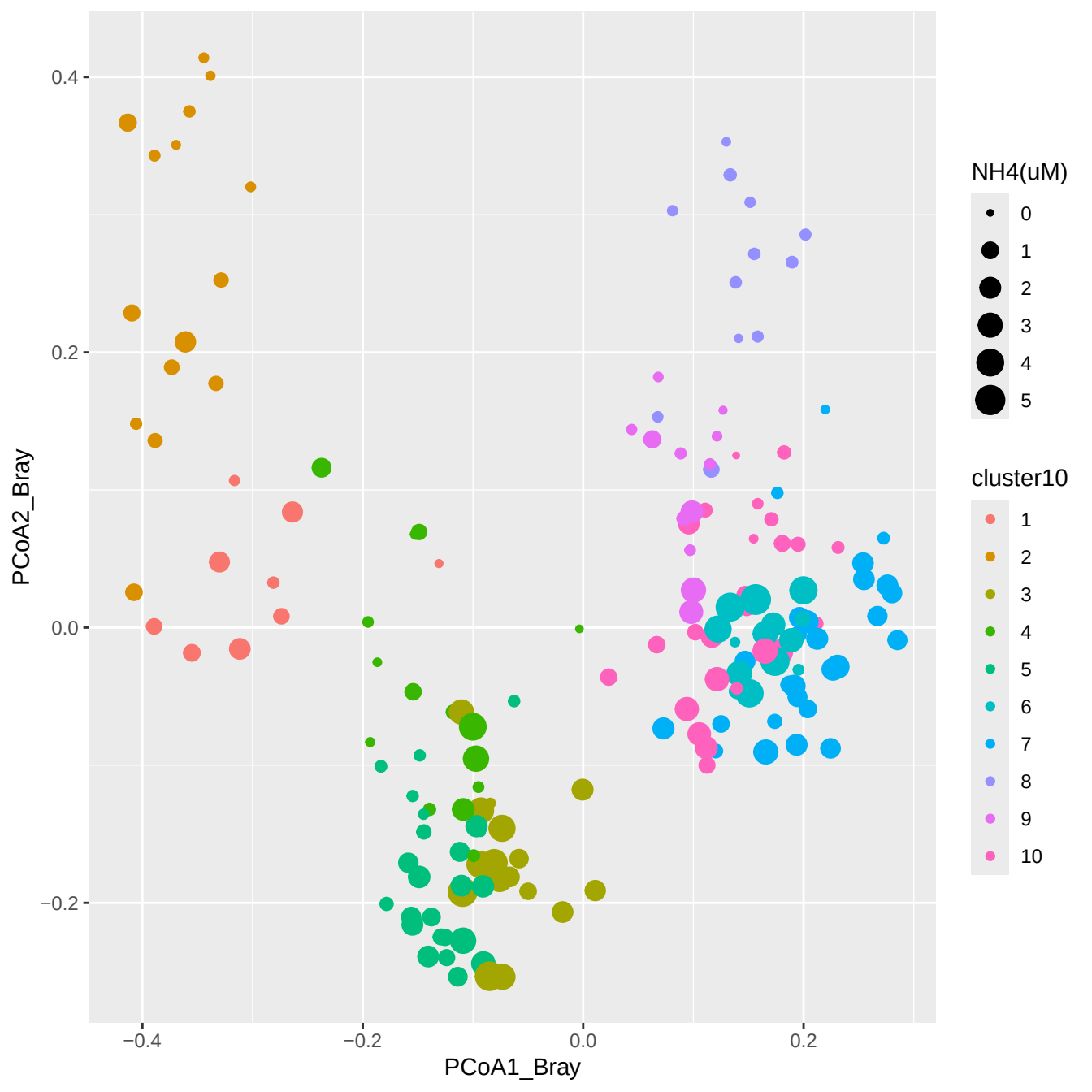

NO3(uM)
NO3(uM)
15
10
5
0
1
2
3
4
5
6
7
8
9
10
cluster10

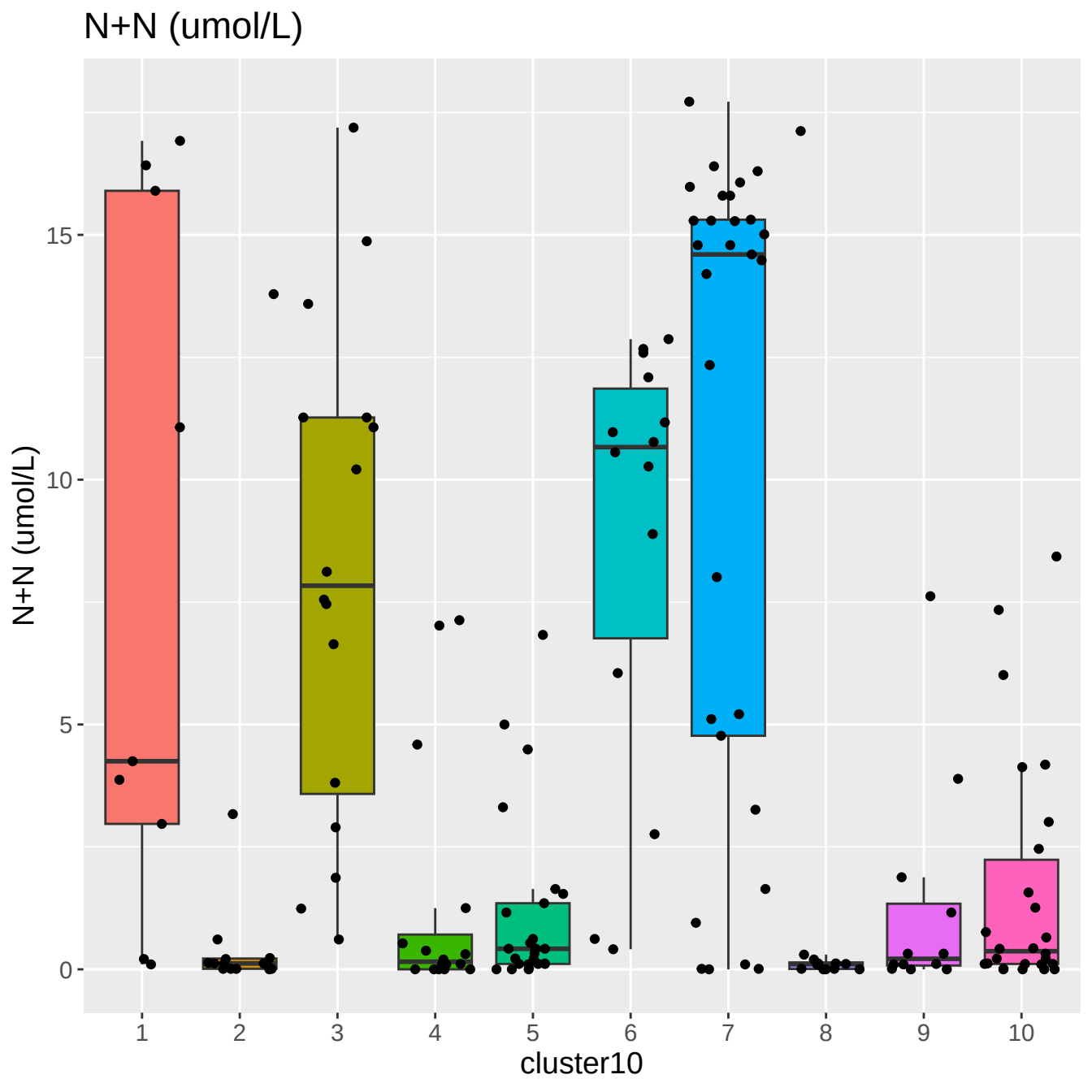

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
NO3(uM)
0
5
10
15

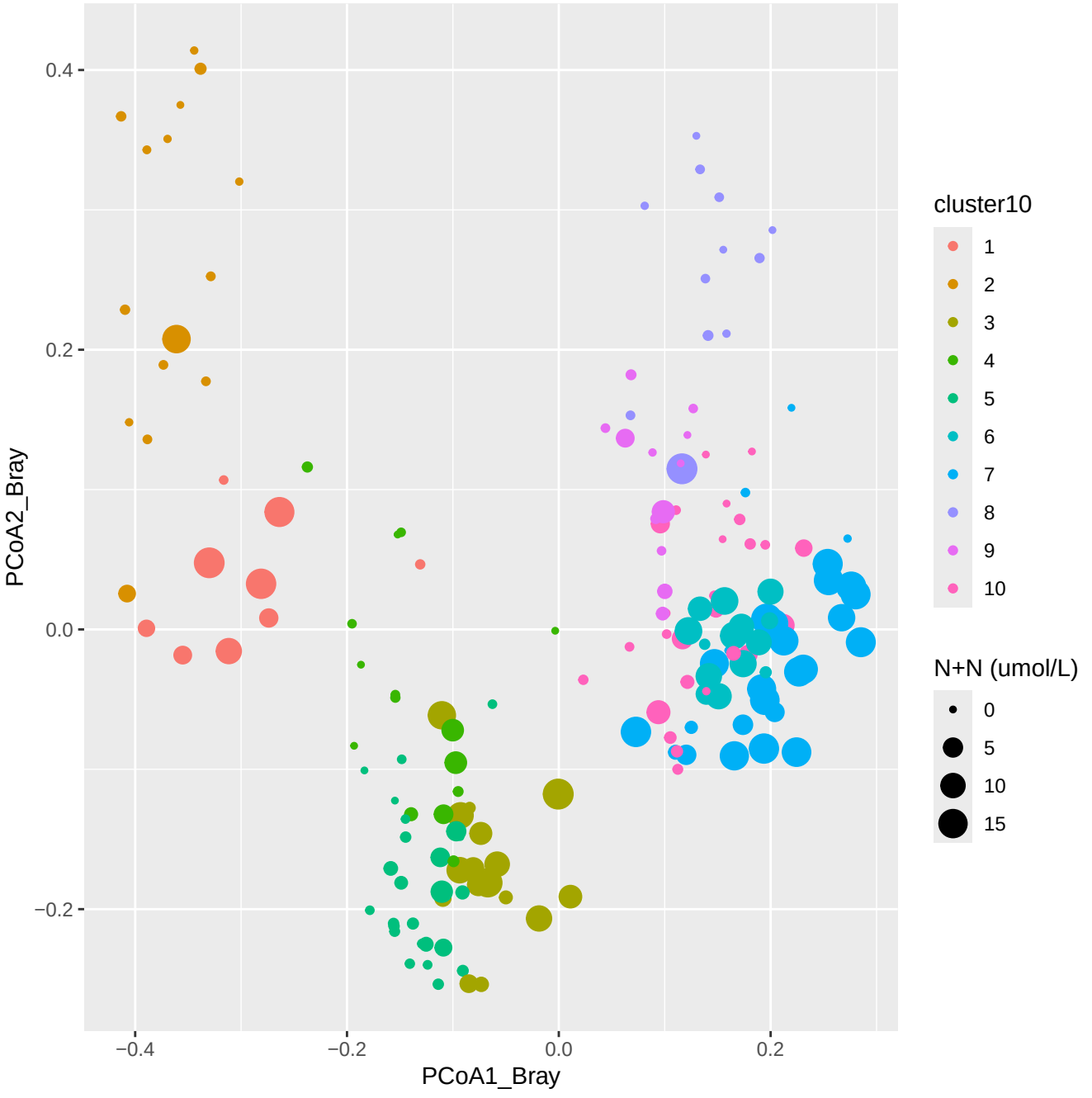

NO2(uM)
NO2(uM)
0.20
0.15
0.10
0.05
0.00
1
2
3
4
5
6
7
8
9
10
cluster10

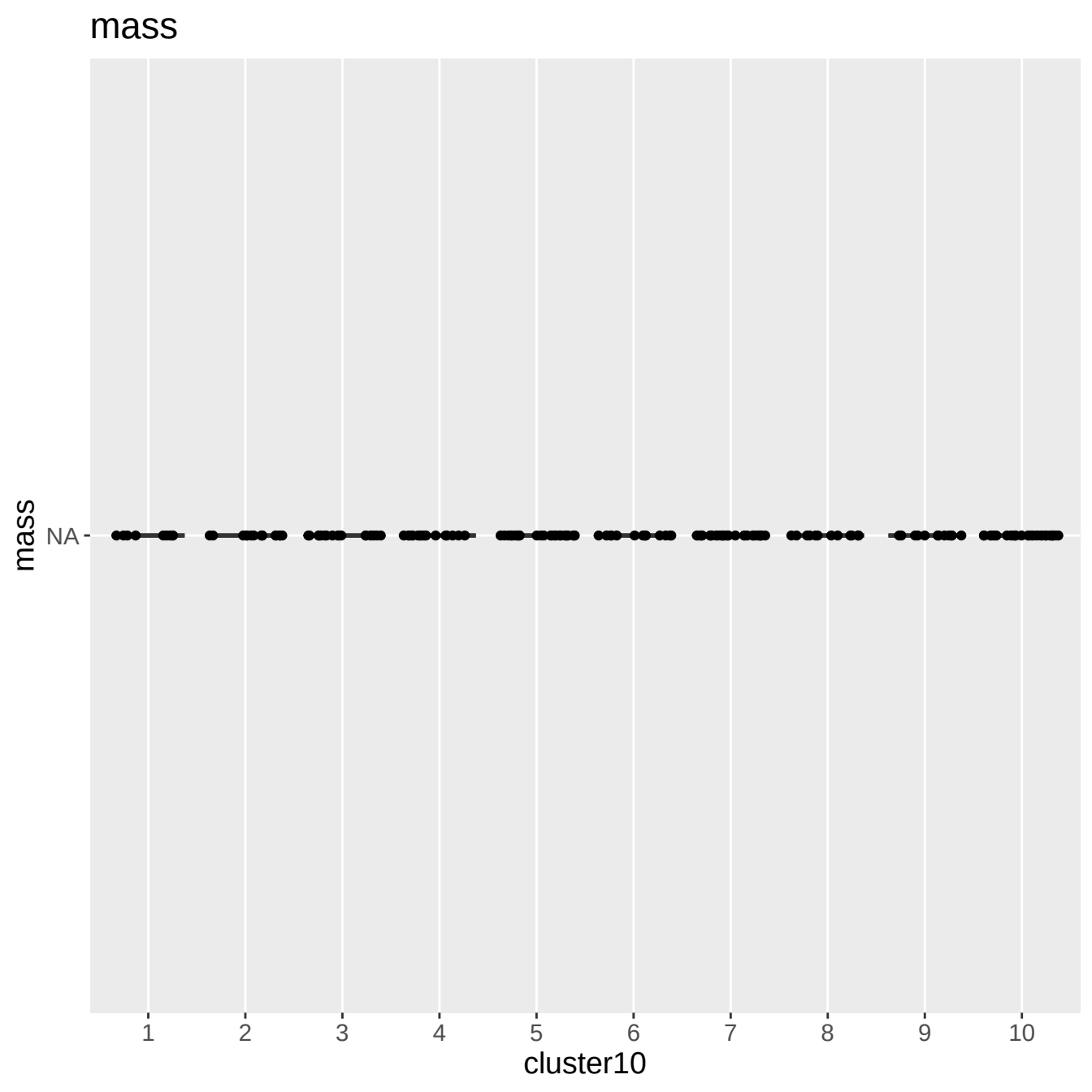

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
NO2(uM)
0.00
0.05
0.10
0.15
0.20


NH4(uM)
NH4(uM)
5
4
3
2
1
0
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
NH4(uM)
0
1
2
3
4
5
cluster10
1
2
3
4
5
6
7
8
9
10


N+N (umol/L)
N+N (umol/L)
15
10
5
0
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
N+N (umol/L)
0
5
10
15


mass
mass
NA
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4
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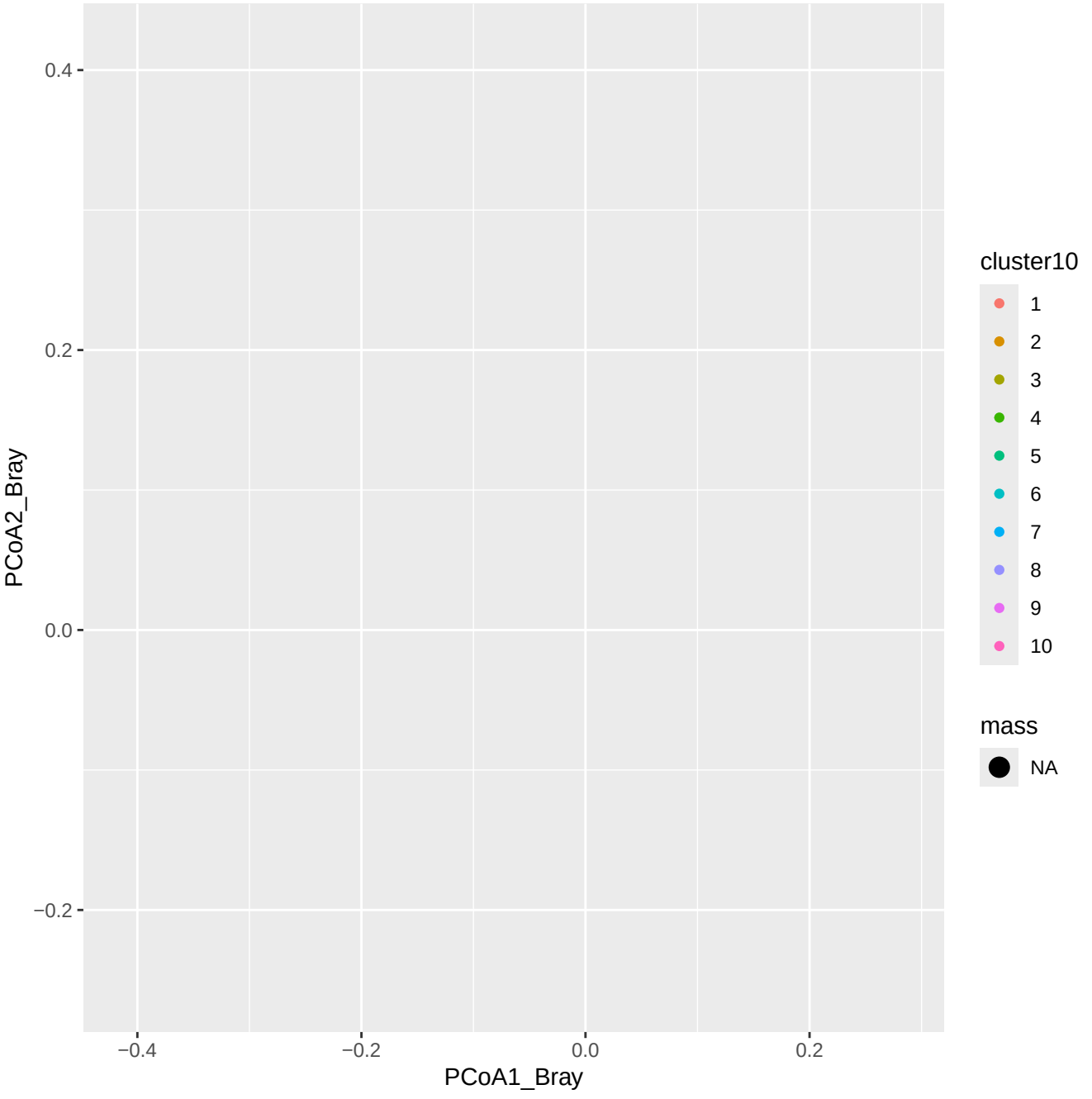
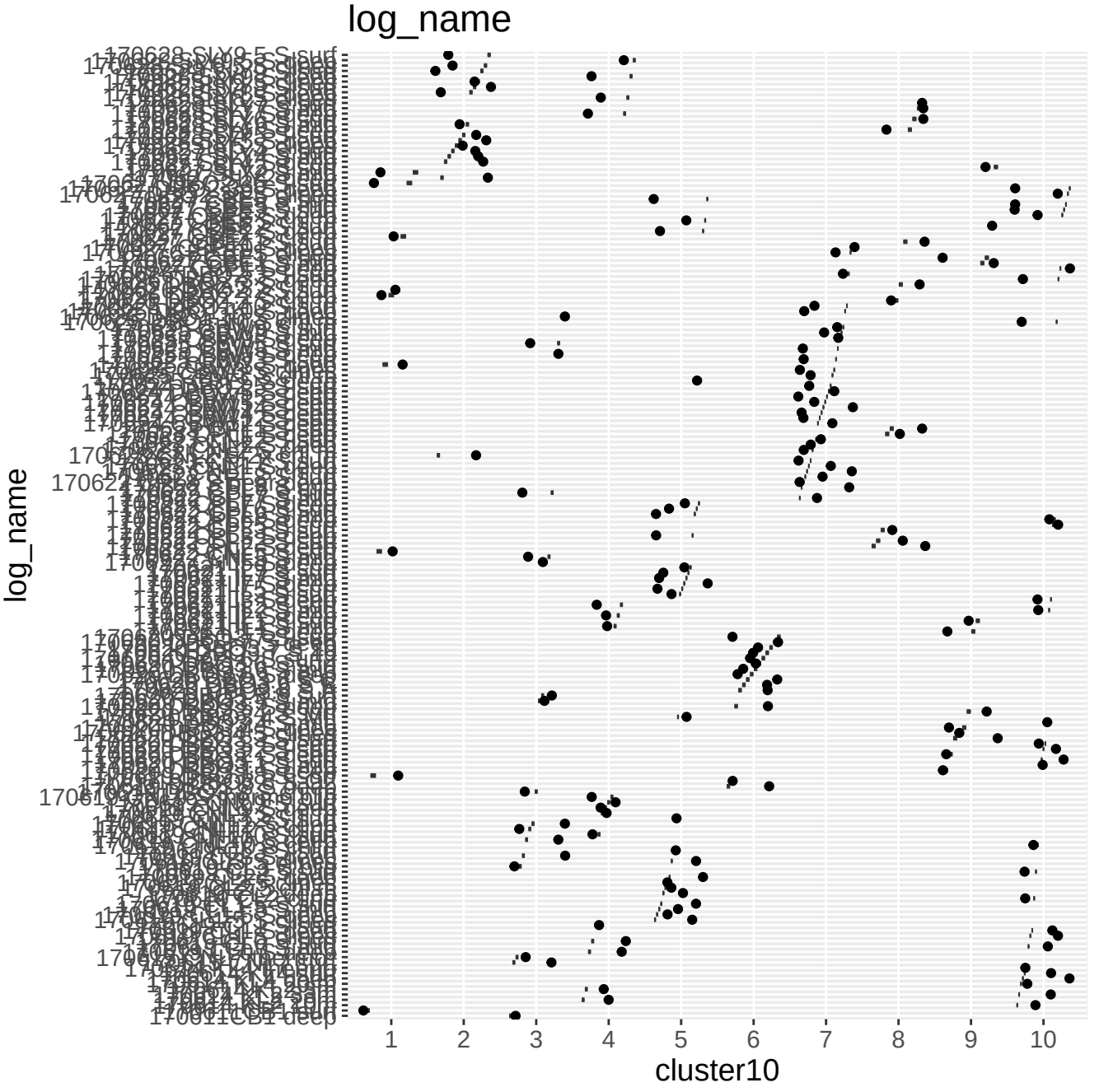
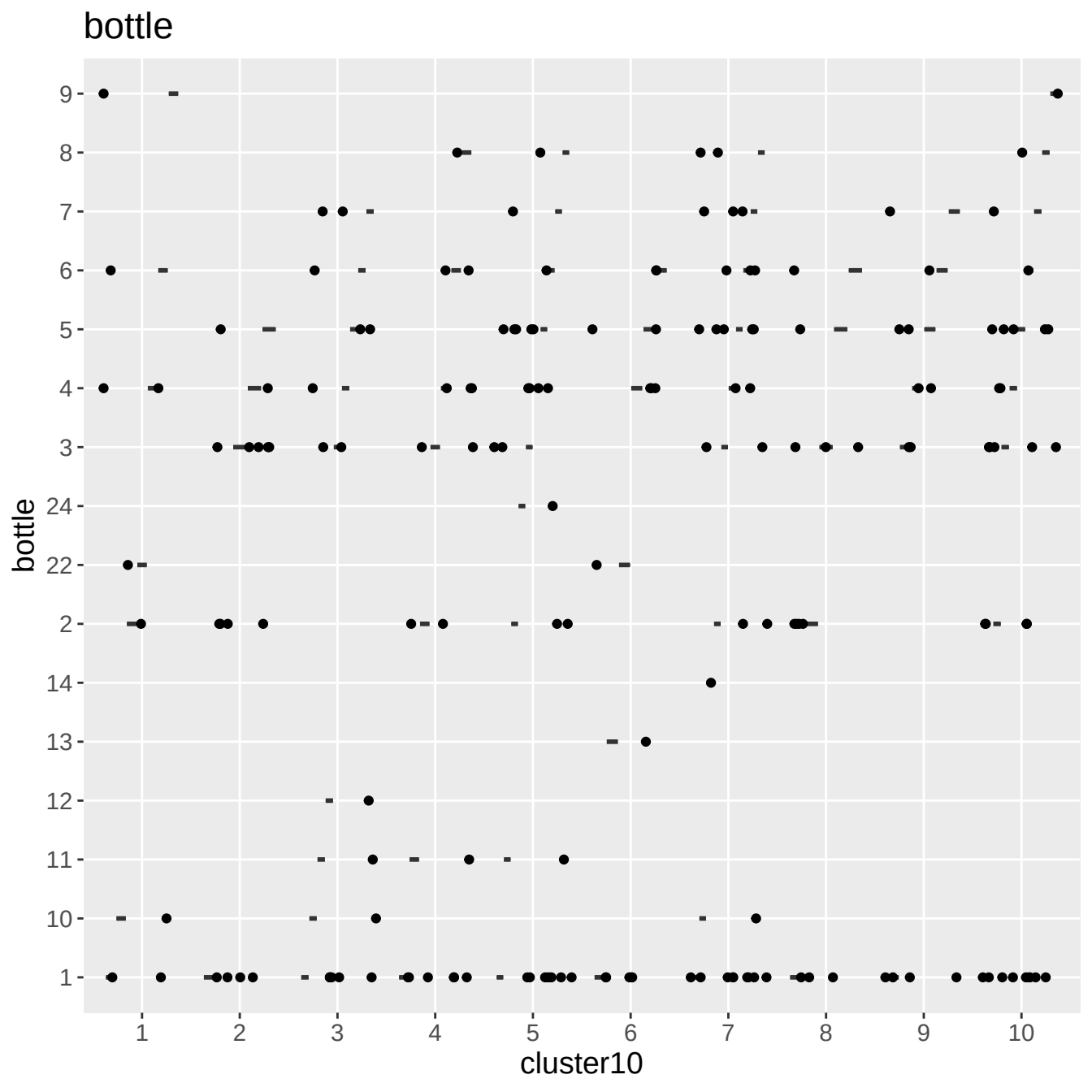
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
2
3
4
5
6
7
8
9
10
mass
NA


log_name
log_name
1
2
3
4
5
6
7
8
9
10
cluster10

blip
m
p
p
p
es
p

- 170620 DBO3.4 S surf
- 170620 DBO3.5 S deep
- 170620 DBO3.5 S mid
- 170620 DBO3.5 S surf
- 170620 DBO3.6 S A
- 170620 DBO3.6 S B
- 170620 DBO3.6 S C
- 170620 DBO3.6 S deep
- 170620 DBO3.6 S mid
- 170620 DBO3.6 S surf
- 170620 DBO3.6 S surf2
- 170620 DBO3.7 S 10
- 170620 DBO3.7 S 20
- 170620 DBO3.7 S deep
- 170620 DBO3.7 S surf
- 170621 IL1 S deep
- 170621 IL1 S mid
- 170621 IL1 S surf
- 170621 IL2 S mid
- 170621 IL2 S surf
- 170621 IL3 S deep
- 170621 IL3 S surf
- 170621 IL5 S surf
- 170621 IL7 S deep
- 170621 IL7 S mid
- 170621 IL7 S surf
- 170621 IL9 S surf
- 170622 CNL5 S deep
- 170622 CNL5 S mid
- 170622 CNL5 S surf
- 170622 CPL2 S deep
- 170622 CPL2 S surf
- 170622 CPL3 S deep
- 170622 CPL3 S surf
- 170622 CPL5 S surf
- 170622 CPL6 S deep
- 170622 CPL6 S surf
- 170622 CPL7 S deep
- 170622 CPL7 S mid
- 170622 CPL7 S surf
- 170622 CPL8 S mid
- 170622 CPL8 S near deep
- 170622 CPL8 S surf
- 170623 CNL1 S chl m
- 170623 CNL1 S deep
- 170623 CNL1 S surf
- 170623 CNL2 S 2x chl m
- 170623 CNL2 S chl m
- 170623 CNL2 S deep
- 170623 CNL2 S surf
- 170623 CPL1 S deep
- 170623 CPL1 S surf
- 170624 CBW12 S deep
- 170624 CBW12 S surf
- 170624 CBW14 S su
- 170624 CBW15 S de
- 170624 CBW15 S su
- 170624 DBO2.5 S de
- 170624 DBO2.5 S su
- 170625 CBE5 S deep
- 170625 CBW3 S chl
- 170625 CBW3 S clin
- 170625 CBW3 S dee
- 170625 CBW3 S surf
- 170625 CBW5 S dee
- 170625 CBW5 S mid
- 170625 CBW5 S surf
- 170625 CBW9 S dee
- 170625 CBW9 S mid
- 170625 CBW9 S surf
- 170625 DBO1.10 S d
- 170625 DBO1.10 S d


bottle
bottle
9
8
7
6
5
4
3
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11
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1
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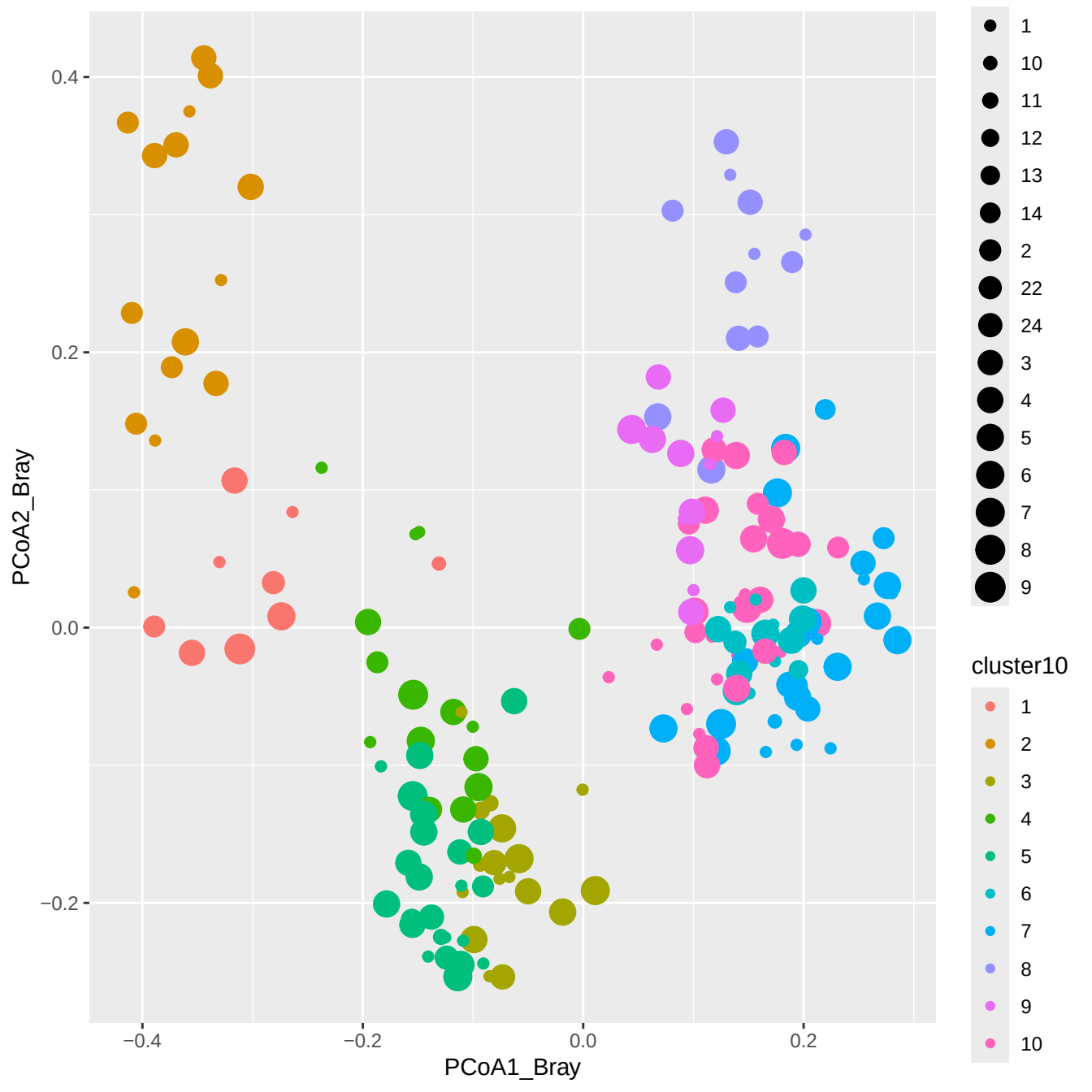
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
1
10
11
12
13
14
2
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5
6
7
8
9
cluster10
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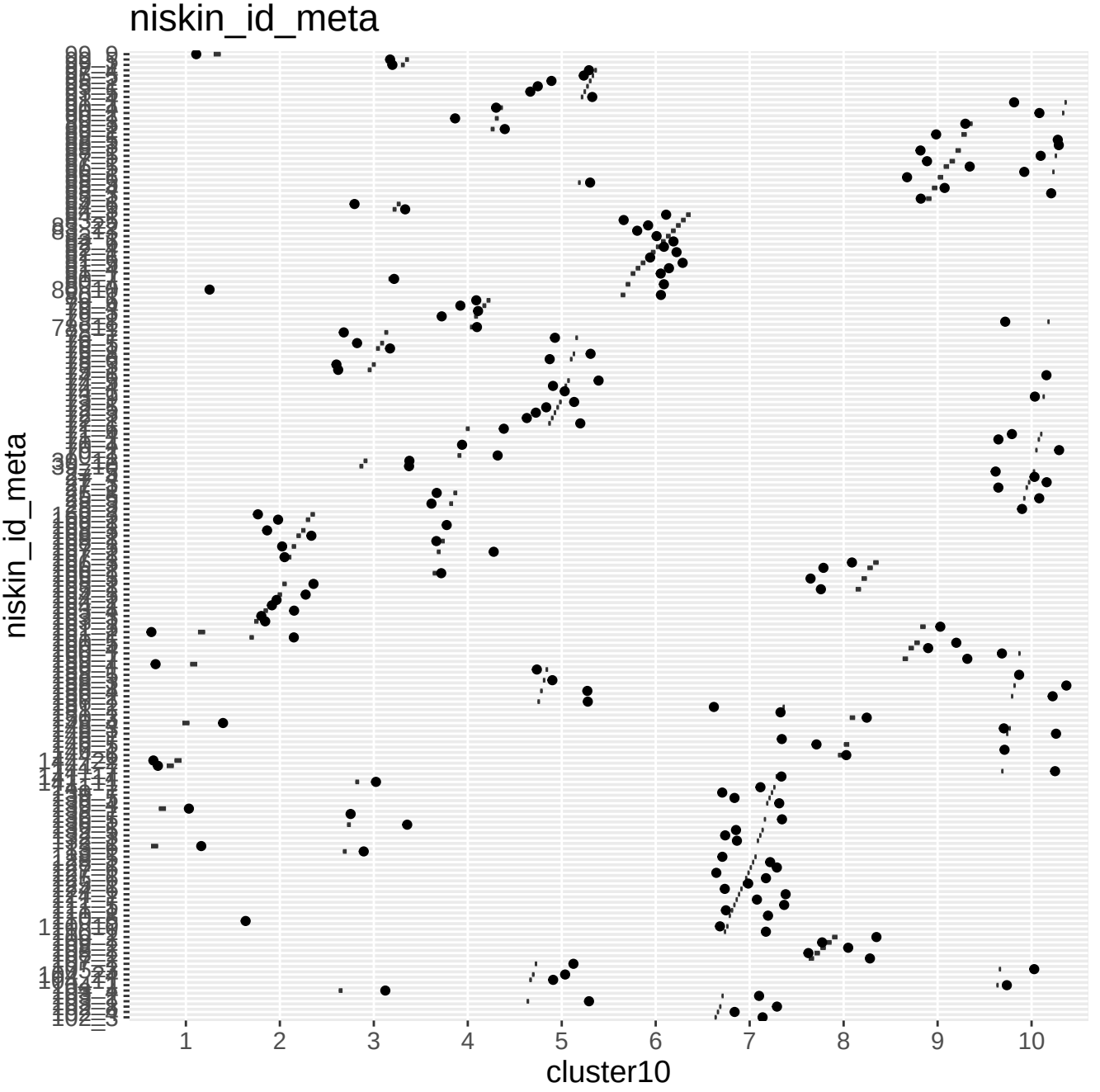
8
9
10


niskin_id_meta
niskin_id_meta
cluster10
1
2
3
4
5
6
7
8
9
10

- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10

niskin_id_meta

- 102_3
- 102_4
- 102_8
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- 103_2
- 103_4
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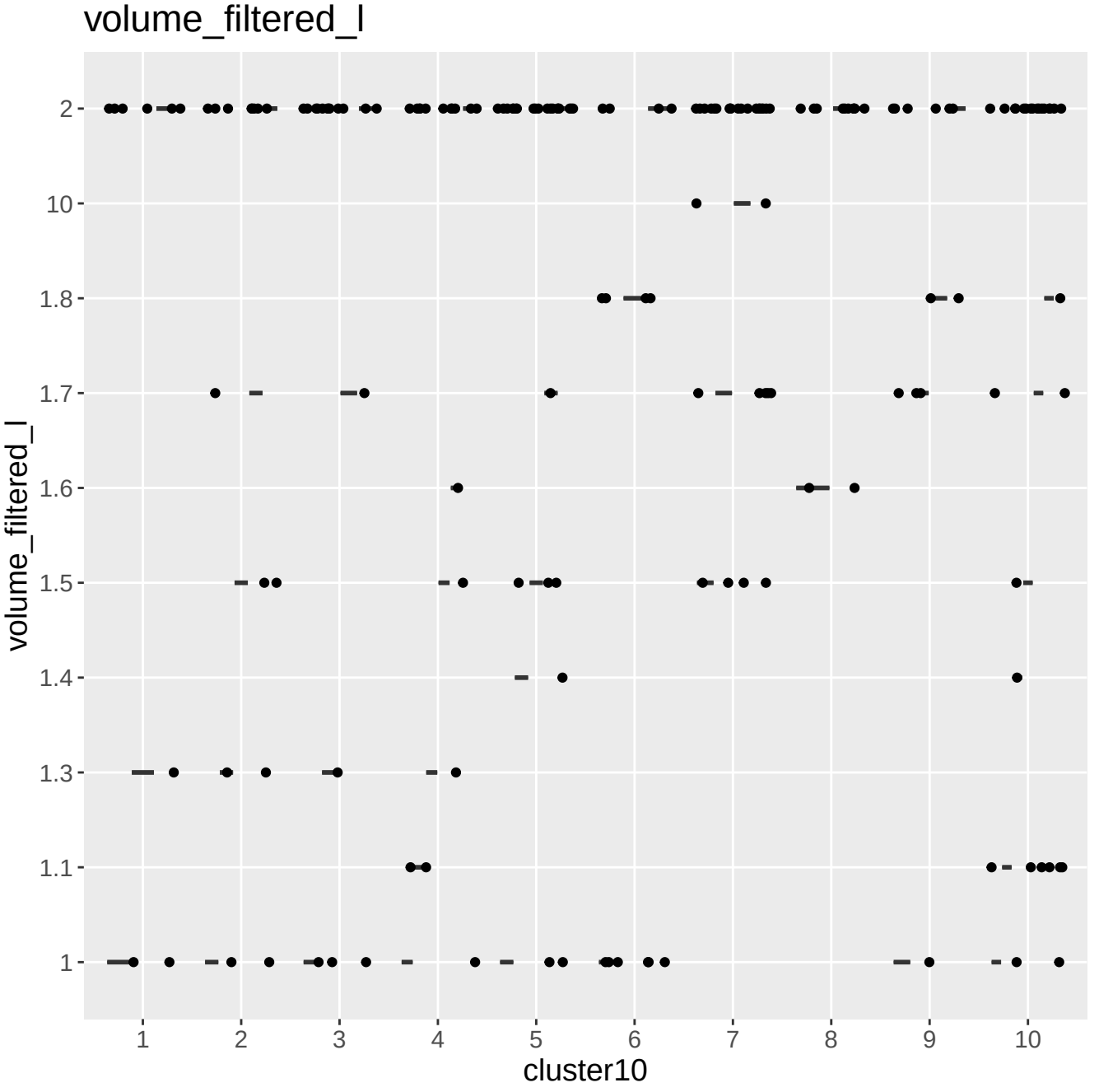
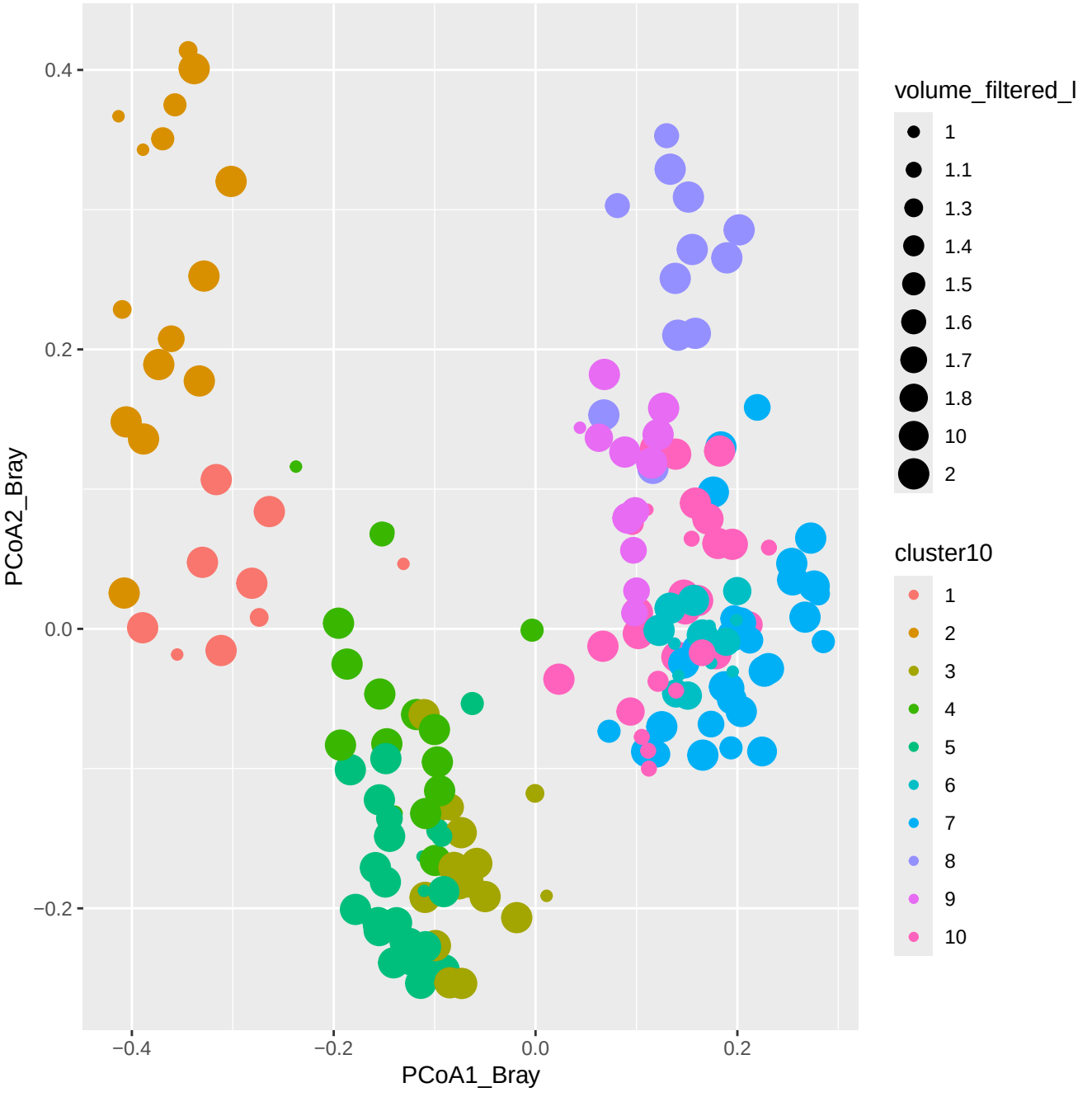
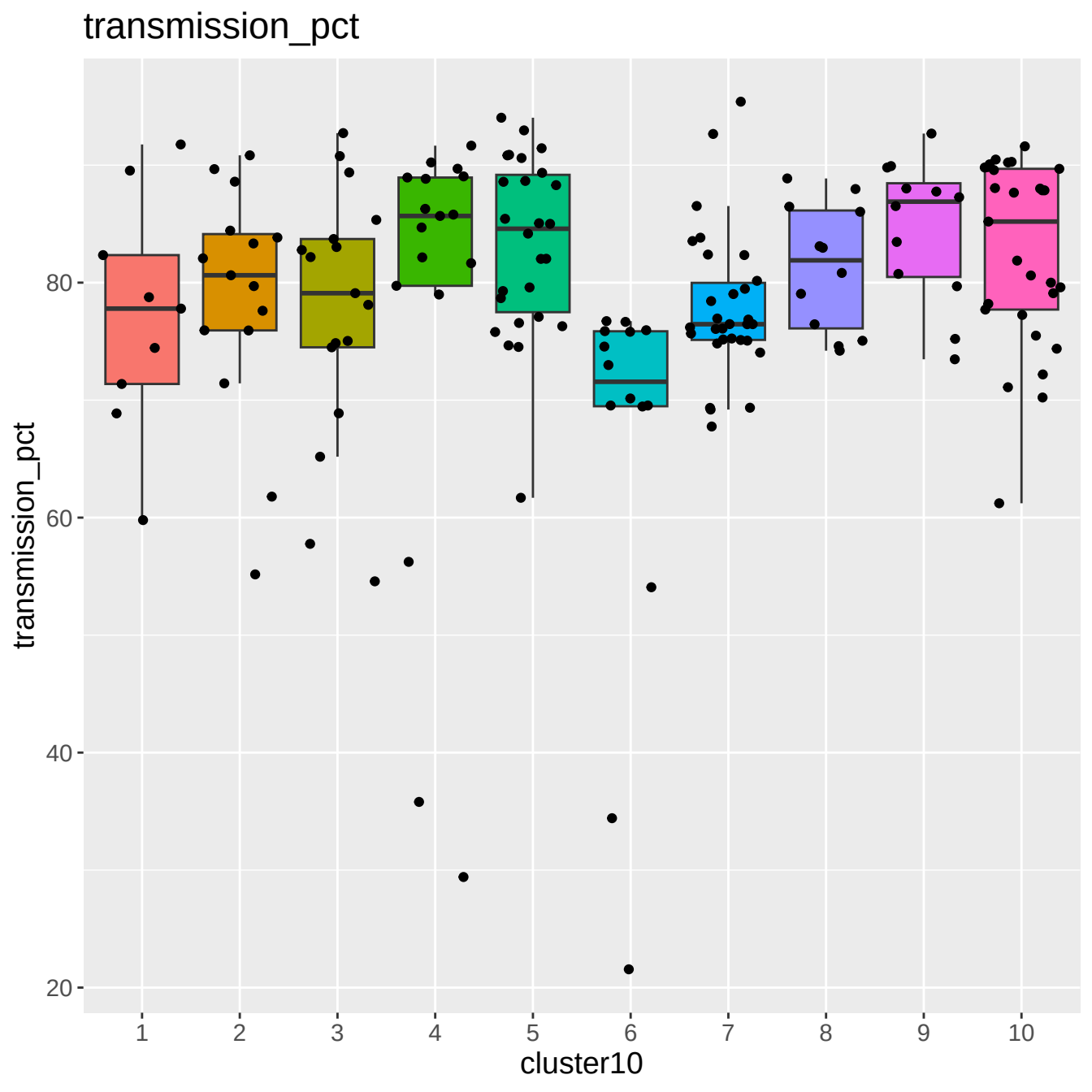
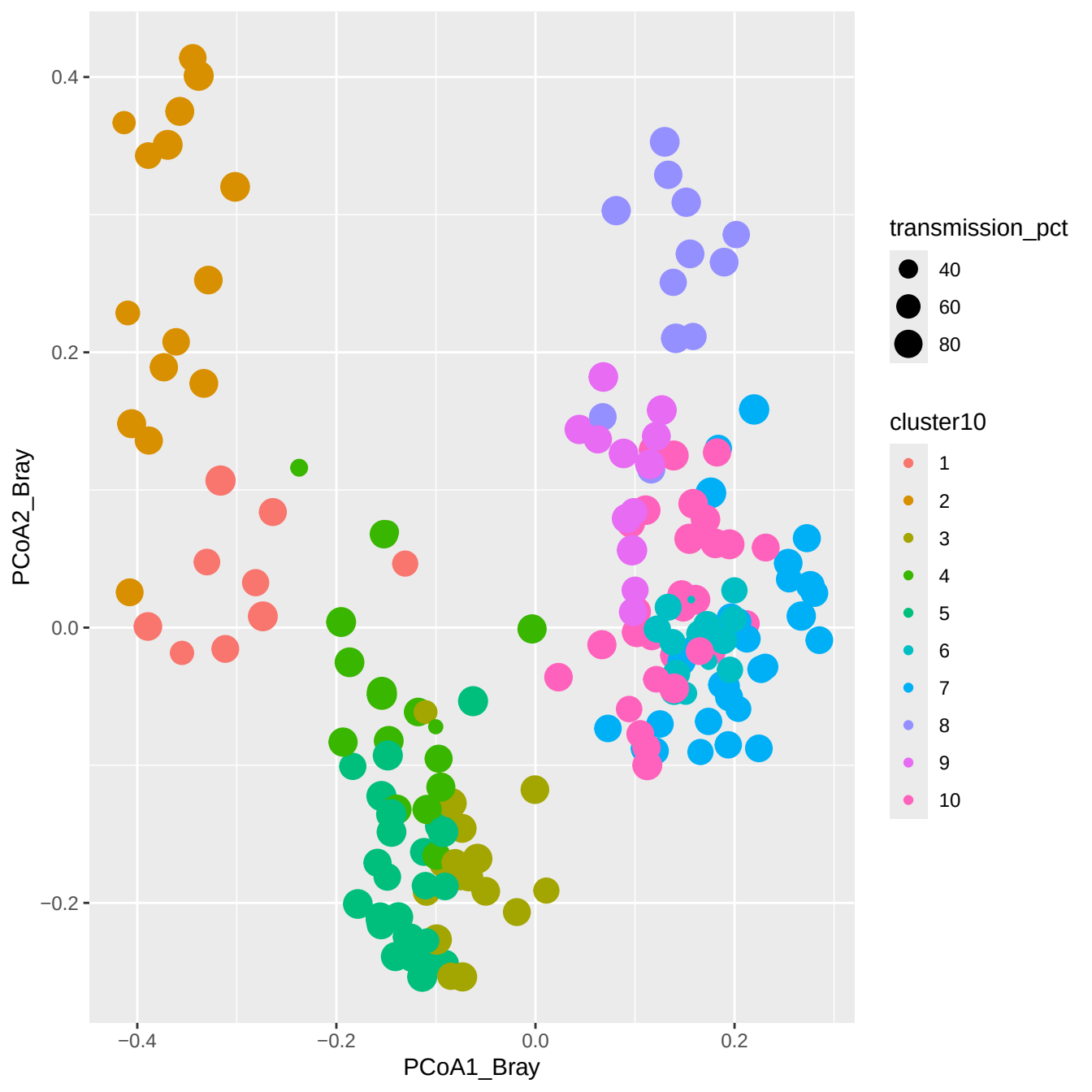
- 90_2


volume_filtered_l
volume_filtered_l
2
10
1.8
1.7
1.6
1.5
1.4
1.3
1.1
1
1
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
volume_filtered_l
1
1.1
1.3
1.4
1.5
1.6
1.7
1.8
10
2
cluster10
1
2
3
4
5
6
7
8
9
10


transmission_pct
transmission_pct
80
60
40
20
1
2
3
4
5
6
7
8
9
10
cluster10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80
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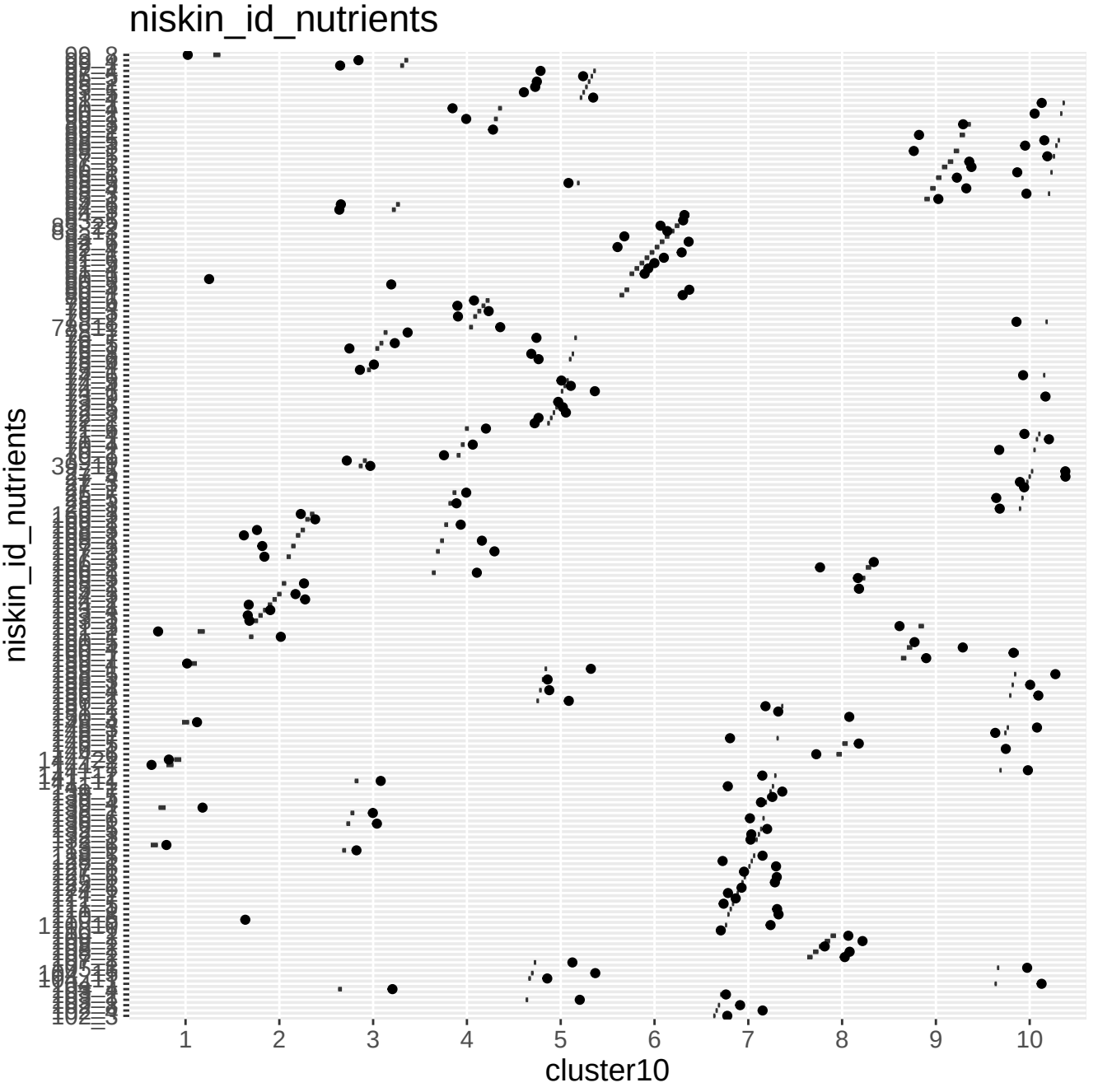
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niskin_id_nutrients
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●	102_8	●	111_5	●	139_4	●	156_1	●	164_2	●	25_5	●	73_9	●	80_1	●	85_3	●	
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●	108_1	●	13_1	●	146_1	●	160_4	●	167_3	●	70_4	●	76_5	●	82_6	●	88_5		
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●	109_1	●	132_1	●	146_7	●	161_1	●	168_2	●	71_4	●	78_1	●	83_13	●	89_2		
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●	110_5	●	136_6	●	150_3	●	163_3	●	169_3	●	72_3	●	79_3	●	84_3	●	90_4		

cluster10

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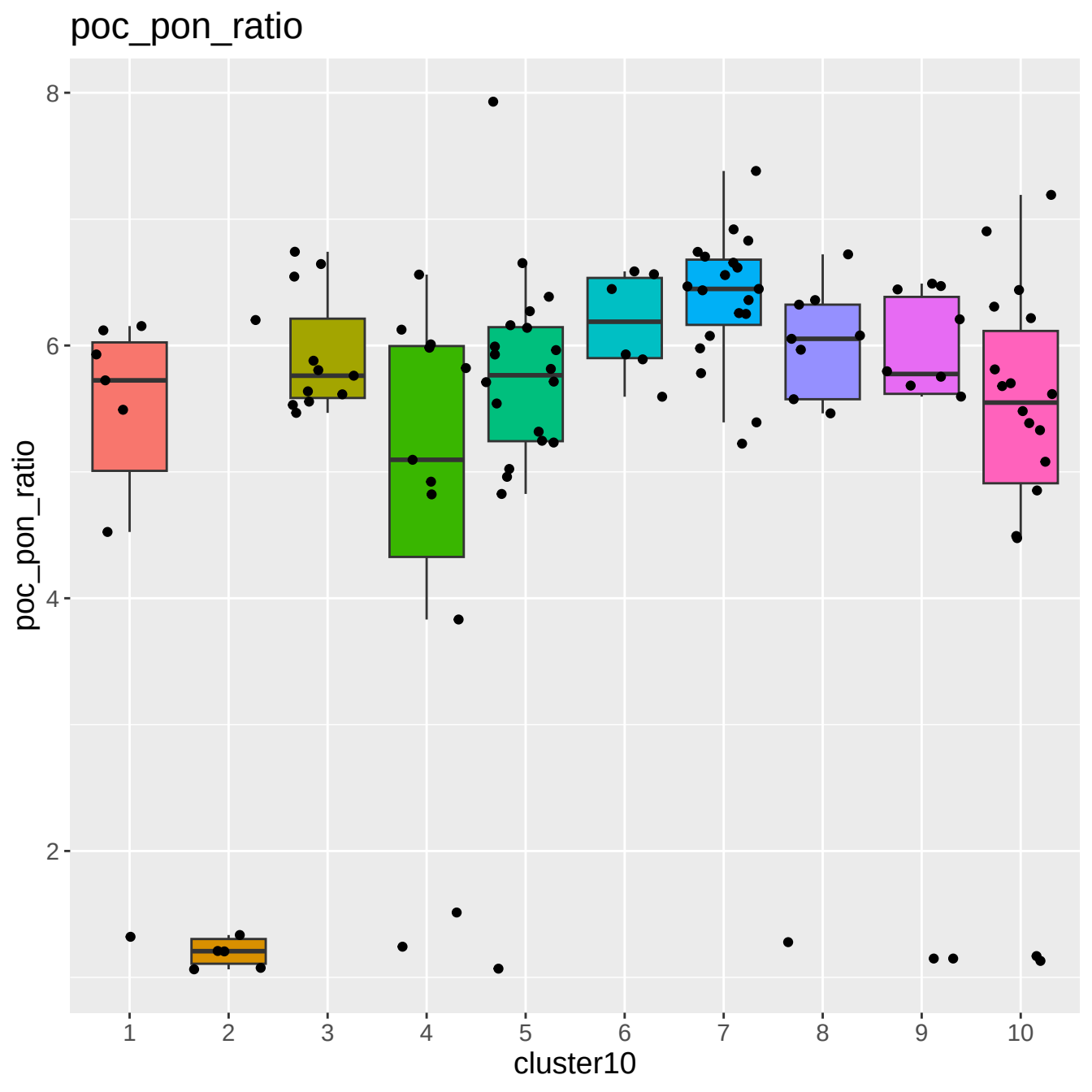
- 9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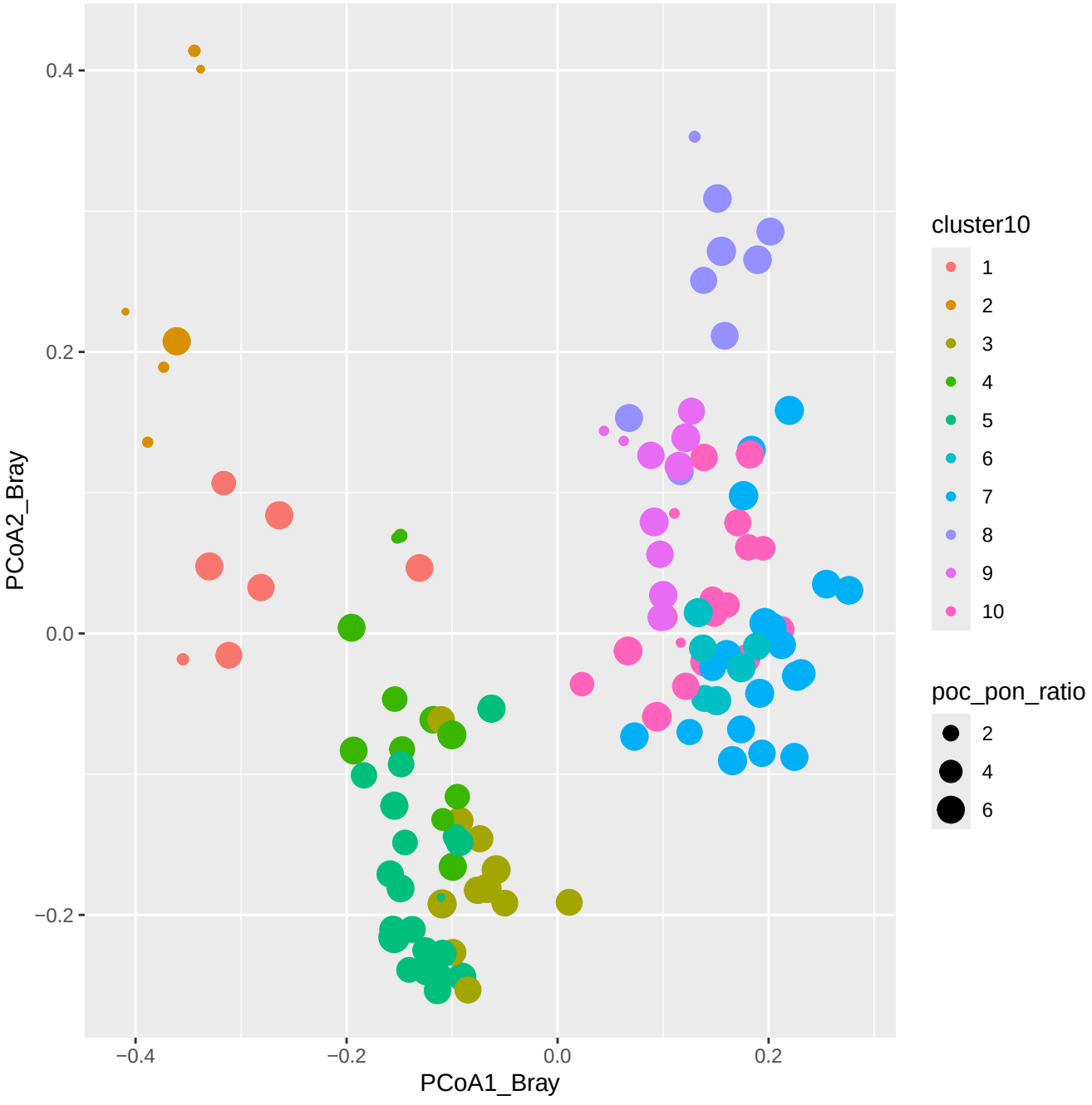
cluster10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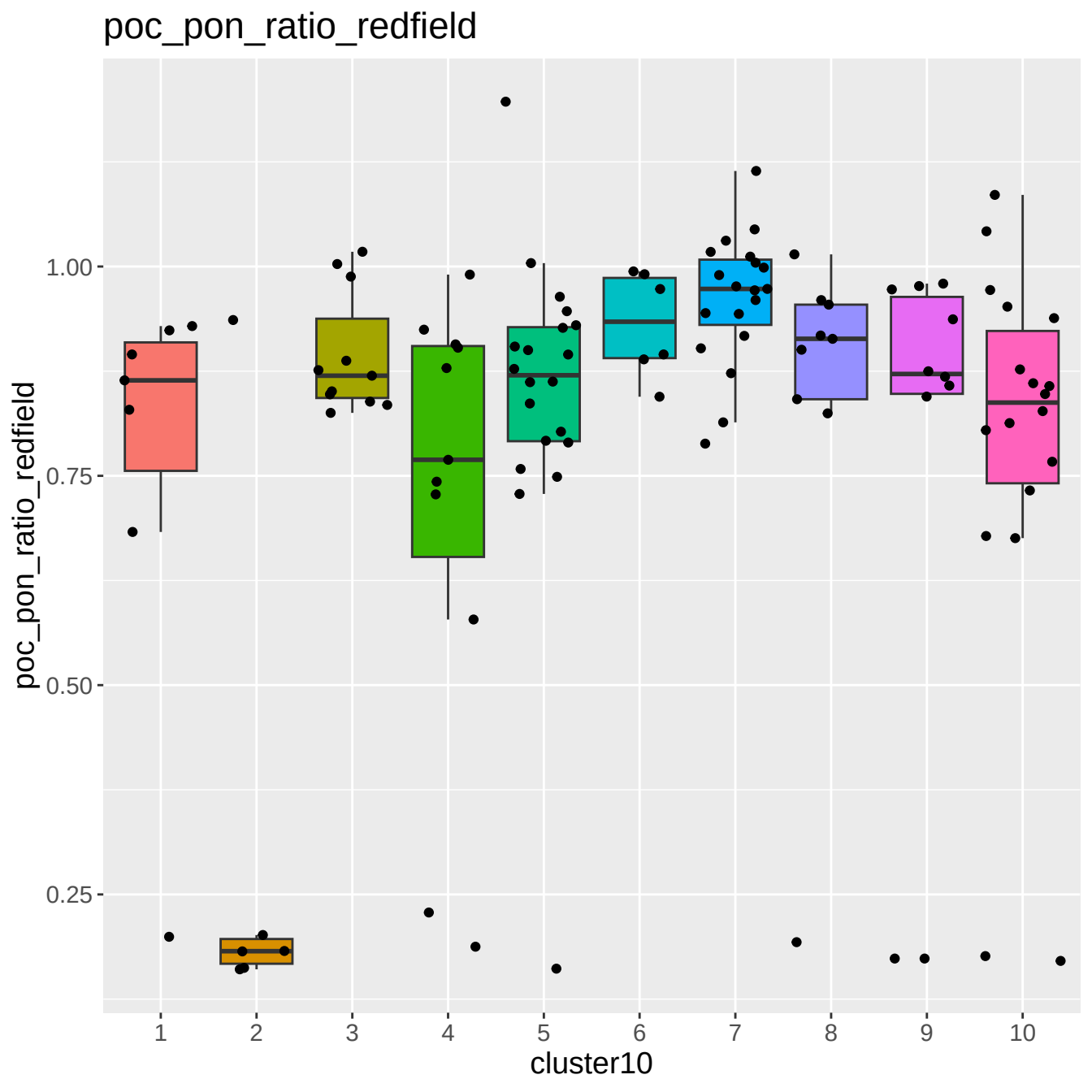
6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poc_pon_ratio_redfield
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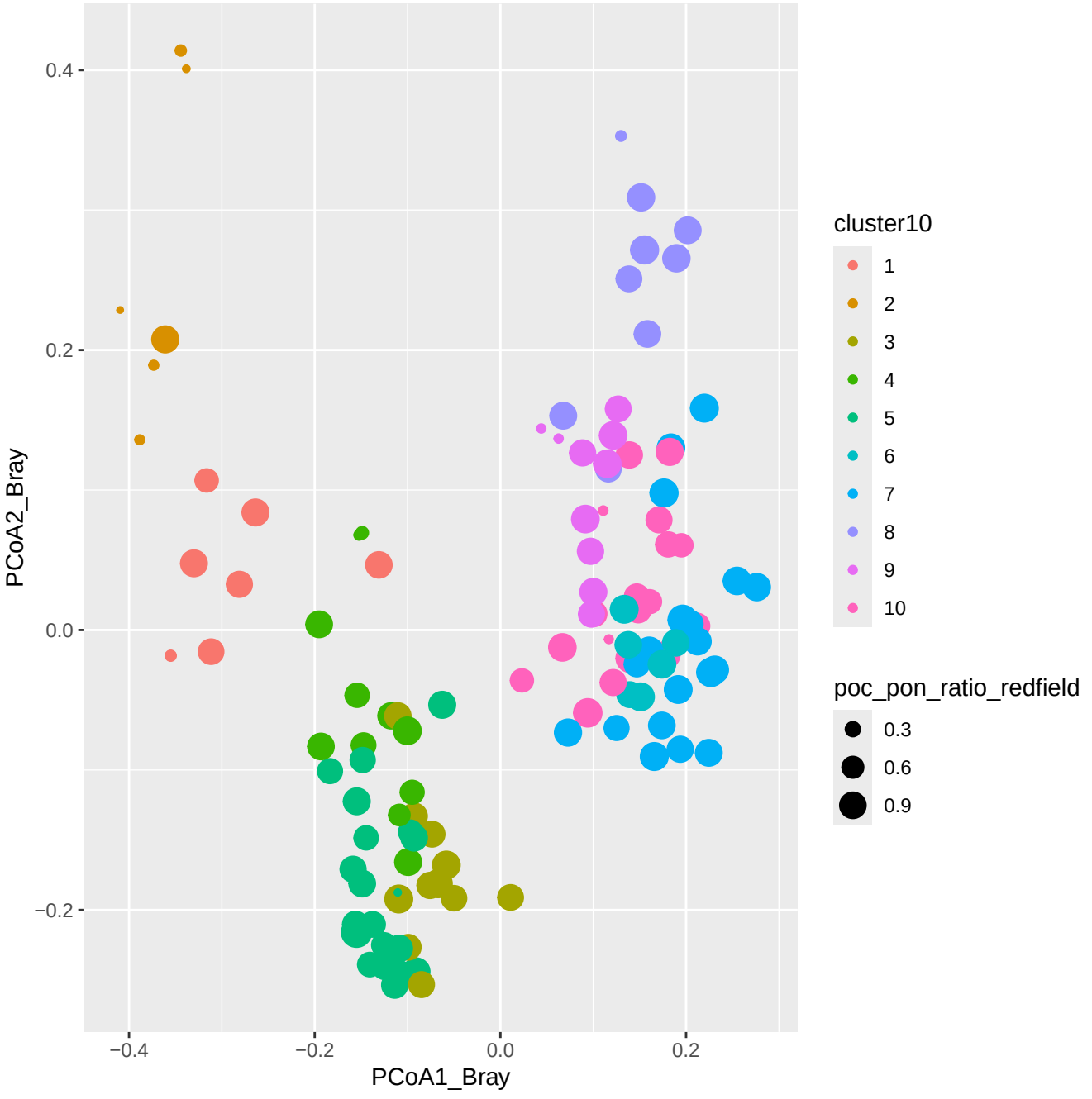
cluster10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PCoA2_Bray
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0.0
0.2
PCoA1_Bray
cluster10
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9
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poc_pon_ratio_redfield
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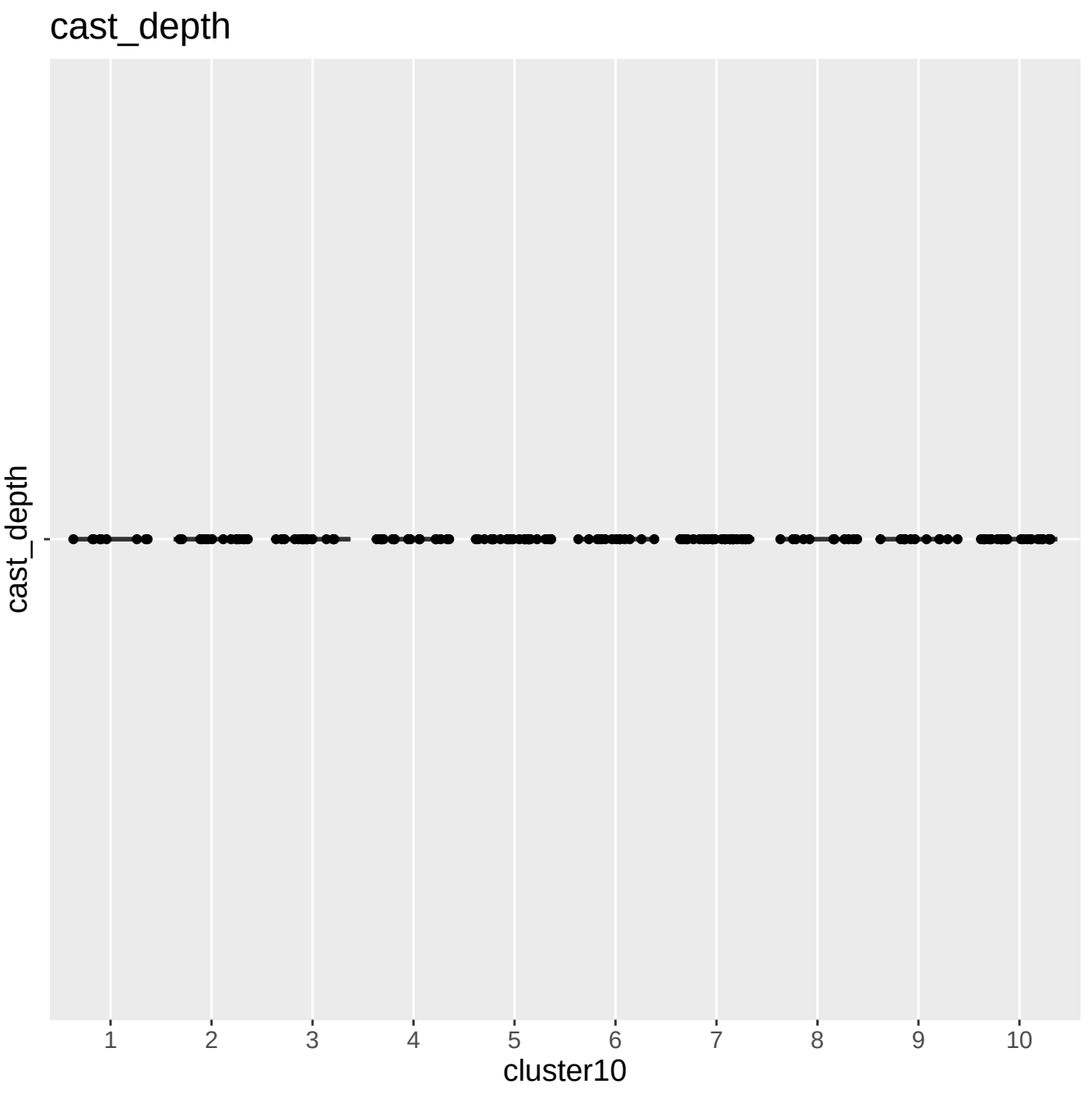
0.9


cast_depth
cast_depth
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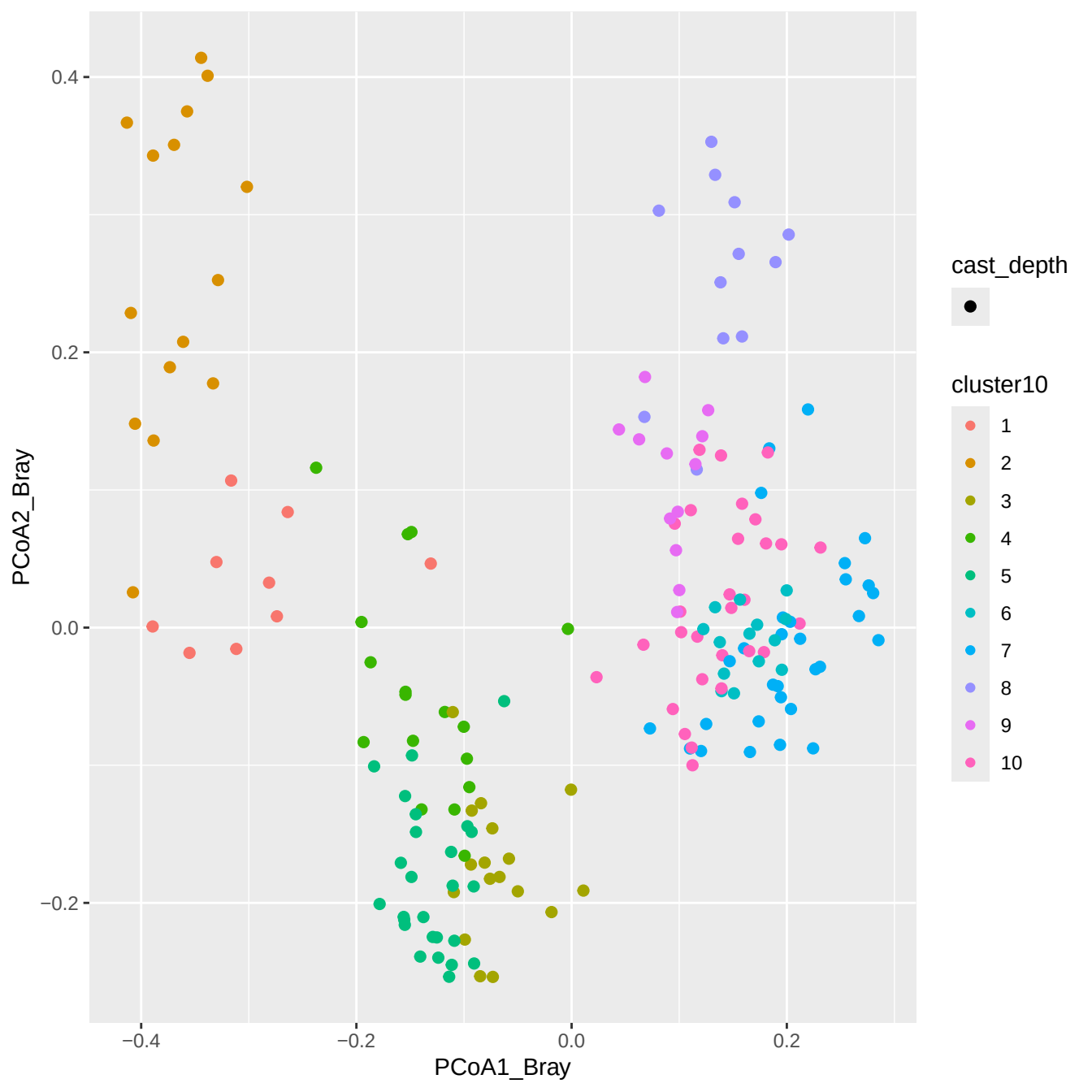
cluster10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PCoA1_Bray
cast_depth
cluster10
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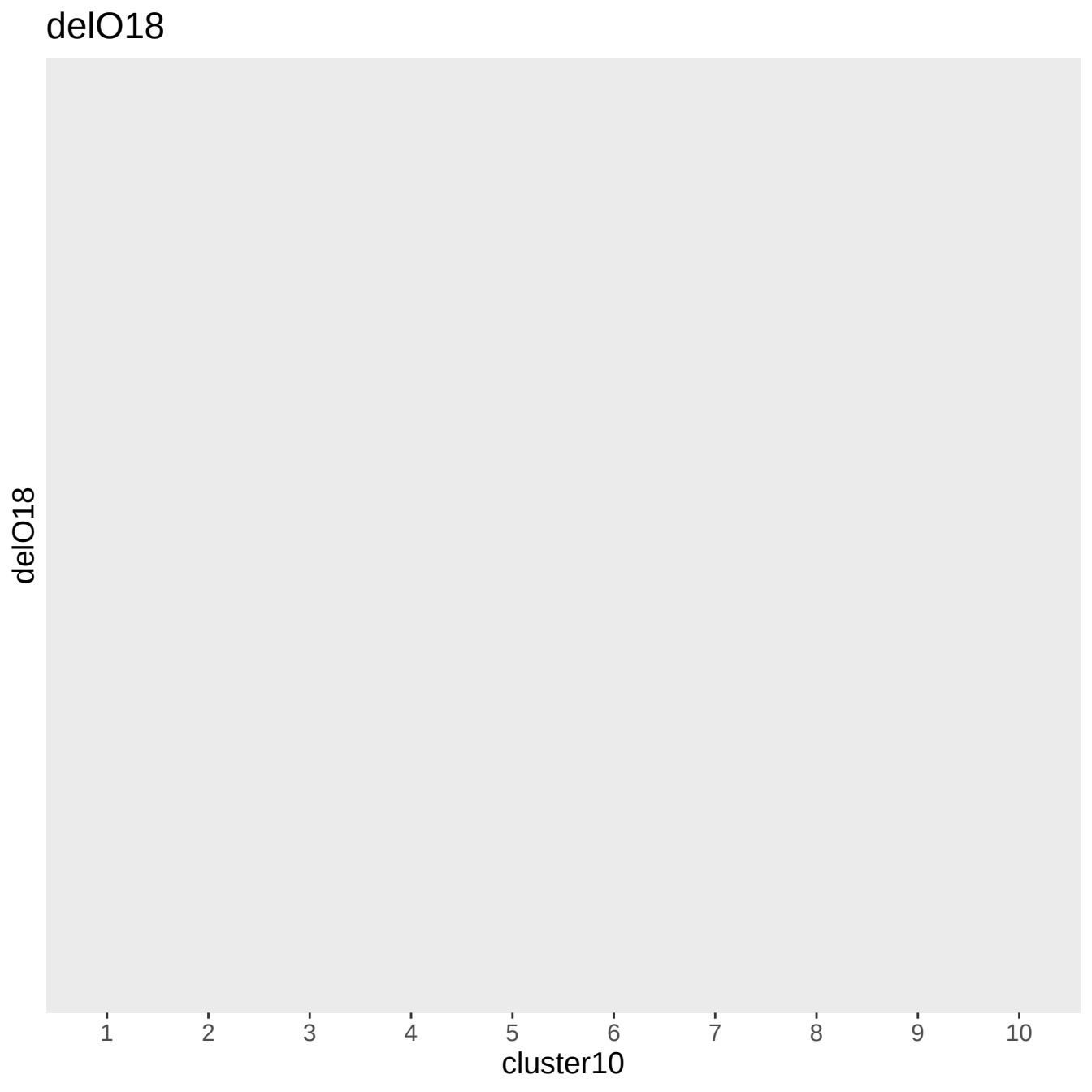
delO18
delO18
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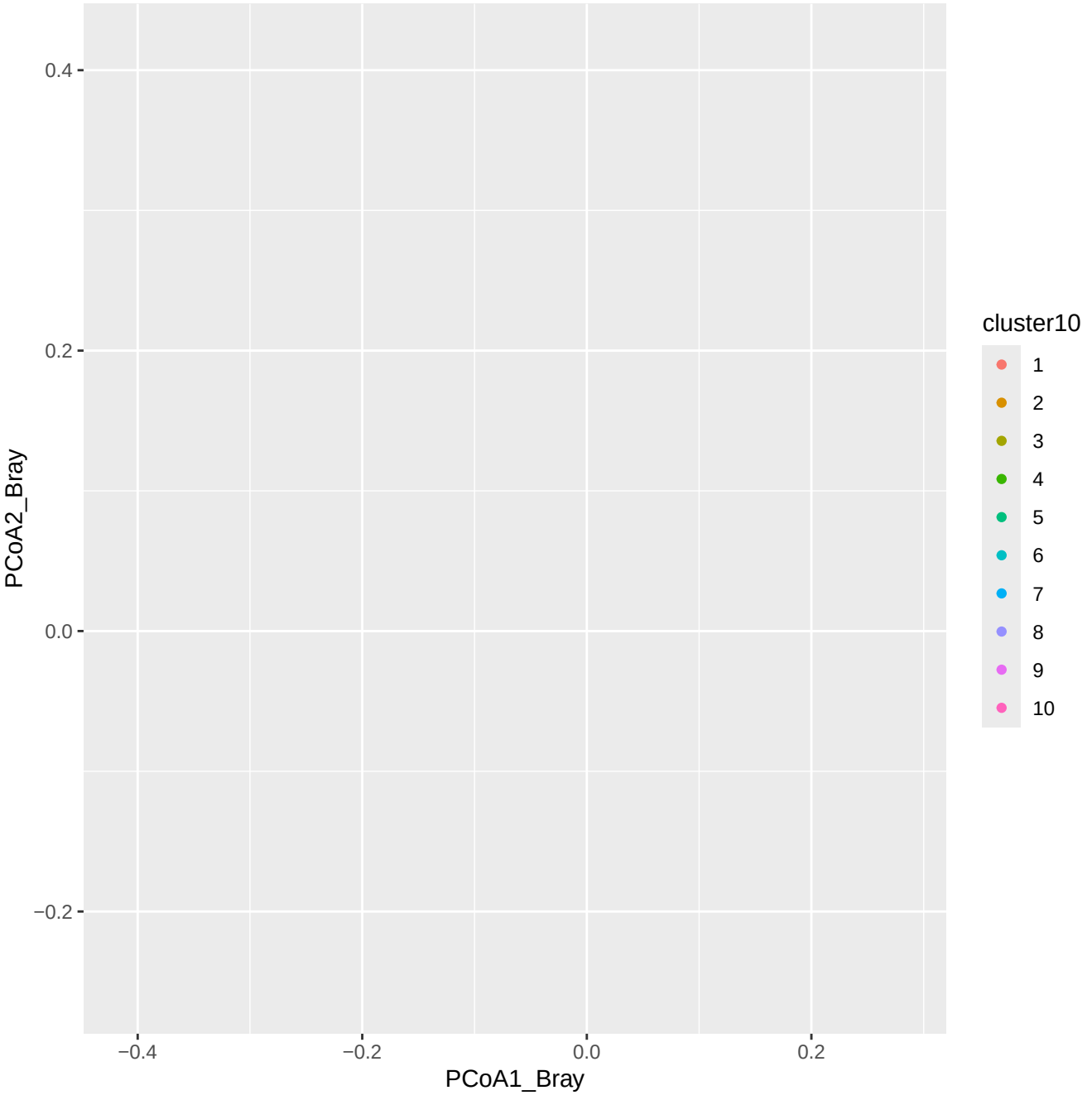
9
10
cluster10


0.4
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PCoA2_Bray
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−0.2
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PCoA1_Bray
cluster10
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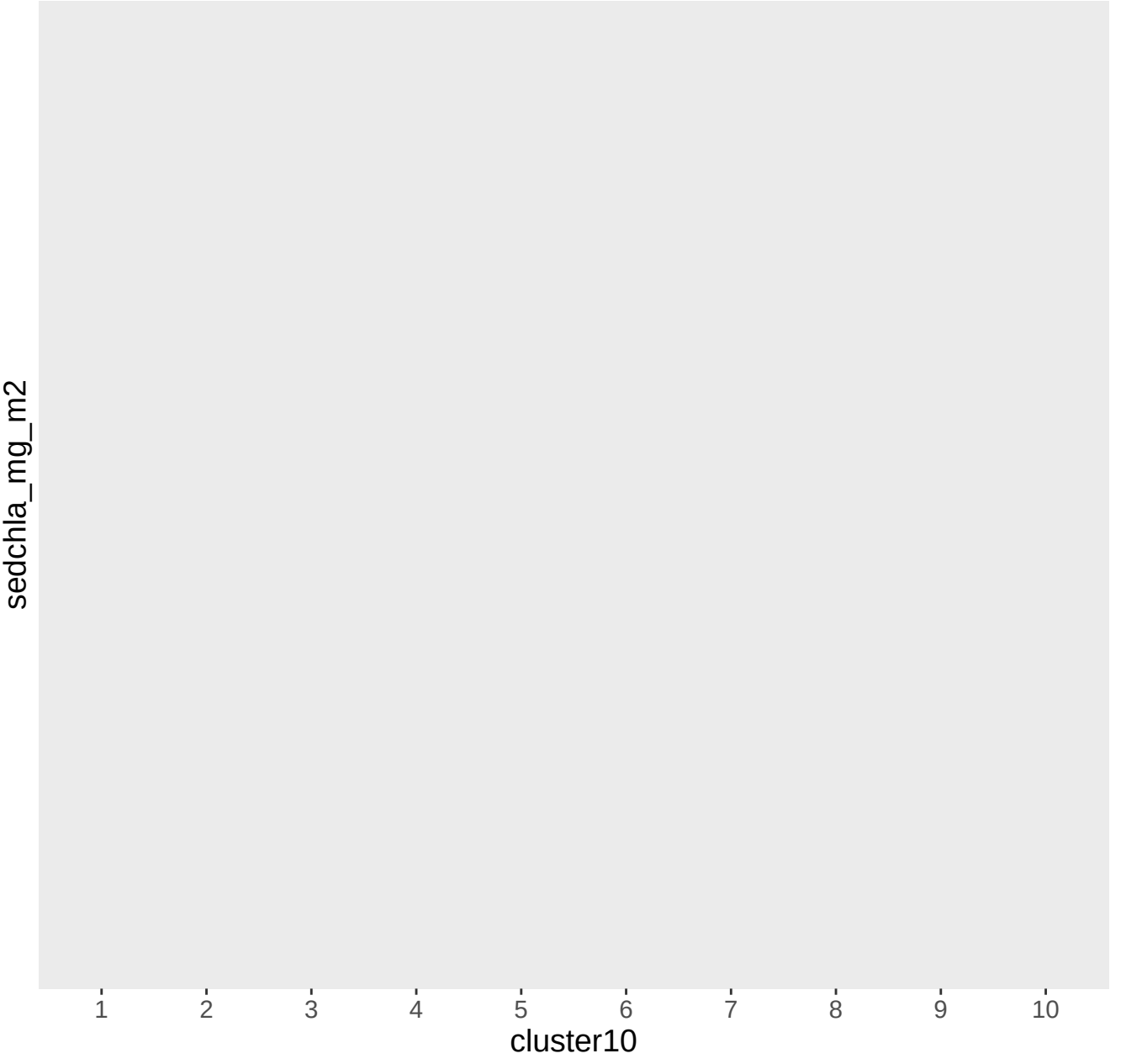
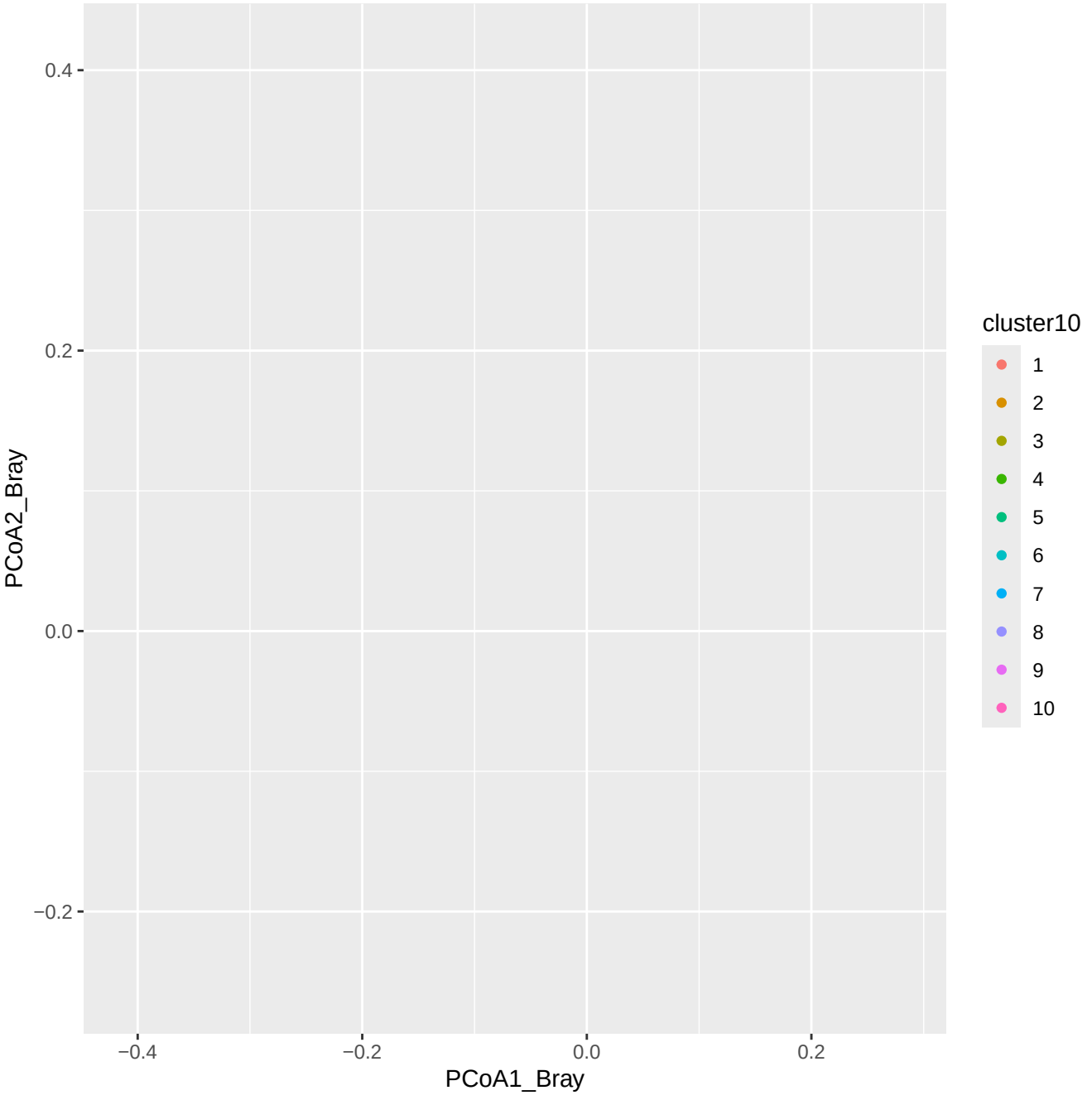
sedchla_mg_m2




0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
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0.2
PCoA1_Bray
cluster10
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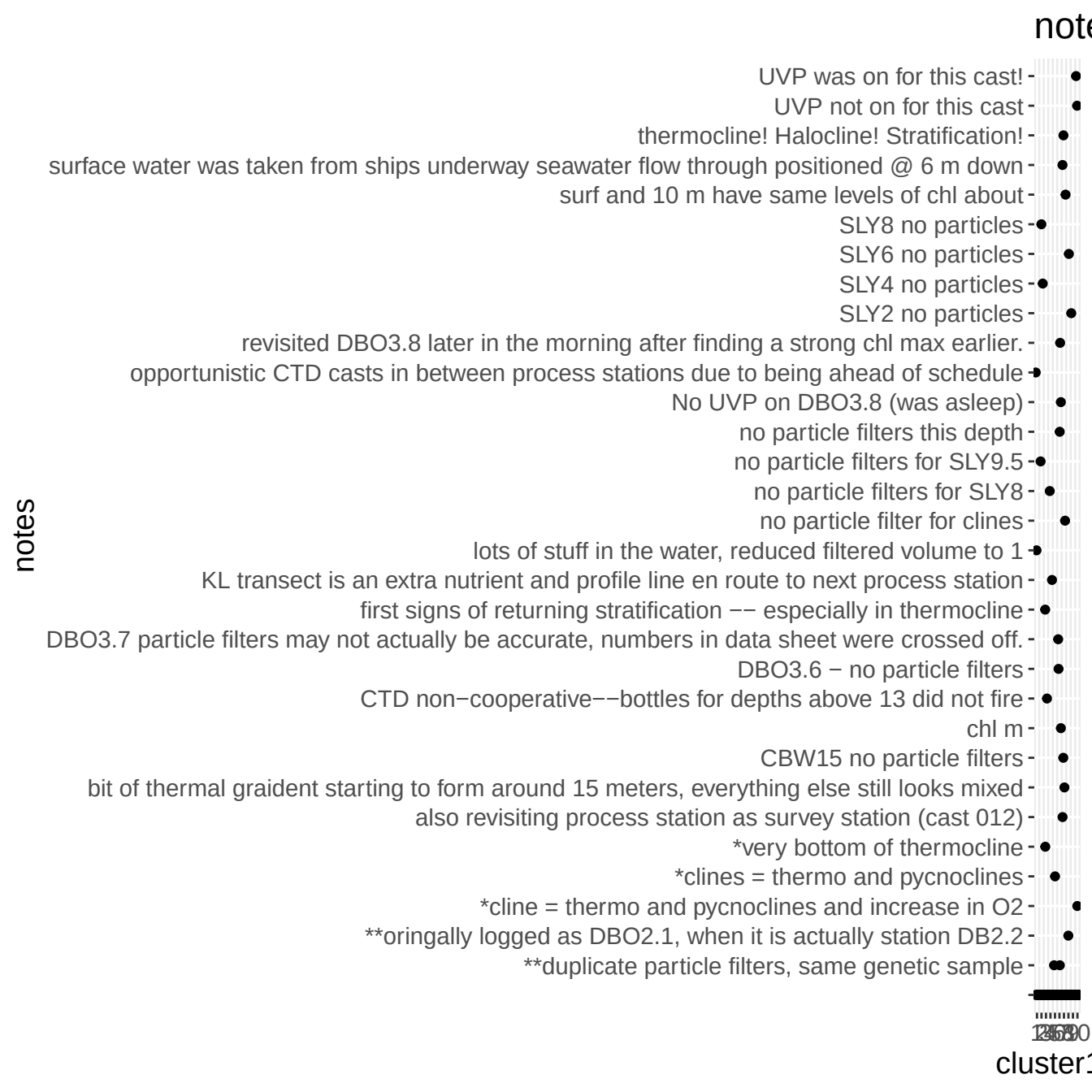
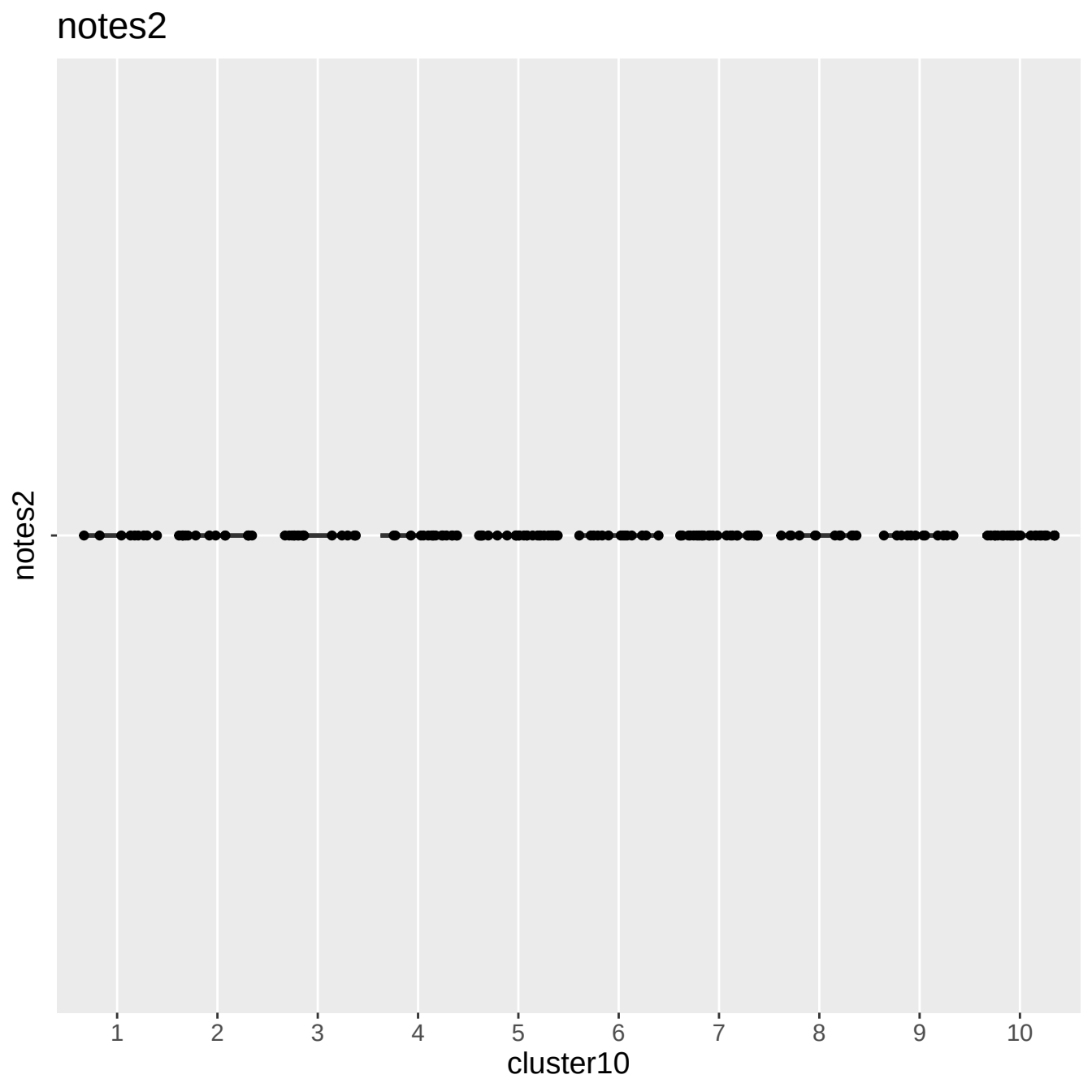

note
UVP was on for this cast!
UVP not on for this cast
thermocline! Halocline! Stratification!
surface water was taken from ships underway seawater flow through positioned @ 6 m down
surf and 10 m have same levels of chl about
SLY8 no particles
SLY6 no particles
SLY4 no particles
SLY2 no particles
revisited DBO3.8 later in the morning after finding a strong chl max earlier.
opportunistic CTD casts in between process stations due to being ahead of schedule
No UVP on DBO3.8 (was asleep)
no particle filters this depth
no particle filters for SLY9.5
no particle filters for SLY8
no particle filter for clines
lots of stuff in the water, reduced filtered volume to 1
KL transect is an extra nutrient and profile line en route to next process station
first signs of returning stratification −− especially in thermocline
DBO3.7 particle filters may not actually be accurate, numbers in data sheet were crossed off.
DBO3.6 − no particle filters
CTD non−cooperative−−bottles for depths above 13 did not fire
chl m
CBW15 no particle filters
bit of thermal graident starting to form around 15 meters, everything else still looks mixed
also revisiting process station as survey station (cast 012)
*very bottom of thermocline
*clines = thermo and pycnoclines
*cline = thermo and pycnoclines and increase in O2
**oringally logged as DBO2.1, when it is actually station DB2.2
**duplicate particle filters, same genetic sample
notes
cluster1

	no particle filter for clines
s, same genetic sample	no particle filters for SLY8
3O2.1, when it is actually station DB2.2	no particle filters for SLY9.5
noclines and increase in O2	no particle filters this depth
cnoclines	No UVP on DBO3.8 (was asleep)
line	opportunistic CTD casts in between process station
tation as survey station (cast 012)	revisited DBO3.8 later in the morning after finding a
tarting to form around 15 meters, everything else still looks mixed	SLY2 no particles
s	SLY4 no particles
	SLY6 no particles
-bottles for depths above 13 did not fire	SLY8 no particles
ers	surf and 10 m have same levels of chl about
nay not actually be accurate, numbers in data sheet were crossed off.	surface water was taken from ships underway seaw
ratification −− especially in thermocline	thermocline! Halocline! Stratification!
utrient and profile line en route to next process station	UVP not on for this cast
reduced filtered volume to 1	UVP was on for this cast!


notes2
notes2
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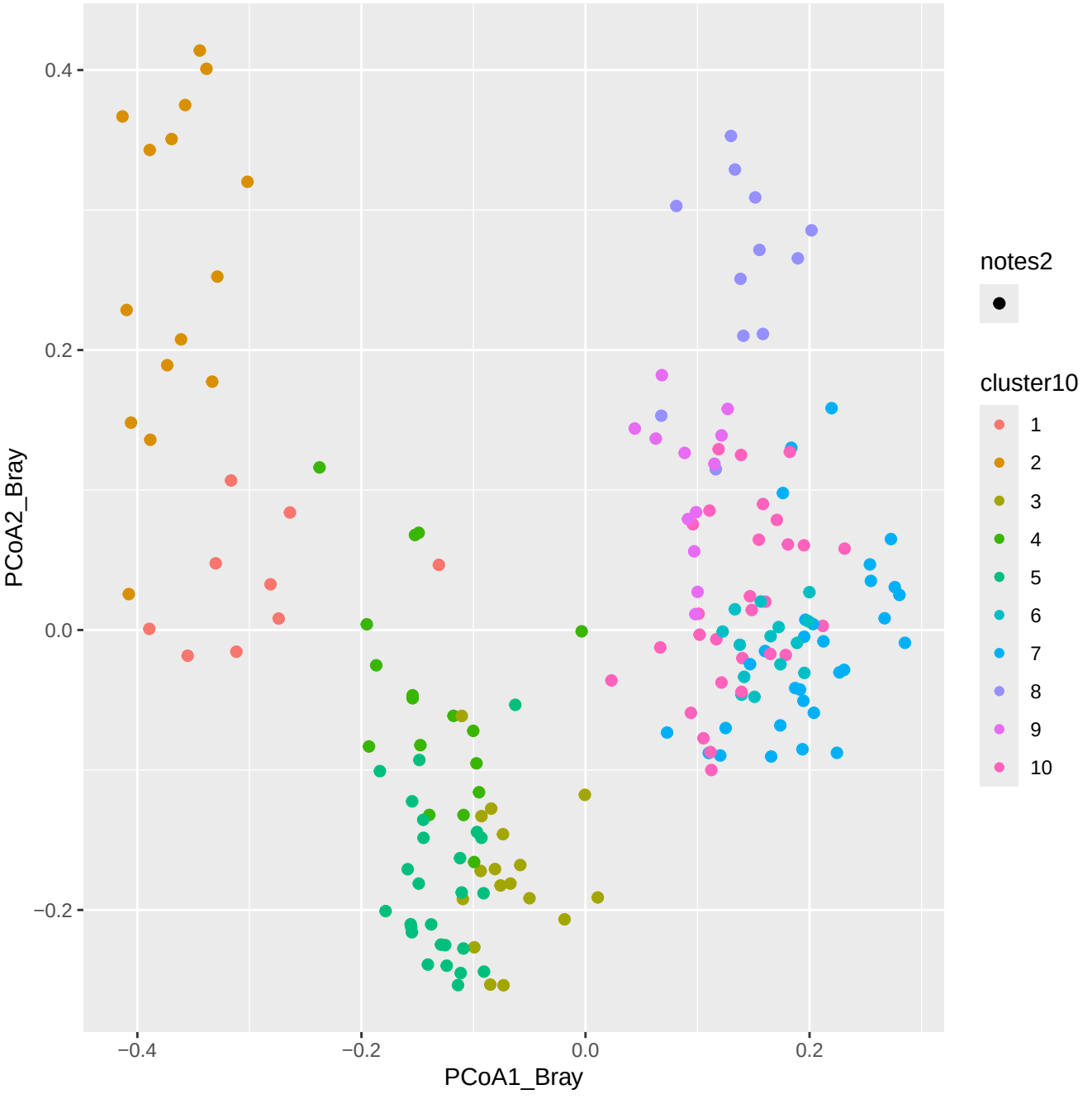
2
3
4
5
6
7
8
9
10
cluster10


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
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PCoA1_Bray
notes2
cluster10
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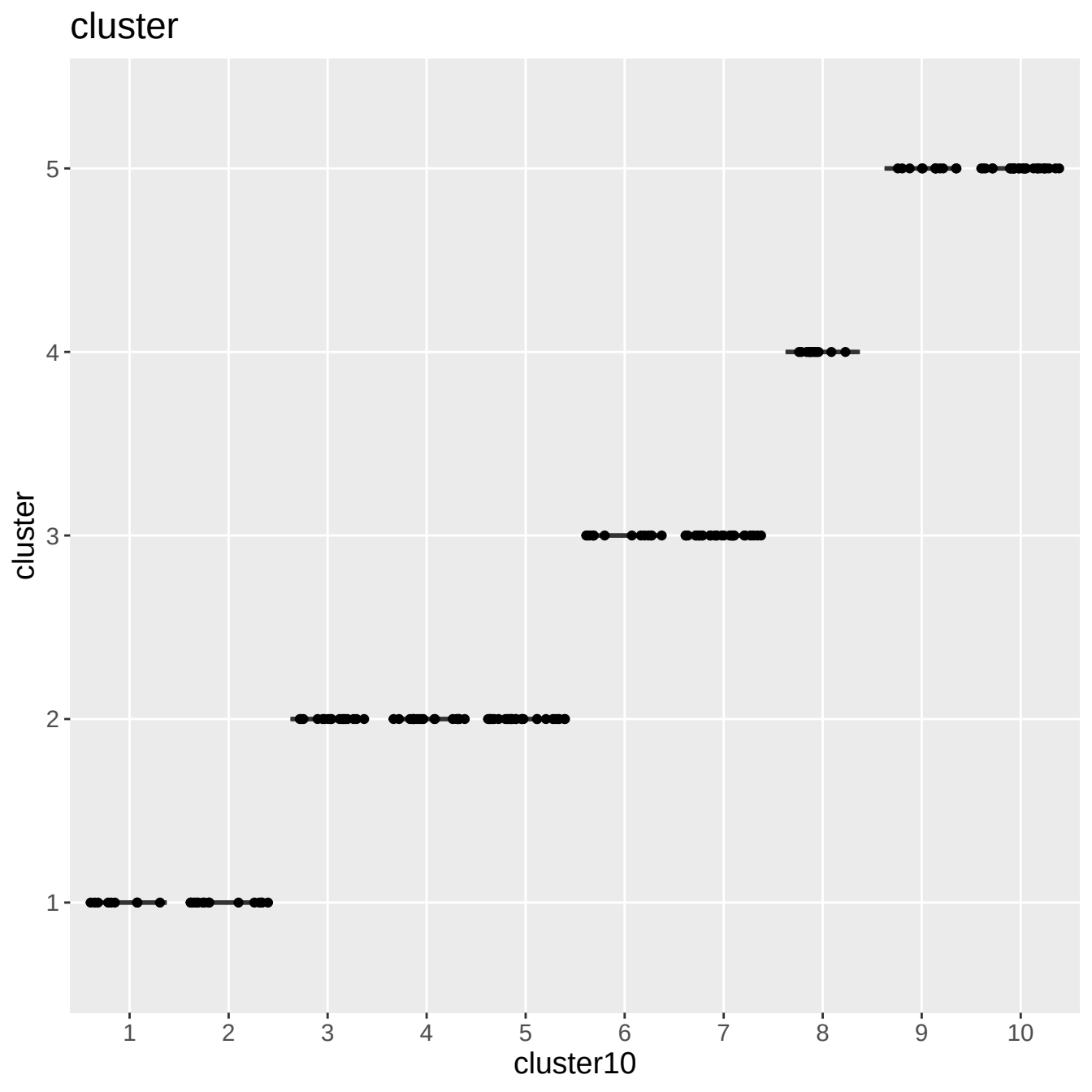
cluster
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2
1
cluster
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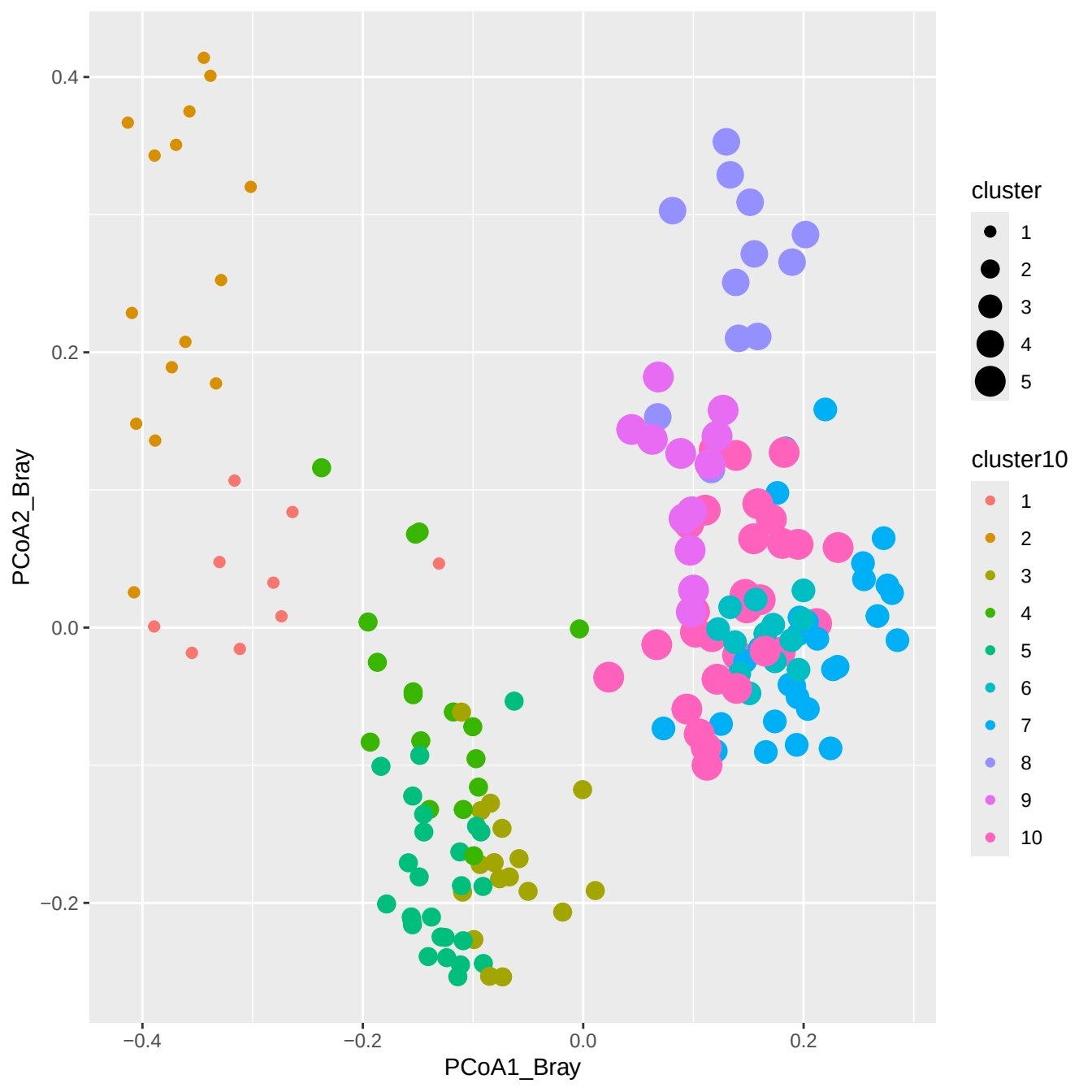
cluster10


0.4
0.2
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−0.2
PCoA2_Bray
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0.2
PCoA1_Bray
cluster
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5
cluster10
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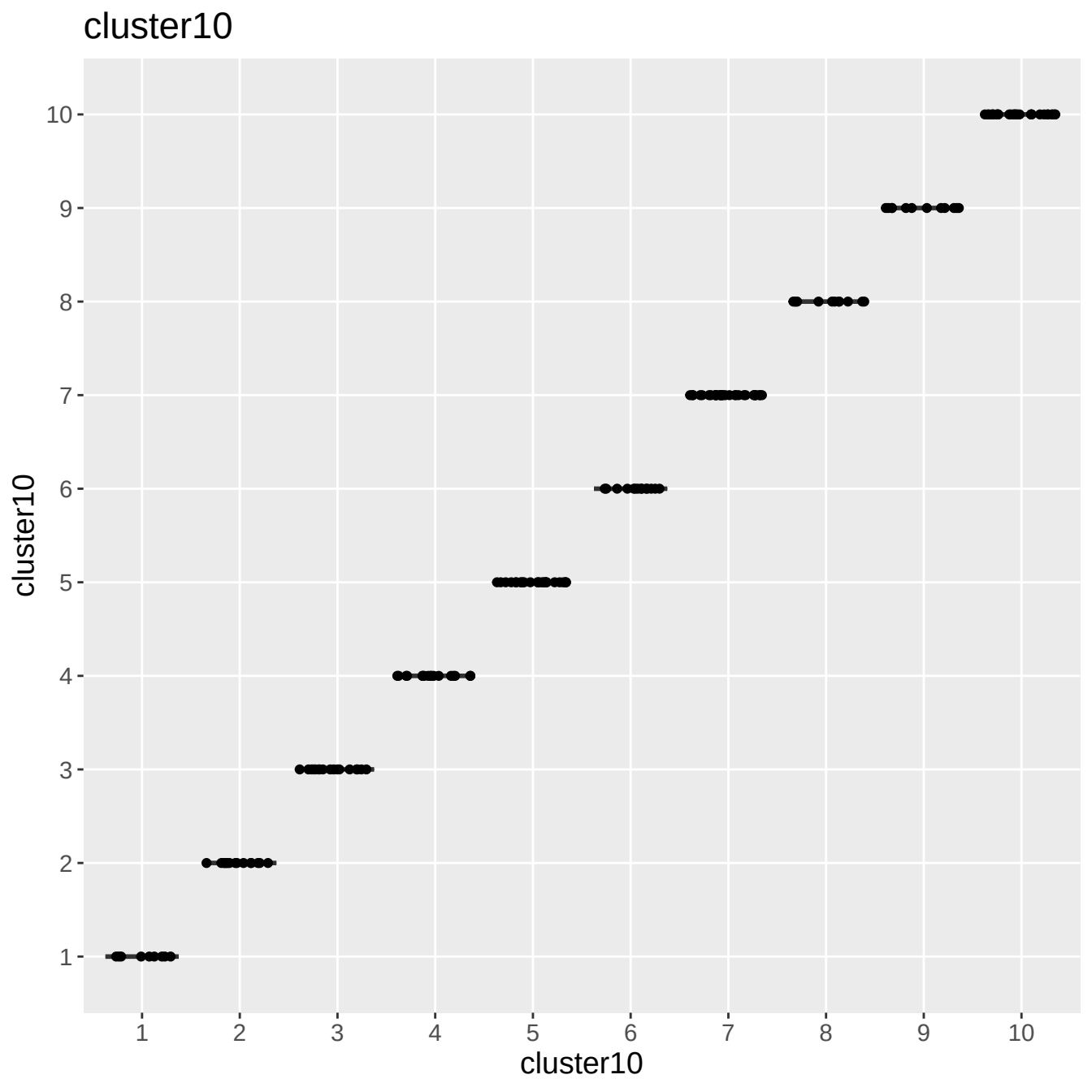
cluster10
cluster10
cluster10
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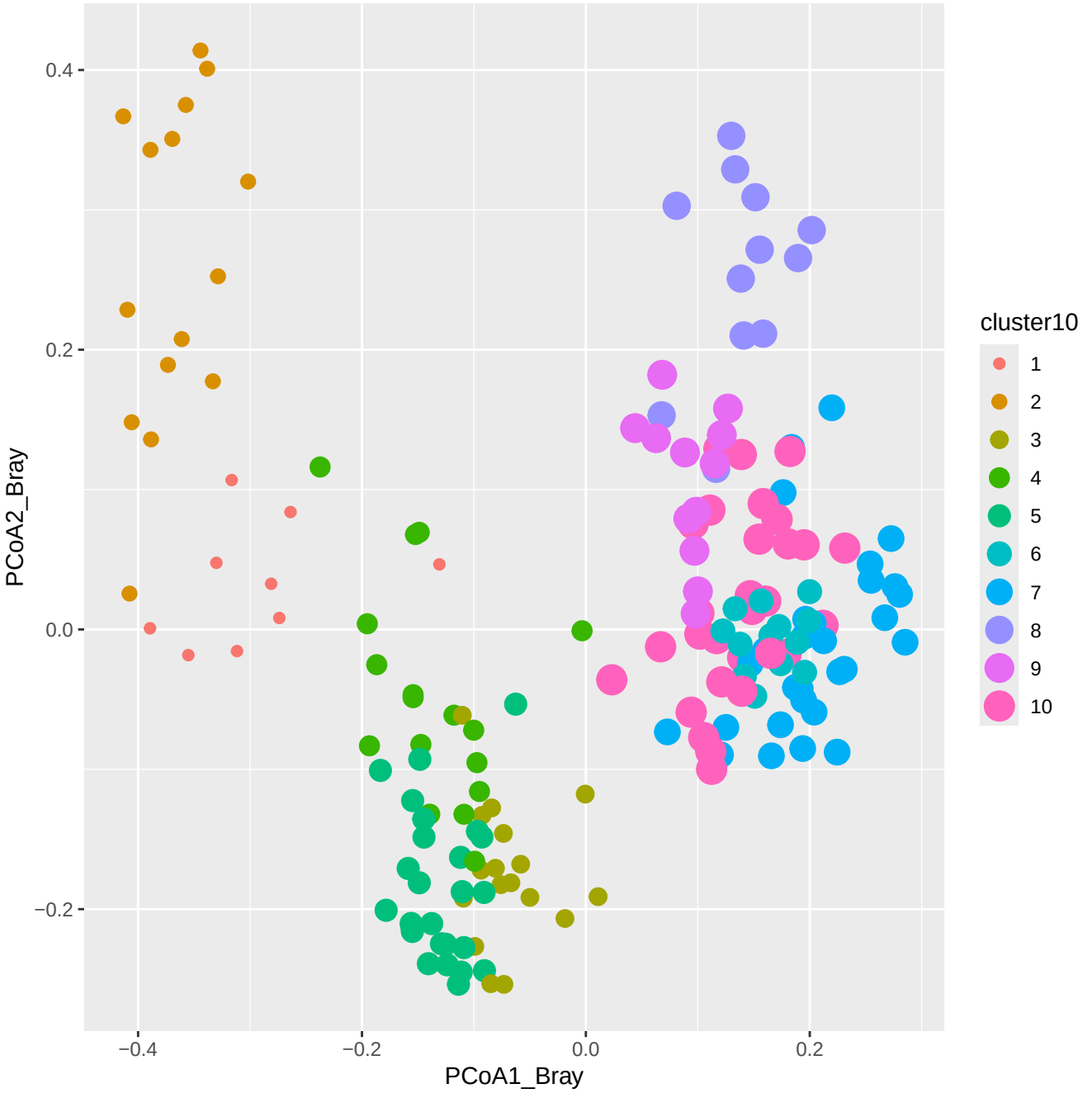
0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster10
1
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